
SS1 MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: NUMBER BASE SYSTEMS

Concept: Understanding How Numbers Are Represented in Different Counting Systems

1. INTRODUCTION: BEYOND BASE 10

Welcome to SS1 Mathematics! Our first topic is the **Number Base System** - the fundamental framework that determines how we write, interpret, and calculate with numbers.

What is a Number Base?

A **number base** (or **radix**) is the number of unique digits, including zero, used to represent numbers in a positional numeral system.

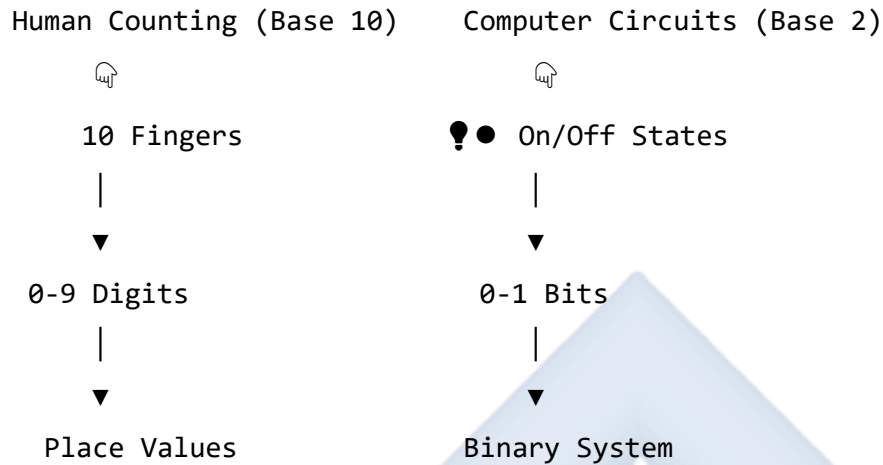
Base	Name	Digits Used	Visual Representation
2	Binary	0, 1	● ○ (On/Off)
8	Octal	0 – 7	
10	Decimal	0 – 9	(Our fingers)
16	Hexadecimal	0 – 9, A – F	(Web colors)

The "Reset" Concept: The base number tells us when counting "resets" to the next place value. For example:

- **Base 10:** After 9 comes **10** (1 group of ten + 0 ones)
 - **Base 2:** After 1 comes **10** (1 group of two + 0 ones)
 - **Base 5:** After 4 comes **10** (1 group of five + 0 ones)
-

Real-World Connections

Diagram 1: Why Different Bases Exist



Practical Applications:

- **Base 10 (Decimal):** Everyday counting, money, measurements
- **Base 2 (Binary):** Computer processors, digital storage
- **Base 16 (Hexadecimal):** Web design (#FF0000 = red), memory addresses
- **Base 60:** Time (60 seconds = 1 minute), angles (60' = 1°)

Core Concept: Place Value Expansion

The General Formula: For any number $(abc.def)_x$ in base x :

$$(a \times x^2) + (b \times x^1) + (c \times x^0) + (d \times x^{-1}) + (e \times x^{-2}) + (f \times x^{-3})$$

Visual Place Value Chart:

Base x Number: (2 4 3 . 1 2)₅

Place Values:	5 ²	5 ¹	5 ⁰	.	5 ⁻¹	5 ⁻²
	25	5	1	.	0.2	0.04

2. CONTENT

A. CONVERSION FROM OTHER BASES TO BASE 10

Method: Polynomial Expansion

Step-by-Step Procedure:

1. Write the number and identify its base
 2. Label each digit with its place value (powers of the base)
 3. Multiply each digit by its place value
 4. Sum all the products
-

Worked Example 1.1: Convert $(243)_5$ to Base 10

Visual Expansion:

$$\begin{array}{ccc} (2 & 4 & 3)_5 \\ \downarrow & \downarrow & \downarrow \\ 2 \times 5^2 & + & 4 \times 5^1 & + & 3 \times 5^0 \\ \downarrow & \downarrow & \downarrow \\ 2 \times 25 & + & 4 \times 5 & + & 3 \times 1 \\ \downarrow & \downarrow & \downarrow \\ 50 & + & 20 & + & 3 & = & 73_{10} \end{array}$$

Calculation Table:

Digit	Place Value	Power	Value	Cumulative
2	25	5^2	$2 \times 25 = 50$	50
4	5	5^1	$4 \times 5 = 20$	70
3	1	5^0	$3 \times 1 = 3$	73

✓ **Answer:** $(243)_5 = 73_{10}$

Worked Example 1.2: Convert $(11011)_2$ to Base 10

Binary Place Value Chart:

1	1	0	1	1 ₂
↓	↓	↓	↓	↓
2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
16	8	4	2	1
↓	↓	↓	↓	↓
16	+ 8	+ 0	+ 2	+ 1 = 27 ₁₀

✓ Answer: $(11011)_2 = 27_{10}$

Worked Example 1.3: Convert $(12.34)_5$ to Base 10

Fractional Expansion:

Integer Part:

$$\begin{array}{rcl} (1 & 2)_5 & \\ \downarrow & \downarrow & \\ 1 \times 5^1 & + & 2 \times 5^0 \\ \downarrow & \downarrow & \\ 1 \times 5 & + & 2 \times 1 \\ \downarrow & \downarrow & \\ 5 & + & 2 = 7 \end{array}$$

Fractional Part:

$$\begin{array}{rcl} (3 & 4)_5 & \\ \downarrow & \downarrow & \\ 3 \times 5^{-1} & + & 4 \times 5^{-2} \\ \downarrow & \downarrow & \\ 3 \times 0.2 & + & 4 \times 0.04 \\ \downarrow & \downarrow & \\ 0.6 & + & 0.16 = 0.76 \end{array}$$

✓ Answer: $(12.34)_5 = 7.76_{10}$

B. CONVERSION FROM BASE 10 TO OTHER BASES

Method 1: Repeated Division (for Integer Part)

Algorithm:

1. Divide the number by the target base
2. Record the remainder
3. Use the quotient for the next division
4. Repeat until quotient becomes 0
5. Read remainders from bottom to top

Diagram: The Division Ladder

Remainders

↑

[1] ← Last division

÷ 2

[1]

÷ 2

[0]

59 → ÷ 2

[1]

÷ 2

[1]

÷ 2

[1] ← First remainder

Worked Example 1.4: Convert 59_{10} to Base 2

Division Process:

Step	Division	Quotient	Remainder	Note
1	$59 \div 2$	29	1	(LSB) Least Significant Bit
2	$29 \div 2$	14	1	
3	$14 \div 2$	7	0	
4	$7 \div 2$	3	1	

Step	Division	Quotient	Remainder	Note
5	$3 \div 2$	1	1	
6	$1 \div 2$	0	1	(MSB) Most Significant Bit

Reading from bottom to top: 111011_2

✓ **Answer: $59_{10} = 111011_2$**

Verification: $1 \times 32 + 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 1 \times 1 = 59$ ✓

Method 2: Repeated Multiplication (for Fractional Part)

Algorithm:

1. Multiply the fraction by the target base
2. Record the integer part of the result
3. Continue with the fractional part
4. Stop when fractional part becomes 0 or pattern repeats

Worked Example 1.5: Convert 0.375_{10} to Base 2

Multiplication Process:

Step	Operation	Integer	Fraction	Binary Digit
1	$0.375 \times 2 = 0.75$	0	0.75	0
2	$0.75 \times 2 = 1.5$	1	0.5	1
3	$0.5 \times 2 = 1.0$	1	0.0	1

Reading from top to bottom: 0.011_2

✓ **Answer: $0.375_{10} = 0.011_2$**

Worked Example 1.6: Convert 26.8_{10} to Base 5

Integer Part (26_{10}):

$$26 \div 5 = 5 \text{ remainder } 1$$

$$5 \div 5 = 1 \text{ remainder } 0$$

$$1 \div 5 = 0 \text{ remainder } 1$$

↑ Read bottom to top: 101_5

Fractional Part (0.8_{10}):

$$0.8 \times 5 = 4.0 \rightarrow \text{integer} = 4, \text{ fraction} = 0$$

→ 0.4_5

✓ **Answer:** $26.8_{10} = 101.4_5$

C. ARITHMETIC OPERATIONS IN DIFFERENT BASES

Golden Rule: All operations follow the same logic as base 10, but the "carry" and "borrow" values equal the base.

Addition in Different Bases

Key Concept: When $\text{sum} \geq \text{base}$, carry over to next column

Worked Example 1.7: Add $243_5 + 134_5$

Step-by-Step Solution:

$$\begin{array}{r} 243_5 \\ + 134_5 \\ \hline \end{array}$$

Column-by-column calculation:

Column	Calculation	Base 5 Conversion	Write	Carry
Units	$3 + 4 = 7$	$7 \div 5 = 1 \text{ remainder } 2$	2	1
Fives	$4 + 3 + 1 = 8$	$8 \div 5 = 1 \text{ remainder } 3$	3	1
Twenty-fives	$2 + 1 + 1 = 4$	$4 < 5$	4	0

✓ **Answer: 432_5**

Verification: $243_5 = 73_{10}$, $134_5 = 44_{10}$, $73 + 44 = 117_{10} = 432_5$ ✓

Subtraction in Different Bases

Key Concept: Borrowing gives you the BASE value, not 10

Worked Example 1.8: Subtract $412_8 - 153_8$

Detailed Borrowing Process:

$$\begin{array}{r} 412_8 \\ - 153_8 \\ \hline \end{array}$$
 Step 1: $2 < 3$, borrow from middle column
 Middle column: 1 becomes 0, units get 8 (base value)
 Now: $2 + 8 = 10$, $10 - 3 = 7$

$$\begin{array}{r} 4012_8 \\ - 153_8 \\ \hline \end{array}$$
 Step 2: $0 < 5$, borrow from left column
 Left column: 4 becomes 3, middle gets 8
 Now: $0 + 8 = 8$, $8 - 5 = 3$

$$\begin{array}{r} 3812_8 \\ - 153_8 \\ \hline \end{array}$$
 Step 3: $3 - 1 = 2$

$$\begin{array}{r} 237_8 \end{array}$$

✓ **Answer: 237_8**

Multiplication in Different Bases

Key Concept: Use multiplication tables in the given base

Worked Example 1.9: Multiply $110_2 \times 11_2$

Long Multiplication:

$$\begin{array}{r} 110_2 \\ \times 11_2 \\ \hline 110 \quad \leftarrow 110 \times 1 \\ + 110 \quad \leftarrow 110 \times 10 \text{ (shifted left)} \\ \hline 10010_2 \end{array}$$

Step Verification:

- In base 2: $1 \times 110 = 110$
- In base 2: $10 \times 110 = 1100$ (equivalent to shifting left)
- Addition: $110 + 1100 = 10010_2$

✓ **Answer: 10010_2**

Decimal Check: $110_2 = 6_{10}$, $11_2 = 3_{10}$, $6 \times 3 = 18_{10} = 10010_2$ ✓

D. APPLICATIONS IN COMPUTING AND DIGITAL SYSTEMS

Diagram: How Computers Use Different Bases

Raw Data: 01011010 10101101 11110000

Grouped (Hex): 5 A A D F 0

Hexadecimal: 5A AD F0

Memory Address: 0x5AADF0

Practical Applications Table:

Base	Primary Use	Real-World Example
Binary (2)	CPU operations, file storage	Computer processors, RAM
Octal (8)	Unix permissions, legacy systems	Linux: <code>chmod 755</code>
Hexadecimal (16)	Memory addresses, color codes	CSS: <code>#FF5733</code> = RGB(255,87,51)

Color Code Example:

FF 57 33₁₆

		└	Red component: 33 ₁₆ = 51 ₁₀
	└		Green component: 57 ₁₆ = 87 ₁₀
└			Blue component: FF ₁₆ = 255 ₁₀

3. REVISION QUESTIONS (WEEK 1)

1. Why do humans naturally use base 10, while computers use base 2?
2. Explain why the number (396)₇ is invalid.
3. Describe the difference between the repeated division and repeated multiplication methods.
4. How does borrowing work in base 8 compared to base 10?
5. Give three real-world applications of hexadecimal numbers.

4. ASSESSMENT (WEEK 1)

EVALUATION (IN-CLASS)

1. Convert (324)₅ to base 10.
2. Convert (101101)₂ to base 10.
3. Convert (127)₁₀ to base 8.
4. Convert (0.625)₁₀ to base 2.
5. Calculate (132)₄ + (211)₄.

ASSIGNMENT (TAKE-HOME)

1. Convert $(2A9)_{16}$ to base 10. (Hint: A = 10, B = 11, C = 12, D = 13, E = 14, F = 15)
 2. Convert $(158)_{10}$ to base 5.
 3. Convert $(44.75)_{10}$ to base 2.
 4. Calculate $(503)_6 - (245)_6$.
 5. Calculate $(23)_5 \times (14)_5$ using long multiplication.
 6. **Challenge:** A computer uses 8-bit binary. What is the largest number it can represent?
What happens if you add 1 to this number?
-

Quick Reference: Base Conversion Cheat Sheet

To Decimal: Expand using place values **From Decimal (Integer):** Repeated division, read remainders upward **From Decimal (Fraction):** Repeated multiplication, read integers downward
Between Binary/Hex: Group binary digits in 4s ($2^3 = 8$, $2^4 = 16$)

Common Bases in Computing:

- 1 byte = 8 bits = 2 hexadecimal digits
 - 1 kibibyte = 1024 bytes = 2^{10} bytes
 - RGB colors: 00 to FF for each component (0-255 in decimal)
-

Answers to Assessment

In-Class:

1. 89
2. 45
3. 177
4. 0.101
5. 1003

Take-Home:

1. 681
2. 1113
3. 101100.11
4. 214
5. 432
6. 255; overflow to 00000000 (0)



WEEK 2: MODULAR ARITHMETIC

Concept: The Mathematics of Cycles and Remainders

1. INTRODUCTION: THE CLOCK AS A MATHEMATICAL MODEL

Imagine a clock. If it's 9 o'clock now, and you add 5 hours, it becomes 2 o'clock, not 14 o'clock. This "wrapping around" behavior is the essence of **Modular Arithmetic**.

Core Idea: In Modular Arithmetic, we are only interested in the **remainder** after division by a fixed number called the **modulus**. The clock's modulus is 12.

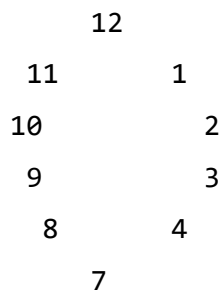
Let's formalize our clock example:

- Normal Arithmetic: $9 + 5 = 14$
- Modular Arithmetic (mod 12): $14 \div 12 = 1$ remainder 2.

So, we write: $9 + 5 \equiv 2 \pmod{12}$ This is read as: "9 plus 5 is congruent to 2, modulo 12."

🕒 Visual Foundation: The Clock & Number Line Analogy

Diagram 1: The Classic Clock



Each hour we move, we cycle around the clock face. After 12, we return to 1.

Diagram 2: The Circular Number Line (Mod 5)

	0	
4		1
3		2

This is a more general "clock" for a modulus of 5. The numbers 0, 1, 2, 3, 4 are equally spaced. If you start at 2 and add 4, you go (2→3→4→0→1), landing on **1**. So, $2 + 4 \equiv 1 \pmod{5}$.

Diagram 3: Linear Number Line Wrapping into a Circle

... -5, 0, 5, 10 ... all map to $\rightarrow 0 \pmod{5}$
 ... -4, 1, 6, 11 ... all map to $\rightarrow 1 \pmod{5}$
 ... -3, 2, 7, 12 ... all map to $\rightarrow 2 \pmod{5}$
 ... -2, 3, 8, 13 ... all map to $\rightarrow 3 \pmod{5}$
 ... -1, 4, 9, 14 ... all map to $\rightarrow 4 \pmod{5}$

Note: Every number on the number line "wraps around" to one of the five residues $\{0, 1, 2, 3, 4\}$ in a modulo 5 system. These groups are called **congruence classes**.

2. CONTENT

A. The Formal Concept of Congruence

Two numbers, **a** and **b**, are said to be **congruent modulo m** if their difference ($a - b$) is exactly divisible by the modulus **m**.

Mathematical Definition: $a \equiv b \pmod{m}$ if and only if $m \mid (a - b)$ (meaning "m divides ($a - b$)").

Example: Is $17 \equiv 5 \pmod{6}$? Check: $17 - 5 = 12$. Is 12 divisible by 6? Yes ($12 \div 6 = 2$). ✓
 Therefore, $17 \equiv 5 \pmod{6}$.

B. Finding the Remainder (The Value Modulo m)

To find **a (mod m)**:

1. Divide **a** by **m**.
2. The **remainder** (which must be between 0 and $m-1$) is your result.

Worked Illustrations

Example	Operation & Calculation	Explanation & Conclusion
25 (mod 4)	$25 \div 4 = 6$ with a remainder of 1	✓ $25 = 6 \times 4 + 1$, so $25 \equiv 1 \pmod{4}$
10 (mod 7)	$10 \div 7 = 1$ with a remainder of 3	✓ $10 = 1 \times 7 + 3$, so $10 \equiv 3 \pmod{7}$

C. Handling Negative Numbers

For negative numbers, we want a **positive remainder**. We achieve this by repeatedly **adding the modulus** until we get a non-negative number less than m .

Example: Find $-14 \pmod{3}$ Think: "What number between 0 and 2 is equivalent to -14 ?" $-14 + 3 = -11$ $-11 + 3 = -8$ $-8 + 3 = -5$ $-5 + 3 = -2$ $-2 + 3 = 1$ ✓ **Answer: $-14 \equiv 1 \pmod{3}$**

Quick Check: $-14 \div 3 = -5$ remainder 1.

3. OPERATIONS IN MODULAR ARITHMETIC

The power of modular arithmetic is that we can perform operations (+, -, $\times$) on the "smaller" remainders instead of the large original numbers.

Key Principle: You can reduce numbers using the modulus *before* or *after* performing operations. The result will be congruent.

Operation	General Rule	Efficient Method
Addition	$(a + b) \bmod m$	Find $(a \bmod m)$ and $(b \bmod m)$, add them, then take mod m .
Subtraction	$(a - b) \bmod m$	Find $(a \bmod m)$ and $(b \bmod m)$, subtract, then take mod m . (If result is negative, add m).
Multiplication	$(a \times b) \bmod m$	Find $(a \bmod m)$ and $(b \bmod m)$, multiply them, then take mod m .

Note: This works because of the distributive properties of modular arithmetic. For example, if $a = mq_1 + r_1$ and $b = mq_2 + r_2$, then $a + b = m(q_1 + q_2) + (r_1 + r_2)$. The term with 'm' disappears when we take mod m, leaving only $(r_1 + r_2) \bmod m$.

4. WORKED EXAMPLES (WITH DEEPER EXPLANATION)

Example 2.1: Evaluate $47 \pmod{6}$

- **Solution:** $47 \div 6 = 7$ remainder **5**.
- ✓ **Answer:** $47 \equiv 5 \pmod{6}$. This means on a "clock" of size 6, 47 o'clock is the same as 5 o'clock.

Example 2.2: Evaluate $(38 + 29) \pmod{3}$

- **Method 1 — Add First:** $38 + 29 = 67$. $67 \div 3 = 22$ remainder **1**.
- **Method 2 — Mod First (More Efficient):** $38 \pmod{3} = 2$ (since $38 \div 3 = 12 \text{ r}2$) $29 \pmod{3} = 2$ (since $29 \div 3 = 9 \text{ r}2$) $(2 + 2) = 4$. $4 \pmod{3} = 1$.
- ✓ **Answer:** **1 (mod 3)**. Both methods agree!

Example 2.3: Compute $(15 \times 22) \pmod{7}$

- **Method 1 — Multiply First:** $15 \times 22 = 330$. $330 \div 7 = 47$ remainder **1**.
- **Method 2 — Mod First (Much Easier!):** $15 \pmod{7} = 1$. $22 \pmod{7} = 1$. $(1 \times 1) = 1$. $1 \pmod{7} = 1$.
- ✓ **Answer:** **1 (mod 7)**. This shows the power of simplification.

Example 2.4: Evaluate $(25 - 12) \pmod{8}$

- **Solution using mod first:** $25 \pmod{8} = 1$. $12 \pmod{8} = 4$. $(1 - 4) = -3$. Since the result is negative, we add the modulus: $-3 + 8 = 5$.
- ✓ **Answer:** **5 (mod 8)**.

Example 2.5: Evaluate $(83 \times 56 + 12) \pmod{9}$

- **Solution using mod first:** $83 \pmod{9}$: $83 \div 9 = 9 \text{ r}2 \rightarrow 2$. $56 \pmod{9}$: $56 \div 9 = 6 \text{ r}2 \rightarrow 2$. $12 \pmod{9}$: $12 \div 9 = 1 \text{ r}3 \rightarrow 3$. Now compute: $(2 \times 2 + 3) = (4 + 3) = 7$. $7 \pmod{9} = 7$.
- ✓ **Answer: 7 (mod 9).**

5. APPLICATIONS TO DAILY LIFE & COMPUTER SCIENCE

A. Days of the Week — A Mod 7 System

We assign: Sun=0, Mon=1, Tue=2, Wed=3, Thu=4, Fri=5, Sat=6. **Q:** If today is Wednesday (3), what day is 90 days from now?

- $(3 + 90) \pmod{7} = 93 \pmod{7}$.
- $93 \div 7 = 13 \text{ remainder } 2$.
- ✓ **Answer: 2 → Tuesday.**

B. Clock Time — Mod 12 or Mod 24

Q: If it's 7:00 PM (19:00) now, what time will it be in 40 hours?

- $(19 + 40) \pmod{24} = 59 \pmod{24}$.
- $59 \div 24 = 2 \text{ remainder } 11$.
- ✓ **Answer: 11:00 AM.**

C. Cryptography & Computer Science

- **ISBN Check Digits:** The last digit of a book's ISBN code is calculated using mod 11 to detect errors in typing or scanning.
- **Cyclic Redundancy Checks (CRCs):** Used in networks and data storage (like on a CD) to detect accidental changes to raw data. It relies heavily on polynomial division modulo 2.
- **Ciphers:** Simple ciphers like the Caesar cipher are just modular arithmetic on the alphabet (mod 26).

6. REVISION QUESTIONS (WEEK 2)

1. What is another name for modular arithmetic? (*Hint: Think about what we are left with.*)
 2. Explain in your own words what $a \equiv b \pmod{m}$ means.
 3. Describe the process for finding the positive value of a negative number like $-17 \pmod{5}$.
 4. In a modulo 7 system, what are the only possible results for any calculation?
 5. Mention three real-life examples of modular arithmetic (beyond clocks and days).
-

7. ASSESSMENT (WEEK 2)

EVALUATION (IN-CLASS)

1. Evaluate $100 \pmod{9}$.
2. Evaluate $-20 \pmod{6}$.
3. Calculate $(45 + 52) \pmod{5}$.
4. Calculate $(14 \times 10) \pmod{4}$.
5. If a cycle repeats every 4 days and today is day 0, what day of the cycle will it be in 100 days?

ASSIGNMENT (TAKE-HOME)

1. Find the simplest positive value of x : a. $x \equiv 55 \pmod{8}$ b. $x \equiv -30 \pmod{7}$
 2. Calculate $(48 - 19) \pmod{3}$.
 3. Evaluate $(12 \times 11 \times 10) \pmod{6}$.
 4. A worker has a 3-day cycle: On, On, Off. If the first "On" day is Monday, what day will the 15th "On" day fall on? (*Hint: The 1st "On" day is day 1. When is the next one?*)
 5. **Challenge Problem:** Find n (where n is between 0 and 4) if $4n \equiv 3 \pmod{5}$. (*Hint: Try $n = 0, 1, 2, 3, 4$ and see which one works.*) This is the beginning of solving modular equations!
-

Answers to Assessment

In-Class:

1. 1
2. 4
3. 2
4. 0
5. 0

Take-Home:

1. a) 7, b) 5
2. 2
3. 0
4. Friday
5. $n = 2$



WEEK 3: INDICES (POWERS AND EXPONENTS)

Concept: The Mathematics of Repeated Multiplication and Exponential Growth

1. INTRODUCTION: THE POWER OF REPETITION

Indices (also called exponents or powers) are mathematical shorthand for repeated multiplication. They help us express very large and very small numbers efficiently.

Basic Terminology:

$$2 \times 2 \times 2 \times 2 \times 2 = 2^5$$

Exponent (Power)

Base

Repeated Multiplication

Read as: "Two raised to the power of five" or "Two to the fifth power"

Real-World Exponential Growth

Diagram 1: Bacterial Growth (Doubling Every Hour)

Hour 0:	•	= 1 = 2^0
Hour 1:	••	= 2 = 2^1
Hour 2:	••••	= 4 = 2^2
Hour 3:	••••••••	= 8 = 2^3
Hour 4:	••••••••••••••	= 16 = 2^4

Diagram 2: Compound Interest Growth

Year 0:	¥1,000 = 1000×2^0
Year 1:	¥2,000 = 1000×2^1
Year 2:	¥4,000 = 1000×2^2

Year 3: ₦8,000 = 1000×2^3

Exponential growth curve

2. CONTENT

A. STANDARD FORM (SCIENTIFIC NOTATION)

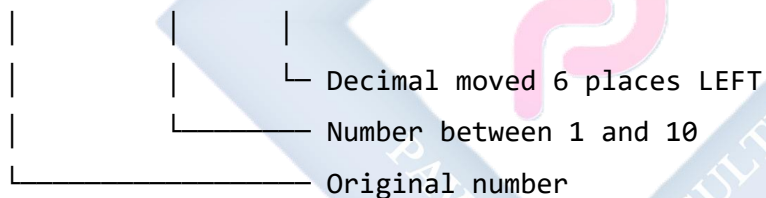
Standard form helps us write extremely large or extremely small numbers efficiently.

Format: $A \times 10^n$ where $1 \leq A < 10$ and n is an integer

Visual Guide: Decimal Point Movement

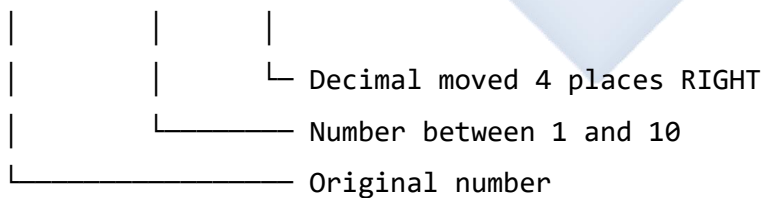
Large Numbers → Positive Exponents:

$$4,750,000 = 4.75 \times 10^6$$



Small Numbers → Negative Exponents:

$$0.00032 = 3.2 \times 10^{-4}$$



Memory Aid:

← LEFT = POSITIVE exponent RIGHT → = NEGATIVE exponent

Large numbers (4,750,000) Small numbers (0.00032)

B. THE SEVEN LAWS OF INDICES

Visual Summary: The Index Law Family

Addition Family (Multiplication)

$$a^m \times a^n = a^{m+n}$$

↓ ↓ ↓

Same base → ADD exponents

Subtraction Family (Division)

$$a^m \div a^n = a^{m-n}$$

↓ ↓ ↓

Same base → SUBTRACT exponents

Multiplication Family (Power of Power)

$$(a^m)^n = a^{m \cdot n}$$

↓ ↓

Multiply the exponents

Special Cases Family:

$$a^0 = 1 \text{ (Zero Power)}$$

$$a^{1/n} = \sqrt[n]{a} \text{ (Root Power)}$$

$$a^{-n} = 1/a^n \text{ (Negative Power)}$$

$$a^{m/n} = (\sqrt[n]{a})^m \text{ (Combined Power)}$$

Detailed Law Breakdown:

Law	Rule	Example	Visual Explanation
1. Multiplication	$a^m \times a^n = a^{m+n}$	$x^5 \times x^3 = x^8$	$x \cdot x \cdot x \cdot x \cdot x \times x \cdot x \cdot x = x^8$
2. Division	$a^m \div a^n = a^{m-n}$	$y^9 \div y^2 = y^7$	Cancel 2 y's: $\frac{y \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y}{y \cdot y} = y^7$
3. Power of Power	$(a^m)^n = a^{m \cdot n}$	$(k^3)^4 = k^{12}$	$(k \cdot k \cdot k) \times (k \cdot k \cdot k) \times (k \cdot k \cdot k) \times (k \cdot k \cdot k) = k^{12}$
4. Zero Power	$a^0 = 1$	$7^0 = 1$	Pattern: $7^3 = 343, 7^2 = 49, 7^1 = 7, 7^0 = 1$
5. Negative Power	$a^{-n} = \frac{1}{a^n}$	$3^{-2} = \frac{1}{9}$	Opposite of positive power

Law	Rule	Example	Visual Explanation
6. Fractional Power	$a^{1/n} = \sqrt[n]{a}$	$64^{1/3} = 4$	"What number cubed equals 64?"
7. Combined Fractional	$a^{m/n} = (\sqrt[n]{a})^m$	$27^{2/3} = 9$	Cube root of 27 = 3, then square = 9

Conceptual Understanding:

Why does $a^0 = 1$?

Pattern Observation:

$$3^4 = 81$$

$$3^3 = 27 \quad (81 \div 3)$$

$$3^2 = 9 \quad (27 \div 3)$$

$$3^1 = 3 \quad (9 \div 3)$$

$$3^0 = 1 \quad (3 \div 3) \leftarrow \text{Continuing the pattern!}$$

Negative Exponents as Reciprocals:

$$2^3 = 8$$

$$2^{-3} = 1/8$$

|

|

↳ Positive
exponent

↳ Reciprocal
of positive

C. SOLVING EXPONENTIAL EQUATIONS

Golden Rule: If $a^x = a^y$ and $a \neq 0, 1$, then $x = y$

Strategy 1: Make Bases the Same

Example: $8 = 2^3$, $27 = 3^3$, $16 = 4^2 = 2^4$

Common conversions:

$$4 = 2^2 \quad 9 = 3^2 \quad 25 = 5^2$$

$$8 = 2^3 \quad 27 = 3^3 \quad 125 = 5^3$$

$$16 = 2^4 \quad 81 = 3^4 \quad 625 = 5^4$$

Strategy 2: Substitution for Quadratic Forms When you have $a^{2x} + b \cdot a^x + c = 0$, let $y = a^x$

3. DETAILED WORKED EXAMPLES

Example 3.1: Simplify $\left(\frac{81}{16}\right)^{-3/4}$

Step-by-Step Solution:

Step 1: Handle negative exponent (flip fraction)

$$(81/16)^{-3/4} = (16/81)^{3/4}$$

Step 2: Apply fractional exponent (root then power)

$$= (\sqrt[4]{16/81})^3$$

Step 3: Calculate fourth roots

$$\sqrt[4]{16} = 2, \sqrt[4]{81} = 3$$

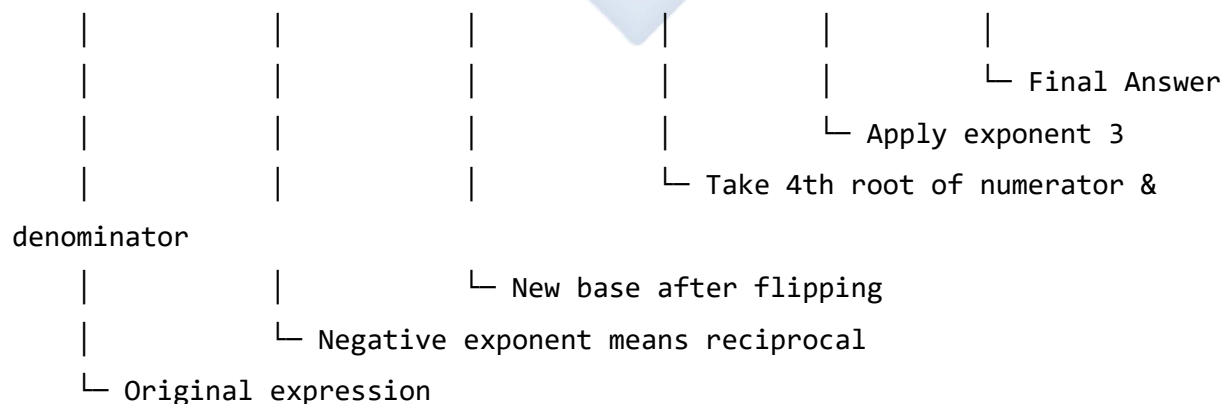
$$= (2/3)^3$$

Step 4: Cube the fraction

$$= 8/27$$

Visual Process:

$$(81/16)^{-3/4} \rightarrow [\text{FLIP}] \rightarrow (16/81)^{3/4} \rightarrow [\text{ROOT}] \rightarrow (2/3)^3 \rightarrow [\text{POWER}] \rightarrow 8/27$$



✓ **Answer: 8/27**

Example 3.2: Simplify $\frac{(x^2y^{-1})^3}{x^4y^5}$

Step-by-Step Solution:

Step 1: Apply power to numerator

$$(x^2y^{-1})^3 = x^6y^{-3}$$

Step 2: Write the full expression

$$= x^6y^{-3} / (x^4y^5)$$

Step 3: Apply division law (subtract exponents)

$$x^{6-4} y^{-3-5} = x^2 y^{-8}$$

Step 4: Handle negative exponent

$$= x^2 / y^8$$

Exponent Arithmetic:

Numerator exponents: x^6y^{-3}

Denominator exponents: x^4y^5

Result: $x^{6-4} y^{-3-5} = x^2 y^{-8}$

✓ **Answer:** x^2/y^8

Example 3.3: Solve $5^x = 125$

Solution:

$$5^x = 125$$

$$5^x = 5^3 \quad \leftarrow \text{Express 125 as power of 5}$$

$$x = 3 \quad \leftarrow \text{Equate exponents (same base)}$$

✓ **Answer:** $x = 3$

Example 3.4: Solve $9^{x-1} = 27^{x+2}$

Step-by-Step Solution:

Step 1: Express both sides with base 3

$$9^{x-1} = (3^2)^{x-1} = 3^{2(x-1)}$$

$$27^{x+2} = (3^3)^{x+2} = 3^{3(x+2)}$$

Step 2: Equate the expressions

$$3^{2(x-1)} = 3^{3(x+2)}$$

Step 3: Equate exponents (same base)

$$2(x - 1) = 3(x + 2)$$

Step 4: Solve the linear equation

$$2x - 2 = 3x + 6$$

$$2x - 3x = 6 + 2$$

$$-x = 8$$

$$x = -8$$

Base Conversion Visualization:

$$9^{x-1} = 27^{x+2}$$

$$\downarrow \quad \downarrow$$

$$(3^2)^{x-1} = (3^3)^{x+2}$$

$$\downarrow \quad \downarrow$$

$$3^{2x-2} = 3^{3x+6}$$

✓ **Answer: $x = -8$**

Example 3.5: Solve $4^x - 3(2^x) - 4 = 0$

Step-by-Step Solution:

Step 1: Recognize that $4^x = (2^x)^2$

Let $y = 2^x$, then $4^x = y^2$

Step 2: Substitute

$$y^2 - 3y - 4 = 0$$

Step 3: Factor the quadratic

$$(y - 4)(y + 1) = 0$$

Step 4: Solve for y

$$y = 4 \text{ or } y = -1$$

Step 5: Back-substitute $y = 2^x$

Case 1: $2^x = 4 \rightarrow 2^x = 2^2 \rightarrow x = 2$

Case 2: $2^x = -1 \rightarrow$ No solution ($2^x > 0$ for all real x)

Substitution Method Visualization:

Original: $4^x - 3(2^x) - 4 = 0$

$$\begin{array}{ccc} | & | & \\ | & \text{└─ This is } y & \\ \text{└─ This is } y^2 & & \end{array}$$

Becomes: $y^2 - 3y - 4 = 0$ ← Much easier to solve!

✓ **Answer: $x = 2$**

4. REVISION QUESTIONS (WEEK 3)

1. Explain the conceptual difference between a^{-n} and $a^{1/n}$ using examples.
2. Convert 0.000507 to standard form and explain each step.
3. Using the pattern method, show why any non-zero number raised to power 0 equals 1.
4. State the golden rule for solving exponential equations and explain why it works.

5. Compare and contrast:

- $(x^2)^3$ vs $x^2 \times x^3$
- Which gives x^5 and which gives x^6 ? Why?

5. ASSESSMENT (WEEK 3)

EVALUATION (IN-CLASS)

1. Write 5.67×10^{-4} as an ordinary number.
2. Simplify $\frac{a^3b^2}{a^5b^{-1}}$
3. Evaluate $100^{1/2} \times 100^0$
4. Evaluate $\left(\frac{64}{9}\right)^{-1/2}$
5. Solve for x: $4^{x+1} = 8$

ASSIGNMENT (TAKE-HOME)

1. Simplify $\sqrt{x^3y^4} \times (x^2y^3)^{-1/2}$
2. Evaluate $\frac{5^n \times 25^{n-1}}{125^{n+1}}$ as a power of 5
3. Solve for k: $16^{2k-3} = 32^{1-k}$
4. Solve for y: $3^{2y} - 10(3^y) + 9 = 0$
5. If $2^x = 5$, find 2^{x+2}

6. BONUS: POWERS OF 2 VISUALIZATION

Table: The Exponential Growth of Powers of 2

Power	Expression	Value	Real-World Equivalent
2^0	1	1	Single bacterium
2^1	2	2	Two coins

Power	Expression	Value	Real-World Equivalent
2^2	2×2	4	Four cardinal directions
2^3	$2 \times 2 \times 2$	8	Bits in a byte (almost)
2^4	$2 \times 2 \times 2 \times 2$	16	Squares on chessboard (one side)
2^5	...	32	Adult human teeth
2^6	...	64	Squares on chessboard (total)
2^7	...	128	ASCII character set size
2^8	...	256	One byte (8 bits)
2^{10}	...	1,024	One kilobyte (in computing)

Negative Powers (The Shrinking Pattern):

Power	Expression	Value	Explanation
2^{-1}	$1 \div 2$	0.5	Half
2^{-2}	$1 \div 4$	0.25	Quarter
2^{-3}	$1 \div 8$	0.125	Eighth

Memory Aid: The Exponent Rules Song

When multiplying, add the powers high,
 When dividing, subtract them by and by,
 A power of a power, multiply you'll see,
 Anything to zero power is one, you'll agree!
 Negative power means flip it right,
 Fractional power is a root, that's the light!

Answers to Assessment

In-Class:

1. 0.000567

2. $\frac{b^3}{a^2}$

3. 10

4. $\frac{3}{8}$

5. $x = \frac{1}{2}$

Take-Home:

1. $x^{1/2}y^{5/2}$

2. 5^{-n-4}

3. $k = \frac{11}{7}$

4. $y = 0$ or $y = 2$

5. 20



WEEK 4: LOGARITHMS

Concept: The Inverse Operation of Exponentiation - Finding the Exponent

1. INTRODUCTION: THE MISSING PIECE IN EXPONENTIAL EQUATIONS

Last week, we studied indices (powers), where we raised a base to an exponent to get a number:

$$10^3 = 1000$$

| | |
| | | $\swarrow$ Result
| $\swarrow$ Exponent
 $\swarrow$ Base

This week, we study the **inverse operation** — logarithms. A logarithm answers the question:

"What power must I raise the base to, in order to get a given number?"

The Inverse Relationship: Logarithms vs. Exponents

Diagram 1: Inverse Operations Comparison

Addition $\leftrightarrow$ Subtraction

$$5 + 3 = 8$$

$$8 - 3 = 5$$

Multiplication $\leftrightarrow$ Division

$$5 \times 3 = 15$$

$$15 \div 3 = 5$$

Exponentiation $\leftrightarrow$ Logarithm

$$10^3 = 1000$$

| | |
| | | $\swarrow$ Result
| $\swarrow$ Exponent
 $\swarrow$ Base

$$\log_{10} 1000 = 3$$

| | |
| | | $\swarrow$ Exponent
| $\swarrow$ Result
 $\swarrow$ Base

Real-World Analogy: The Detective Story

CRIME SCENE: $10^3 = 1000$

We know: Base = 10, Result = 1000

MYSTERY: What was the exponent?

LOGARITHM DETECTIVE: "The exponent was 3!"

$$\log_{10} 1000 = 3$$

2. CONTENT

A. FORMAL DEFINITION OF LOGARITHMS

Mathematical Definition: If $y = a^x$, then $x = \log_a y$

Where:

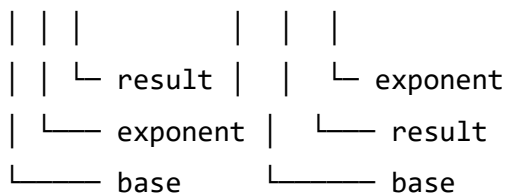
- $a \rightarrow$ **base** (must be positive and $\neq 1$)
- $y \rightarrow$ **argument** (the result in exponential form)
- $x \rightarrow$ **logarithm** (the exponent we're looking for)

Memory Aid: "The logarithm is the power!"

Visual: Converting Between Forms

Exponential Form $\rightarrow$ Logarithmic Form

$$a^x = y \iff \log_a y = x$$



Examples 4.1: Converting Between Forms

Exponential Form	Logarithmic Form	Verbal Explanation
$10^2 = 100$	$\log_{10} 100 = 2$	"10 to what power gives 100? The answer is 2"
$3^4 = 81$	$\log_3 81 = 4$	"3 to what power gives 81? The answer is 4"
$5^{-1} = \frac{1}{5}$	$\log_5 \left(\frac{1}{5}\right) = -1$	"5 to what power gives 1/5? The answer is -1"
$16^{1/2} = 4$	$\log_{16} 4 = \frac{1}{2}$	"16 to what power gives 4? The answer is 1/2"

Fundamental Logarithm Identities

Identity 1: $\log_a a = 1$

Why? Because $a^1 = a$

Examples:

$$\log_{10} 10 = 1 \quad (\text{since } 10^1 = 10)$$

$$\log_5 5 = 1 \quad (\text{since } 5^1 = 5)$$

$$\log_x x = 1 \quad (\text{for any valid base } x)$$

Identity 2: $\log_a 1 = 0$

Why? Because $a^0 = 1$

Examples:

$$\log_{10} 1 = 0 \quad (\text{since } 10^0 = 1)$$

$$\log_7 1 = 0 \quad (\text{since } 7^0 = 1)$$

$$\log_x 1 = 0 \quad (\text{for any valid base } x)$$

Memory Aid: "Base to the power of the log equals the argument"

$$a^{\log_a y} = y$$

B. CHARACTERISTIC AND MANTISSA

Diagram 2: Anatomy of a Logarithm

$$\log_{10} 5000 = 3.6990$$

$\begin{array}{r} | \quad ||| \\ | \quad ||| \quad \text{Last decimal digit} \\ | \quad || \quad \text{Thousandths} \\ | \quad | \quad \text{Hundredths} \\ | \quad \quad \text{Tenths} \\ \text{Characteristic (integer part)} \\ \text{Mantissa (decimal part)} \end{array}$

Understanding Characteristic from Standard Form:

Number	Standard Form	Characteristic	Reasoning
5000	5.0×10^3	3	Power of 10 is +3
0.00345	3.45×10^{-3}	-3	Power of 10 is -3

Bar Notation for Negative Characteristics:

Instead of writing -3.5378, we write: $3\bar{.}5378$

Read as: "Bar three point five three seven eight"

Examples:

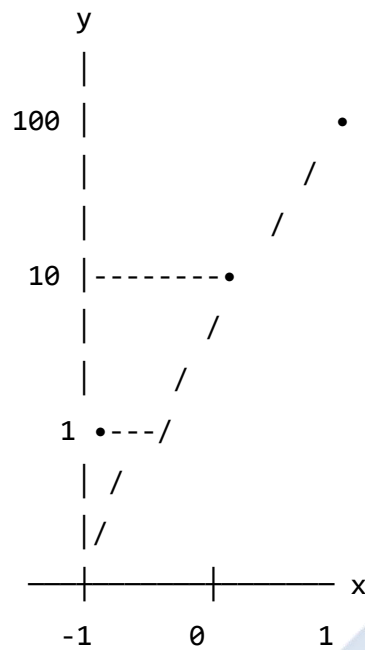
$$-2.8549 = 2\bar{.}8549$$

$$-1.2345 = 1\bar{.}2345$$

$$-0.9876 = 0\bar{.}9876$$

C. GRAPHS: EXPONENTIAL vs. LOGARITHMIC FUNCTIONS

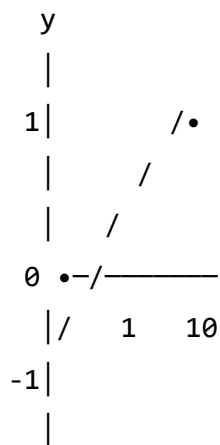
Graph 1: Exponential Function $y = 10^x$



Key Features of $y = 10^x$:

- Passes through (0, 1) because $10^0 = 1$
- Never touches the x-axis (horizontal asymptote at $y = 0$)
- Rapid growth for positive x
- Slow decay for negative x

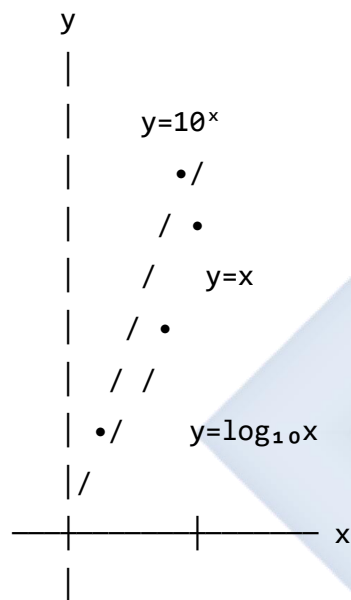
Graph 2: Logarithmic Function $y = \log_{10} x$



Key Features of $y = \log_{10} x$:

- Passes through (1, 0) because $\log_{10} 1 = 0$
- Never touches the y-axis (vertical asymptote at $x = 0$)
- Slow growth for $x > 1$
- Undefined for $x \leq 0$

Graph 3: Inverse Relationship



The graphs are **mirror images** across the line $y = x$!

D. USING LOGARITHM AND ANTILOGARITHM TABLES

Method 1: Finding Logarithms

Step-by-Step Procedure:

1. Convert number to standard form: $N = A \times 10^n$
2. Characteristic = n
3. Use log table with the digits of A to find mantissa
4. Combine: $\log N = n.\text{mantissa}$

Worked Example 4.2: Find $\log_{10} 43.82$

Step 1: Standard Form

$$43.82 = 4.382 \times 10^1 \rightarrow \text{Characteristic} = 1$$

Step 2: Use Log Table

Look up 4.382 in log table:

- Find row 43, column 8 $\rightarrow 0.6415$
- Find difference for 2 (last digit) $\rightarrow 0.0002$
- Mantissa = $0.6415 + 0.0002 = 0.6417$

Step 3: Combine

$$\log_{10} 43.82 = 1.6417$$

Visual Table Lookup:

Log Table Extract:

	0	1	2	...	8	9
43	.6335	.6345	.6355	...	.6415	.6425
	↑			↑		
Row 43				Column 8 → 0.6415		
				Difference for 2 → +0.0002		
				Total mantissa = 0.6417		

Worked Example 4.3: Find $\log_{10} 0.0716$

Step 1: Standard Form

$$0.0716 = 7.16 \times 10^{-2} \rightarrow \text{Characteristic} = -2$$

Step 2: Use Log Table

Look up 7.16 in log table:

- Row 71, column 6 → 0.8549
- Mantissa = 0.8549

Step 3: Combine with Bar Notation

$$\log_{10} 0.0716 = 2.8549$$

Method 2: Finding Antilogarithms

Step-by-Step Procedure:

1. Separate characteristic and mantissa
 2. Use antilog table with mantissa to find the digits
 3. Use characteristic to place decimal point
-

Worked Example 4.4: Find $\text{antilog}(3.9042)$

Step 1: Separate

Characteristic = 3, Mantissa = 0.9042

Step 2: Use Antilog Table

Look up 0.9042 in antilog table:

- Row 90, column 4 → 8017
- Difference for 2 → +4 → 8021
- Digits = 8021

Step 3: Place Decimal Point

Characteristic = 3 → Decimal after 3 digits from left

8021 → 802.1? Wait, characteristic 3 means: $A \times 10^3$

So: $8.021 \times 10^3 = 8021$

✓ **Answer: 8021**

Worked Example 4.5: Find $\text{antilog}(1.6609)$

Step 1: Separate

Characteristic = -1, Mantissa = 0.6609

Step 2: Use Antilog Table

Look up 0.6609:

- Row 66, column 0 $\rightarrow$ 4571
- Difference for 9 $\rightarrow$ +9 $\rightarrow$ 4580
- Digits = 4580 $\rightarrow$ 4.580

Step 3: Place Decimal Point

Characteristic = -1 $\rightarrow A \times 10^{-1} = 4.580 \times 10^{-1} = 0.4580$

✓ **Answer: 0.4580**

Memory Aid: Table Usage Flowchart

Finding LOG:

Number $\rightarrow$ Standard Form



Characteristic (from power)



Use A in log table $\rightarrow$ Mantissa



Combine $\rightarrow$ Final answer

Finding ANTILOG:

Logarithm $\rightarrow$ Separate



Use mantissa in antilog table



Get digits



Use characteristic $\rightarrow$ Place decimal

3. REVISION QUESTIONS (WEEK 4)

1. Explain why logarithms and exponents are considered inverse operations, using a real-world analogy.
 2. If $\log_b x = y$, write this in exponential form and explain what each variable represents.
 3. Why is $\log_5 5 = 1$ and $\log_5 1 = 0$? Use the definition of logarithms to explain.
 4. Describe the difference between characteristic and mantissa. How do you find each?
 5. Find the characteristics of:
 - $\log_{10} 5432$
 - $\log_{10} 0.5432$ Explain your reasoning.
-

4. ASSESSMENT (WEEK 4)

EVALUATION (IN-CLASS)

1. Write $2^5 = 32$ in logarithmic form.
2. Write $\log_3 \left(\frac{1}{9}\right) = -2$ in exponential form.
3. Solve for x: $\log_x 49 = 2$
4. Using log tables, find $\log_{10} 2.87$
5. Using antilog tables, find the antilog of 1.4502

ASSIGNMENT (TAKE-HOME)

1. Solve for x: $\log_4 x = 3$
2. Solve for x: $\log_3 27 = x$
3. Using log tables, find:
 - $\log_{10} 602.4$
 - $\log_{10} 0.000199$
4. Using antilog tables, find:
 - $\text{antilog}(4.8872)$
 - $\text{antilog}(2.1119)$

5. Consider the graph $y = 10^x$:

- What happens to y as x becomes a very large negative number (e.g., -100)?
- Does y ever become zero? Explain why or why not.

5. REAL-WORLD APPLICATIONS OF LOGARITHMS

Application 1: Richter Scale (Earthquake Magnitude)

Richter Scale is logarithmic: Each whole number increase = $10\times$ energy

Magnitude 3 $\rightarrow 10^3$ joules

Magnitude 4 $\rightarrow 10^4$ joules (10 times more energy!)

Magnitude 5 $\rightarrow 10^5$ joules (100 times more than magnitude 3!)

Application 2: pH Scale (Acidity)

$\text{pH} = -\log_{10}[\text{H}^+]$ where $[\text{H}^+]$ is hydrogen ion concentration

$\text{pH } 3 \rightarrow [\text{H}^+] = 10^{-3} = 0.001$

$\text{pH } 4 \rightarrow [\text{H}^+] = 10^{-4} = 0.0001$ (10 times less acidic!)

Application 3: Sound Intensity (Decibels)

$\text{dB} = 10 \cdot \log_{10}(I/I_0)$ where I is sound intensity

Whisper: 30 dB $\rightarrow 10^3$ times reference intensity

Normal talk: 60 dB $\rightarrow 10^6$ times reference intensity

Memory Poem:

Logarithms find the power, that's their special dower,

Inverse of exponents, their incredible power!

Characteristic tells us where the decimal points tower,

Mantissa gives the digits, making numbers not sour!

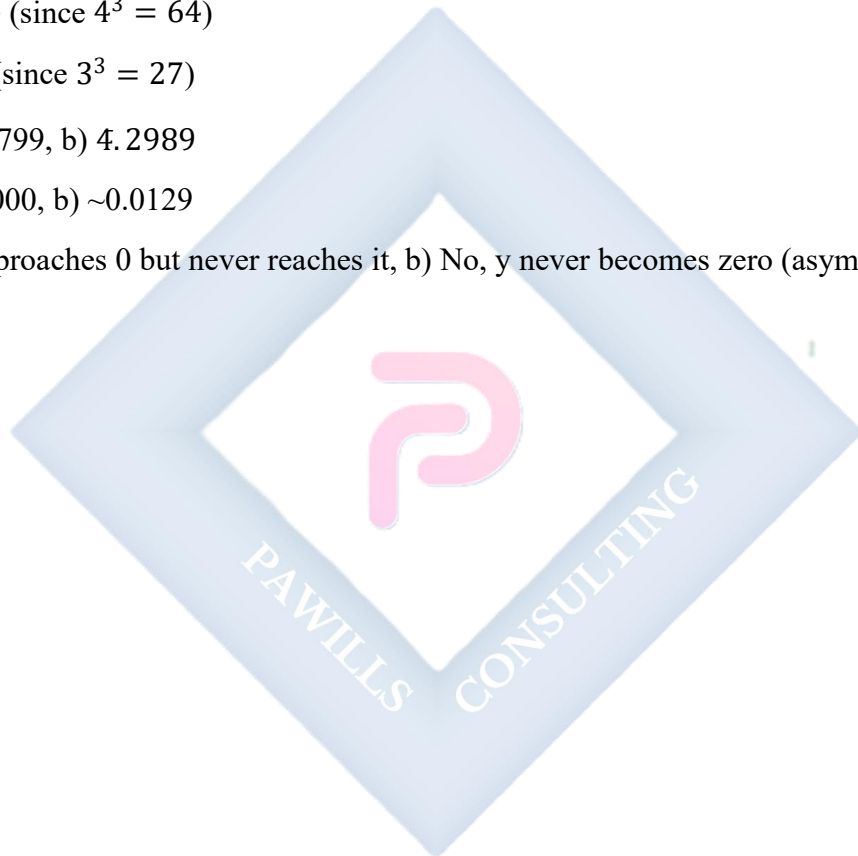
Answers to Assessment

In-Class:

1. $\log_2 32 = 5$
2. $3^{-2} = \frac{1}{9}$
3. $x = 7$ (since $7^2 = 49$)
4. Approximately 0.4579 (from tables)
5. Approximately 28.21 (from antilog tables)

Take-Home:

1. $x = 64$ (since $4^3 = 64$)
2. $x = 3$ (since $3^3 = 27$)
3. a) ~ 2.7799 , b) 4.2989
4. a) ~ 77000 , b) ~ 0.0129
5. a) y approaches 0 but never reaches it, b) No, y never becomes zero (asymptotic behavior)



WEEK 6: EFFECTIVE USE OF LOGARITHM TABLES

Concept: Using Logarithms as a Human Calculator for Complex Operations

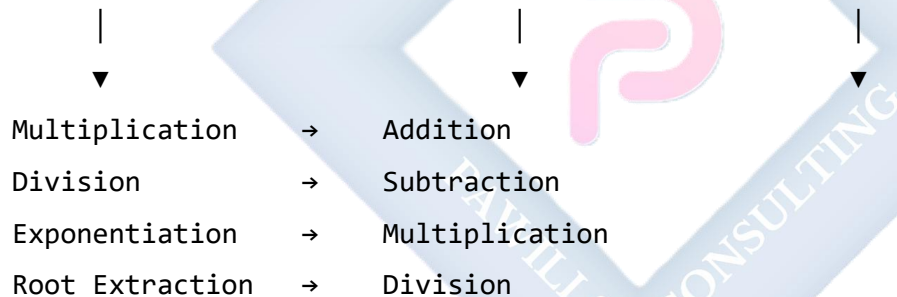
1. INTRODUCTION: THE PRE-CALCULATOR SUPERPOWER

Before digital calculators, scientists, engineers, and mathematicians used logarithm tables as their "human computers" to solve complex calculations efficiently.

Why Logarithms Transform Mathematics

Diagram 1: The Logarithmic Transformation Machine

Complex Operations → [LOGARITHM MACHINE] → Simple Operations



The Magic of Logarithmic Simplification:

Operation	Direct Calculation	Using Logarithms	Efficiency Gain
Multiplication	$32.5 \times 7.2 = ?$	$\log(32.5) + \log(7.2)$	Addition vs. complex multiplication
Division	$145.6 \div 8.3 = ?$	$\log(145.6) - \log(8.3)$	Subtraction vs. long division
Power	$(2.8)^5 = ?$	$5 \times \log(2.8)$	Multiplication vs. repeated multiplication

Operation	Direct Calculation	Using Logarithms	Efficiency Gain
Root	$\sqrt[3]{125} = ?$	$(1/3) \times \log(125)$	Division vs. trial and error

Real-World Historical Context

Before Calculators (1600s-1970s):

- **Scientists:** Used log tables for astronomical calculations
- **Engineers:** Designed bridges and buildings using logarithmic slide rules
- **Accountants:** Computed compound interest and amortization schedules

Visual: The Logarithmic Slide Rule

[Slide Rule Diagram]



Align scales to multiply: position 2 above 3, read 6 below

2. CONTENT

A. THE FOUR LAWS OF LOGARITHMIC CALCULATIONS

Law 1: Multiplication Law

$$\log_a(M \times N) = \log_a M + \log_a N$$

Visual Proof:

If $M = a^x$ and $N = a^y$, then:

$$M \times N = a^x \times a^y = a^{x+y}$$

$$\text{Therefore: } \log_a(M \times N) = x + y = \log_a M + \log_a N$$

Law 2: Division Law

$$\log_a \left(\frac{M}{N} \right) = \log_a M - \log_a N$$

Visual Proof:

If $M = a^x$ and $N = a^y$, then:

$$M/N = a^x / a^y = a^{x-y}$$

$$\text{Therefore: } \log_a(M/N) = x - y = \log_a M - \log_a N$$

Law 3: Power Law

$$\log_a(M^p) = p \times \log_a M$$

Visual Proof:

If $M = a^x$, then:

$$M^p = (a^x)^p = a^{p \times x}$$

$$\text{Therefore: } \log_a(M^p) = px = p \times \log_a M$$

Law 4: Root Law

$$\log_a(\sqrt[n]{M}) = \frac{1}{n} \times \log_a M$$

Visual Proof:

If $M = a^x$, then:

$$\sqrt[n]{M} = M^{1/n} = (a^x)^{1/n} = a^{x/n}$$

$$\text{Therefore: } \log_a(\sqrt[n]{M}) = x/n = (1/n) \times \log_a M$$

Memory Aid: "MAD ROOT"

- **M**ultiplication $\rightarrow$ **A**ddition
- **D**ivision $\rightarrow$ **S**ubtraction
- **R**OOT $\rightarrow$ Division of log

B. STEP-BY-STEP PROCEDURE FOR LOGARITHMIC CALCULATIONS

Flowchart: The Logarithmic Calculation Process

Start with complex expression



Break into components



Create log table for each number



Apply logarithmic laws



Perform simple arithmetic on logs



Use antilog table to convert back



Final answer!

Step 1: Expression Analysis Identify all components and operations in your expression.

Step 2: Logarithm Table Setup Create a systematic working table:

Number	Logarithm	Notes
[Value 1]	[Lookup from table]	[Any special handling]
[Value 2]	[Lookup from table]	[Any special handling]

Step 3: Law Application Apply the appropriate logarithmic laws based on operations.

Step 4: Final Computation Combine the logarithmic results and use antilog tables.

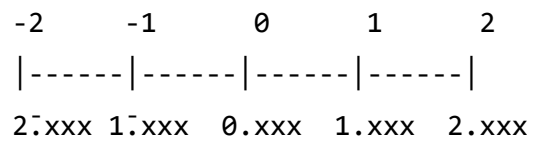
C. MASTERING BAR NOTATION (NEGATIVE CHARACTERISTICS)

Understanding Bar Notation:

$$1\bar{.}3979 = -1 + 0.3979 = -0.6021$$

$$2\bar{.}8543 = -2 + 0.8543 = -1.1457$$

Visual: The Bar Notation Number Line



Operation Rules with Bar Logarithms:

1. Addition of Bar Logs:

$$\begin{aligned} 2\bar{.}3 + 1\bar{.}5 &= ? \\ &= (-2 + 0.3) + (-1 + 0.5) \\ &= -3 + 0.8 \\ &= 3\bar{.}8 \end{aligned}$$

2. Subtraction of Bar Logs:

$$\begin{aligned} 2\bar{.}3 - 1\bar{.}5 &= ? \\ &= (-2 + 0.3) - (-1 + 0.5) \\ &= -2 + 0.3 + 1 - 0.5 \\ &= -1 - 0.2 \\ &= -1.2 = 2\bar{.}8 \quad (\text{since } -1.2 = -2 + 0.8) \end{aligned}$$

3. Multiplication of Bar Logs:

$$\begin{aligned} 2\bar{.}7 \times 3 &= ? \\ &= (-2 + 0.7) \times 3 \\ &= -6 + 2.1 \\ &= -4 + 0.1 \quad (\text{since } -6 + 2.1 = -3.9 = -4 + 0.1) \\ &= 4\bar{.}1 \end{aligned}$$

4. Division of Bar Logs:

$$3\bar{.}4 \div 2 = ?$$

$$3\bar{.}4 = -3 + 0.4 = -4 + 1.4 \quad (\text{borrow 1 from characteristic})$$

$$= (-4 + 1.4) \div 2$$

$$= -2 + 0.7$$

$$= 2\bar{.}7$$

Golden Rule: Never let the mantissa become negative! Always borrow from the characteristic to keep the mantissa positive.

3. DETAILED WORKED EXAMPLES

Example 6.1: Multiplication and Division

Evaluate: $\frac{38.4 \times 8.6}{15.2}$

Step-by-Step Solution:

Step 1: Logarithmic Transformation

$$\log\left(\frac{38.4 \times 8.6}{15.2}\right) = \log(38.4) + \log(8.6) - \log(15.2)$$

Step 2: Table Lookup and Setup

Number	Logarithm	Calculation
38.4	1.5843	From log table (384)
8.6	0.9345	From log table (860)
15.2	1.1818	From log table (152)

Step 3: Apply Logarithmic Laws

Numerator: $1.5843 + 0.9345 = 2.5188$

Complete: $2.5188 - 1.1818 = 1.3370$

Step 4: Antilog Conversion

Final log = 1.3370

Characteristic = 1, Mantissa = 0.3370

From antilog table: 0.3370 → 2173

Result = $2.173 \times 10^1 = 21.73$

Verification: $\frac{38.4 \times 8.6}{15.2} = \frac{330.24}{15.2} = 21.73 \checkmark$

✓ **Answer: 21.73**

Example 6.2: Powers and Roots

Evaluate: $\sqrt[3]{\frac{1.65^2}{29.4}}$

Step-by-Step Solution:

Step 1: Logarithmic Transformation

$$\log \left(\sqrt[3]{\frac{1.65^2}{29.4}} \right) = \frac{1}{3} \times [2 \times \log(1.65) - \log(29.4)]$$

Step 2: Table Lookup and Setup

Number	Logarithm	Calculation
1.65	0.2175	From log table
1.65^2	0.4350	2×0.2175
29.4	1.4683	From log table

Step 3: Apply Logarithmic Laws

Numerator: $2 \times \log(1.65) = 0.4350$

Difference: $\log(1.65^2) - \log(29.4) = 0.4350 - 1.4683$

Step 4: Handle Negative Result

$$0.4350 - 1.4683 = -1.0333$$

Express with bar notation: $1\bar{.9667}$

(Since $-1.0333 = -2 + 0.9667$)

Step 5: Apply Cube Root

$$1\bar{.9667} \div 3 = ?$$

$$1\bar{.9667} = -1 + 0.9667 = -2 + 1.9667 \text{ (borrow 1)}$$

$$\text{Now divide: } (-2 + 1.9667) \div 3 = -0.6667 + 0.6556 = 1\bar{.6556}$$

Step 6: Antilog Conversion

$$\text{Final log} = 1\bar{.6556}$$

Mantissa $0.6556 \rightarrow 4525$ (from antilog table)

$$\text{Characteristic } -1 \rightarrow 4.525 \times 10^{-1} = 0.4525$$

✓ **Answer: 0.4525**

Example 6.3: Real-World Application - Compound Interest

Problem: How many years will it take ₦50,000 to grow to ₦70,000 at 8% per year?

Compound Interest Formula:

$$A = P(1 + r)^n$$

Where:

- A = Final amount (₦70,000)
- P = Principal (₦50,000)
- r = Interest rate (8% = 0.08)
- n = Number of years

Step-by-Step Solution:

Step 1: Set Up Equation

$$70,000 = 50,000(1.08)^n$$

$$1.4 = (1.08)^n$$

Step 2: Apply Logarithms

$$\log(1.4) = n \times \log(1.08)$$

$$n = \frac{\log(1.4)}{\log(1.08)}$$

Step 3: Table Lookup

Number	Logarithm
1.4	0.1461
1.08	0.0334

Step 4: Calculate

$$n = \frac{0.1461}{0.0334} = 4.374$$

Step 5: Interpret Result

$$n \approx 4.37 \text{ years} \approx 4 \text{ years and } 4.5 \text{ months}$$

✓ **Answer: 4.4 years**

4. TROUBLESHOOTING AND COMMON PITFALLS

Pitfall 1: Negative Mantissa

Incorrect: $2.3 - 1.8 = 1.5$ (wrong!) **Correct:** $2.3 - 1.8 = 2.5$ (borrow to keep mantissa positive)

Pitfall 2: Decimal Point Placement

Rule: Characteristic determines decimal placement:

- Positive characteristic: Decimal after (characteristic + 1) digits from left
- Negative characteristic: Decimal before first non-zero digit

Pitfall 3: Bar Notation Arithmetic

Always convert: $2.7 = -2 + 0.7 = -1.3$ **Never write:** $2.7 \neq -2.7$

5. REVISION QUESTIONS

1. How do logarithms transform multiplication into addition? Provide a mathematical proof.
 2. Explain the process of handling division with bar notation logarithms.
 3. Why must we never let the mantissa become negative during calculations?
 4. Describe the step-by-step process for calculating $\sqrt[3]{125}$ using logarithms.
 5. How would you use logarithms to calculate compound interest for irregular time periods?
-

6. ASSESSMENT

EVALUATION (IN-CLASS)

1. State the logarithmic law for:
 - (a) $\log(M \times N)$
 - (b) $\log(M^p)$
2. Set up the log table to calculate $A = 5.2 \times 9.8$
3. Set up the log table to calculate $B = \sqrt{5.2}$
4. Calculate 2×3.6100
5. Calculate $5.2000 \div 4$

ASSIGNMENT (TAKE-HOME)

Use 4-figure logarithm tables for all calculations.

1. Evaluate $\frac{403.2 \times 0.056}{10.8}$

2. Evaluate $\sqrt{45.6}$
 3. Evaluate $(1.34)^5$
 4. Evaluate $\frac{\sqrt[3]{29.8 \times (4.1)^2}}{8.05}$ (set up a full Number/Logarithm table)
 5. Find n if $1000 = 200(1.05)^n$
-

7. HISTORICAL PERSPECTIVE AND MODERN APPLICATIONS

The Invention of Logarithms

- **1614:** John Napier publishes "Mirifici Logarithmorum Canonis Descriptio"
- **1620:** Edmund Gunter creates the logarithmic scale
- **1632:** William Oughtred invents the slide rule

Modern Applications

- **Computer Science:** Logarithmic time complexity ($O(\log n)$)
- **Finance:** Compound interest and annuities
- **Science:** pH scale, Richter scale, decibel measurements
- **Engineering:** Signal processing and control systems

Memory Poem:

When numbers multiply, and you want to cry,
Use logs to add, and reach for the sky!
Division's a pain, but logs make it sane,
Just subtract them out, and knowledge you'll gain!

Powers and roots, once complex and deep,
Become simple arithmetic, secrets to keep.
With tables in hand, you're a math wizard grand,
The master of numbers throughout the land!

Answers to Assessment

In-Class:

1. (a) $\log(M \times N) = \log M + \log N$, (b) $\log(M^p) = p \times \log M$
2. $\log A = \log 5.2 + \log 9.8$
3. $\log B = \frac{1}{2} \times \log 5.2$
4. $2 \times 3.6100 = 2.2200$
5. $5.2000 \div 4 = 2.8000$

Take-Home:

1. Approximately 2.09
2. Approximately 6.75
3. Approximately 4.30
4. Approximately 2.45
5. Approximately 32.99 years

WEEK 8: SIMPLE EQUATIONS AND VARIATIONS

Concept: Mastering Algebraic Manipulation and Proportional Relationships

1. INTRODUCTION: THE LANGUAGE OF MATHEMATICS

Algebraic manipulation is the foundation of all advanced mathematics and scientific reasoning. This week, we unlock two powerful tools that appear throughout STEM fields.

Real-World Applications

Diagram 1: Formulas in Everyday Life

SCIENCE & ENGINEERING



$$F = ma \text{ (Physics)}$$

$$v = u + at \text{ (Motion)}$$

$$PV = nRT \text{ (Chemistry)}$$

FINANCE & ECONOMICS



$$A = P(1 + r)^n \text{ (Compound Interest)}$$

$$I = Prt \text{ (Simple Interest)}$$

$$C = \pi d \text{ (Circumference)}$$

GEOMETRY & MEASUREMENT



$$A = \pi r^2 \text{ (Circle area)}$$

$$V = \frac{1}{3}\pi r^2 h \text{ (Cone volume)}$$

COMPUTER SCIENCE



$$y = mx + c \text{ (Linear algorithms)}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \text{ (Distance)}$$

The Power of Rearrangement: Being able to manipulate formulas means you can:

- Calculate any variable when others are known
 - Solve real-world problems efficiently
 - Understand relationships between quantities
 - Prepare for advanced mathematics and sciences
-

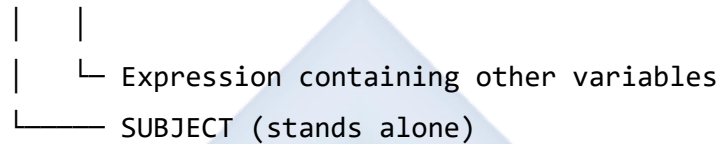
2. CONTENT

A. CHANGE OF SUBJECT OF FORMULA

Definition: The **subject** of a formula is the variable that stands alone on one side of the equation (typically the left side).

Visual: Identifying the Subject

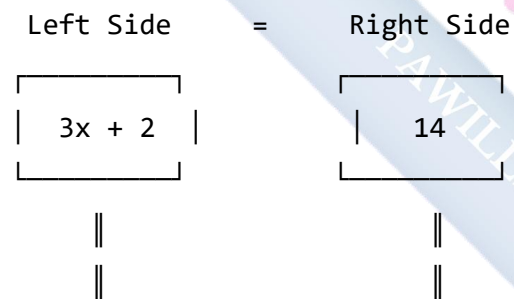
Original formula: $A = \pi r^2$



Changing the subject means rearranging the formula to make a different variable the subject.

The Golden Rule: Balance Principle

Visual: The Equation as a Balance Scale



Must remain balanced!

Operation: Subtract 2 from both sides



Operation: Divide both sides by 3



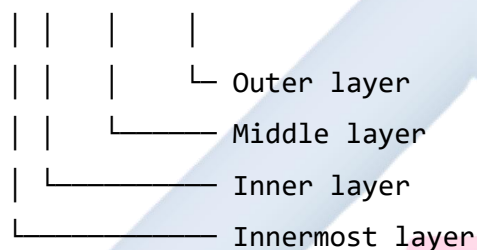
$$\begin{array}{|c|} \hline x \\ \hline \end{array} \quad \begin{array}{|c|} \hline 4 \\ \hline \end{array}$$

Key Principle: Whatever operation you perform on one side, you MUST perform on the other side to maintain balance.

Reverse BODMAS Strategy

Diagram 2: The "Unwrapping" Analogy

Original: $2(x + 3) = 14$



UNWRAPPING ORDER (Reverse BODMAS):

1. Remove outer layer ($\div 2$): $(x + 3) = 7$
2. Remove middle layer (-3): $x = 4$
3. Innermost layer revealed: $x = 4$

Systematic Approach:

Step	Operation to Undo	Example	Action
1	Brackets	$2(x + 3) = 14$	Expand or factor
2	Orders (powers/roots)	$x^2 = 16$	Square root both sides
3	Division/Multiplication	$3x = 12$	Divide by 3
4	Addition/Subtraction	$x + 5 = 9$	Subtract 5
5	Subject isolated	$x = 4$	Solution found

Operation Inverses Reference Table

Operation	Inverse Operation	Visual Example
$+ a$	$- a$	$x + 5 = 8 \rightarrow x = 8 - 5$
$- a$	$+ a$	$x - 3 = 7 \rightarrow x = 7 + 3$
$\times a$	$\div a$	$4x = 20 \rightarrow x = 20 \div 4$
$\div a$	$\times a$	$x/2 = 6 \rightarrow x = 6 \times 2$
x^2	$\sqrt{x}$	$x^2 = 25 \rightarrow x = \sqrt{25}$
$\sqrt{x}$	x^2	$\sqrt{x} = 3 \rightarrow x = 3^2$
$1/x$	Reciprocal	$1/x = 2 \rightarrow x = 1/2$

DETAILED WORKED EXAMPLES

Example 8.1: Handling Brackets

Make x the subject of: $a(x + b) = c$

Step-by-Step Solution:

Step 1: Divide both sides by a (undo multiplication)

$$\frac{a(x + b)}{a} = \frac{c}{a}$$

$$x + b = c/a$$

Step 2: Subtract b from both sides (undo addition)

$$x + b - b = c/a - b$$

$$x = c/a - b$$

Step 3: Common denominator (optional simplification)

$$x = (c - ab)/a$$

Visual Process:

$$a(x + b) = c$$

$$\downarrow [\div a]$$

$$x + b = c/a$$

$$\downarrow [-b]$$

$$x = c/a - b$$

✓ **Answer:** $x = \frac{c-ab}{a}$

Example 8.2: Dealing with Roots

Make L the subject of: $T = 2\pi\sqrt{\frac{L}{g}}$

Step-by-Step Solution:

Step 1: Divide by 2π (undo multiplication)

$$T/(2\pi) = \sqrt{L/g}$$

Step 2: Square both sides (undo square root)

$$[T/(2\pi)]^2 = L/g$$

$$T^2/(4\pi^2) = L/g$$

Step 3: Multiply by g (undo division)

$$L = gT^2/(4\pi^2)$$

Visual Process:

$$T = 2\pi\sqrt{L/g}$$

$$\downarrow [\div 2\pi]$$

$$T/(2\pi) = \sqrt{L/g}$$

$$\downarrow [^2]$$

$$T^2/(4\pi^2) = L/g$$

$$\downarrow [\times g]$$

$$L = gT^2/(4\pi^2)$$

✓ **Answer:** $L = \frac{gT^2}{4\pi^2}$

Real-World Connection: This is the formula for the period of a pendulum!

Example 8.3: Variable on Both Sides

Make x the subject of: $y = \frac{x+1}{x-2}$

Step-by-Step Solution:

Step 1: Multiply both sides by (x - 2)

$$y(x - 2) = x + 1$$

Step 2: Expand the left side

$$xy - 2y = x + 1$$

Step 3: Collect x terms on one side

$$xy - x = 1 + 2y$$

Step 4: Factor out x

$$x(y - 1) = 1 + 2y$$

Step 5: Divide by (y - 1)

$$x = (1 + 2y)/(y - 1)$$

Visual Process:

$$x + 1$$

$$y = \frac{\quad}{\quad}$$

$$x - 2$$

$$\downarrow [x(x-2)]$$

$$y(x - 2) = x + 1$$

$$\downarrow [\text{Expand}]$$

$$xy - 2y = x + 1$$

↓ [Rearrange]

$$xy - x = 1 + 2y$$

↓ [Factor]

$$x(y - 1) = 1 + 2y$$

↓ [$\div(y-1)$]

$$x = (1 + 2y)/(y - 1)$$

✓ **Answer:** $x = \frac{1+2y}{y-1}$

Example 8.4: Powers and Denominators

Make x the subject of: $y = \frac{3x^2+1}{2}$

Step-by-Step Solution:

Step 1: Multiply both sides by 2

$$2y = 3x^2 + 1$$

Step 2: Subtract 1

$$2y - 1 = 3x^2$$

Step 3: Divide by 3

$$(2y - 1)/3 = x^2$$

Step 4: Take square root (remember $\pm$)

$$x = \pm\sqrt{(2y - 1)/3}$$

Important Note: When taking square roots, we include $\pm$ because both positive and negative values satisfy the equation.

✓ **Answer:** $x = \pm\sqrt{\frac{2y-1}{3}}$

B. SUBSTITUTION INTO FORMULAS

Definition: Substitution means replacing variables with given numerical values to calculate a result.

Process:

1. Write the formula
2. Replace each variable with its given value
3. Follow BODMAS rules to compute
4. Include appropriate units

Example 8.5: Volume of a Cylinder

Given: $V = \pi r^2 h$, find V when $\pi = \frac{22}{7}$, $r = 7$, $h = 5$

Step-by-Step Solution:

$$\begin{aligned} V &= \pi r^2 h \\ &= (22/7) \times (7)^2 \times 5 \\ &= (22/7) \times 49 \times 5 \\ &= 22 \times 7 \times 5 \\ &= 22 \times 35 \\ &= 770 \end{aligned}$$

Calculation Check:

$$\begin{aligned} (22/7) \times 49 &= 22 \times 7 = 154 \\ 154 \times 5 &= 770 \end{aligned}$$

✓ **Answer:** $V = 770$

Example 8.6: Compound Interest

Given: $A = P(1 + rt)$, find A when $P = 2000$, $r = 0.05$, $t = 3$

Step-by-Step Solution:

$$\begin{aligned}A &= P(1 + rt) \\&= 2000 \times (1 + 0.05 \times 3) \\&= 2000 \times (1 + 0.15) \\&= 2000 \times 1.15 \\&= 2300\end{aligned}$$

Financial Interpretation: ₦2,000 invested at 5% simple interest for 3 years grows to ₦2,300

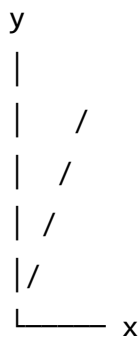
✓ **Answer:** $A = 2300$

C. VARIATION: MATHEMATICAL RELATIONSHIPS

Concept: Variation describes how one quantity changes in relation to another.

Diagram 3: The Four Types of Variation

DIRECT VARIATION



"More x → More y"

INVERSE VARIATION



"More x → Less y"

JOINT VARIATION

y varies with both
x and z together

COMBINED VARIATION

Mix of direct and
inverse relationships

Variation Types Reference Table

Type	Statement	Symbolic Form	Formula	Real-World Example
Direct	"y varies directly as x"	$y \propto x$	$y = kx$	Distance vs. Time (constant speed)
Inverse	"y varies inversely as x"	$y \propto 1/x$	$y = k/x$	Pressure vs. Volume (Boyle's Law)
Joint	"y varies jointly as x and z"	$y \propto xz$	$y = kxz$	Area of rectangle ($A = l \times w$)
Combined	Mixed direct and inverse	$y \propto x^2/\sqrt{z}$	$y = kx^2/\sqrt{z}$	Gravitational force

Solving Variation Problems: 5-Step Method

Step 1: Write the proportional statement **Step 2:** Convert to equation using constant k **Step 3:** Use given data to find k **Step 4:** Write the complete formula **Step 5:** Solve for unknown values

DETAILED VARIATION EXAMPLES

Example 8.7: Direct Variation

Problem: y varies directly as the square of x. When $x = 3$, $y = 18$. Find y when $x = 5$.

Step-by-Step Solution:

Step 1: Write proportional statement

$$y \propto x^2$$

Step 2: Convert to equation

$$y = kx^2$$

Step 3: Find k using given data

$$18 = k \times (3)^2$$

$$18 = 9k$$

$$k = 2$$

Step 4: Write complete formula

$$y = 2x^2$$

Step 5: Solve for unknown

$$\text{When } x = 5: y = 2 \times (5)^2 = 2 \times 25 = 50$$

Graphical Insight: This is a parabola opening upward

✓ **Answer:** $y = 50$

Example 8.8: Inverse Variation

Problem: R varies inversely as $\sqrt{V}$. When $R = 10$, $V = 25$, find V when $R = 5$.

Step-by-Step Solution:

Step 1: Write proportional statement

$$R \propto 1/\sqrt{V}$$

Step 2: Convert to equation

$$R = k/\sqrt{V}$$

Step 3: Find k using given data

$$10 = k/\sqrt{25}$$

$$10 = k/5$$

$$k = 50$$

Step 4: Write complete formula

$$R = 50/\sqrt{V}$$

Step 5: Solve for unknown

$$5 = 50/\sqrt{V}$$

$$\sqrt{V} = 50/5 = 10$$

$$V = 10^2 = 100$$

Real-World Connection: Electrical resistance vs. conductor diameter

✓ **Answer:** $V = 100$

Example 8.9: Joint Variation

Problem: y varies jointly as x and z . When $x = 2$, $z = 3$, $y = 24$, find y when $x = 4$, $z = 6$.

Step-by-Step Solution:

Step 1: Write proportional statement

$$y \propto xz$$

Step 2: Convert to equation

$$y = kxz$$

Step 3: Find k using given data

$$24 = k \times 2 \times 3$$

$$24 = 6k$$

$$k = 4$$

Step 4: Write complete formula

$$y = 4xz$$

Step 5: Solve for unknown

$$\text{When } x = 4, z = 6: y = 4 \times 4 \times 6 = 96$$

Observation: Both x and z doubled, so y quadrupled ($2 \times 2 = 4$ times)

✓ **Answer:** $y = 96$

Example 8.10: Combined Variation

Problem: y varies directly as x^2 and inversely as $\sqrt{z}$. When $x = 4$, $z = 9$, $y = 8$, find y when $x = 6$, $z = 16$.

Step-by-Step Solution:

Step 1: Write proportional statement

$$y \propto x^2/\sqrt{z}$$

Step 2: Convert to equation

$$y = kx^2/\sqrt{z}$$

Step 3: Find k using given data

$$8 = k \times (4)^2 / \sqrt{9}$$

$$8 = k \times 16 / 3$$

$$8 = 16k/3$$

$$k = 8 \times 3/16 = 24/16 = 1.5$$

Step 4: Write complete formula

$$y = 1.5x^2/\sqrt{z}$$

Step 5: Solve for unknown

$$\text{When } x = 6, z = 16: y = 1.5 \times (6)^2 / \sqrt{16}$$

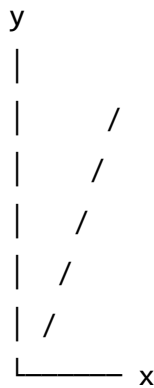
$$y = 1.5 \times 36 / 4 = 54 / 4 = 13.5$$

✓ **Answer:** $y = 13.5$

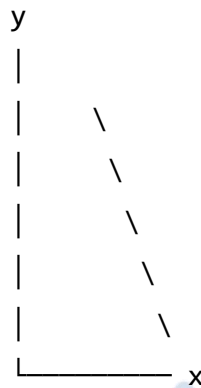
3. GRAPHICAL REPRESENTATION OF VARIATION

Diagram 4: Variation Graphs

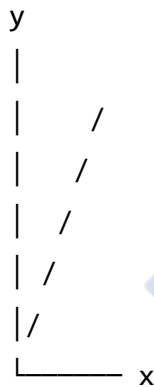
DIRECT: $y = kx$



INVERSE: $y = k/x$



DIRECT SQUARE: $y = kx^2$



INVERSE SQUARE: $y = k/x^2$



4. REAL-WORLD APPLICATIONS TABLE

Situation	Variation Type	Formula	Explanation
Speed	Direct	$d = st$	Double time → Double distance
Boyle's Law	Inverse	$P \propto 1/V$	Half volume → Double pressure
Gravity	Inverse Square	$F \propto 1/r^2$	Double distance → $\frac{1}{4}$ force
Circle Area	Direct Square	$A \propto r^2$	Double radius → $4\times$ area
Light Intensity	Inverse Square	$I \propto 1/d^2$	Double distance → $\frac{1}{4}$ brightness
Electrical Resistance	Inverse	$R \propto 1/A$	Double wire area → Half resistance

5. REVISION QUESTIONS

1. Define "subject of a formula" and explain why changing the subject is useful.
 2. State the golden rule for equation manipulation and illustrate with an example.
 3. Convert these statements to equations:
 - "y varies directly as the cube of x"
 - "P varies inversely as the square root of t"
 - "A varies jointly as b and h"
 4. Explain the difference between direct and inverse variation using real-world examples.
 5. If y doubles when x doubles, what type of variation is this? What if y halves when x doubles?
-

6. ASSESSMENT

EVALUATION (IN-CLASS)

1. Make C the subject of: $F = \frac{9}{5}C + 32$
2. Make r the subject of: $A = P(1 + rt)$
3. If $y \propto x$ and $y = 10$ when $x = 2$, find y when $x = 7$
4. If $y \propto 1/x$ and $y = 5$ when $x = 4$, find y when $x = 10$
5. If $P \propto L^2$ and L is doubled, what happens to P?

ASSIGNMENT (TAKE-HOME)

1. Make h the subject of: $A = 2\pi r(r + h)$
2. Make x the subject of: $T = \sqrt{\frac{ax}{b-x}}$
3. If $V = \frac{1}{3}\pi r^2 h$, find V when $\pi = 22/7$, $r = 3$, $h = 7$
4. F varies inversely as d^2 . When $d = 2$ cm, $F = 5$ N. Find F when $d = 5$ cm
5. Z varies directly as x^2 and inversely as y. When $x = 2$, $y = 3$, $Z = 8$. Find Z when $x = 3$, $y = 2$

EXTENSION (CHALLENGE PROBLEMS)

1. Make x the subject of: $\frac{1}{x} + \frac{1}{y} = \frac{1}{a}$
 2. If P varies directly as q and inversely as the square of r, find P when q = 8, r = 2, given P = 12 when q = 6, r = 3
 3. Make m the subject of: $V = \sqrt{\frac{F}{m}}$
-

7. COMMON PITFALLS AND TIPS

Pitfall 1: Forgetting to perform operations on both sides **Tip:** Always visualize the balance scale

Pitfall 2: Incorrect order of operations **Tip:** Use reverse BODMAS systematically

Pitfall 3: Mishandling square roots **Tip:** Remember $\pm$ when taking square roots

Pitfall 4: Confusing direct and inverse variation **Tip:** Direct = "same direction", Inverse = "opposite direction"

Memory Poem:

Equations are like scales, keep them balanced and true,
What you do to one side, to the other side too!

Reverse BODMAS is the key, to set variables free,
Unwrap like a present, and the answer you'll see!

Variation shows relationships, how things change and grow,
Direct, inverse, joint, combined - their patterns you will know!

Answers to Assessment

In-Class:

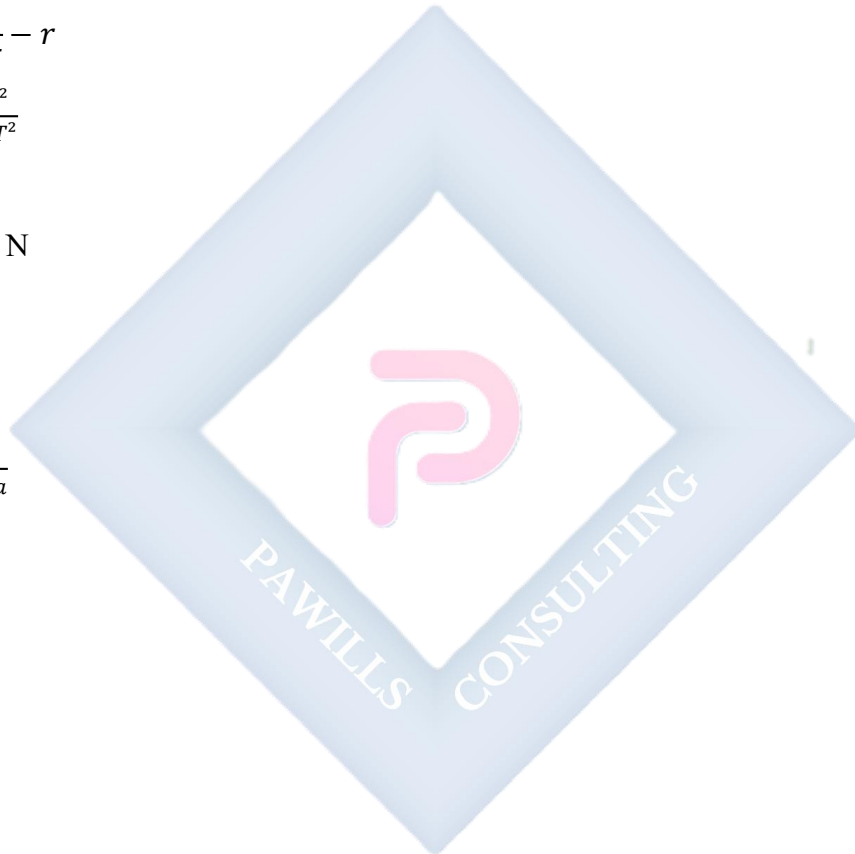
1. $C = \frac{5}{9}(F - 32)$
2. $r = \frac{A/P-1}{t}$
3. $y = 35$
4. $y = 2$
5. P becomes 4 times larger

Take-Home:

1. $h = \frac{A}{2\pi r} - r$
2. $x = \frac{bT^2}{a+T^2}$
3. $V = 66$
4. $F = 0.8 \text{ N}$
5. $Z = 18$

Extension:

1. $x = \frac{ay}{y-a}$
2. $P = 54$
3. $m = \frac{F}{v^2}$



WEEK 9: QUADRATIC EQUATIONS I

Concept: Mastering the Art of Solving Second-Degree Equations Through Factorization

1. INTRODUCTION: FROM LINEAR TO QUADRATIC

The Evolution of Equations

Linear Equations (Previous Knowledge):

Equation: $2x + 3 = 11$

Solution: $x = 4$

Characteristic: Straight line graph, one solution

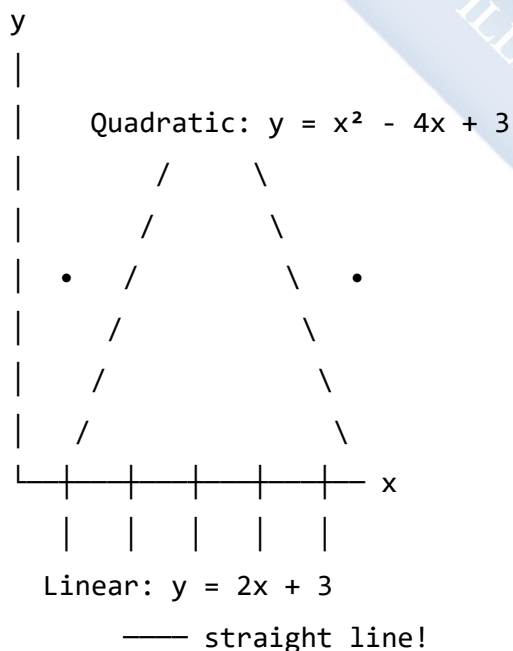
Quadratic Equations (New Frontier):

Equation: $x^2 - 4x + 3 = 0$

Solutions: $x = 1$ or $x = 3$

Characteristic: Curved parabola, up to two solutions

Visual Comparison: Linear vs. Quadratic



Formal Definition

A **quadratic equation** is an equation where the highest power of the variable is 2.

Standard Form:

$$ax^2 + bx + c = 0$$

Where:

- a, b, c are real numbers (constants)
- $a \neq 0$ (if $a = 0$, it becomes linear)

Examples:

- $x^2 - 5x + 6 = 0 \rightarrow a = 1, b = -5, c = 6$
 - $2x^2 + 3x - 4 = 0 \rightarrow a = 2, b = 3, c = -4$
 - $4x^2 - 9 = 0 \rightarrow a = 4, b = 0, c = -9$
-

Real-World Applications

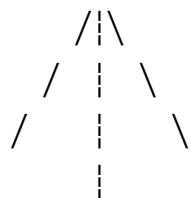
Diagram 1: Quadratic Equations in Daily Life

SPORTS & PROJECTILES



Ball trajectory:

$$h = -5t^2 + 20t$$



GEOMETRY & DESIGN



BUSINESS & ECONOMICS



Profit maximization:

$$P = -2x^2 + 100x - 500$$



└ Maximum profit point

SCIENCE & ENGINEERING



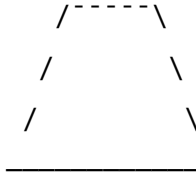
Area calculations:

$$A = x(x+3) = x^2 + 3x$$



Bridge design:

Structural analysis



Key Insight: Quadratic equations model situations where change accelerates or decelerates, unlike linear equations that model constant change.

2. CONTENT

A. REVISION: THE CONCEPT OF FACTORIZATION

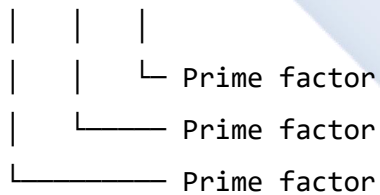
Definition: Factorization is the process of breaking down an expression into simpler parts (factors) that multiply together to give the original expression.

Visual: Factorization as Decomposition

Original: 30

↓

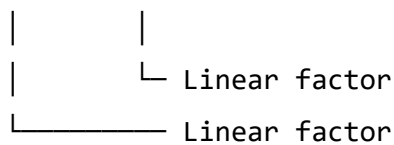
Factors: $2 \times 3 \times 5$



Original: $x^2 + 7x + 12$

↓

Factors: $(x + 3)(x + 4)$



Why Factorize?

- Simplifies complex expressions
- Helps solve equations
- Reveals hidden patterns and relationships

B. FACTORIZATION OF QUADRATIC EXPRESSIONS

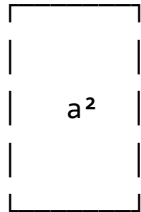
Case 1: Difference of Two Squares ($b = 0$)

Pattern Recognition:

$$a^2 - b^2 = (a - b)(a + b)$$

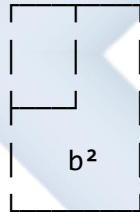
Visual Proof: Geometric Interpretation

Area = a^2



→

Remove b^2



→

Rearrange



Total area: $a^2 - b^2$

$$= (a-b)(a+b)$$

Examples:

Example 1.1: Factorize $x^2 - 49$

$$x^2 - 49 = x^2 - 7^2 = (x - 7)(x + 7)$$

Example 1.2: Factorize $4y^2 - 25$

$$4y^2 - 25 = (2y)^2 - 5^2 = (2y - 5)(2y + 5)$$

Example 1.3: Factorize $9 - 16z^2$

$$9 - 16z^2 = 3^2 - (4z)^2 = (3 - 4z)(3 + 4z)$$

Case 2: Trinomials with $a = 1$

Form: $x^2 + bx + c$

Strategy: The Number Pair Method We need two numbers that:

- **Add** to give b (the coefficient of x)
- **Multiply** to give c (the constant term)

Visual: The Factor Diamond

Product: c

/ \

/ \

? ?

\ /

\ /

Sum: b

Step-by-Step Process:

Example 2.1: Factorize $x^2 + 7x + 12$

Step 1: Identify $b = 7$, $c = 12$

Step 2: Find number pairs that multiply to 12:

$(1, 12), (2, 6), (3, 4)$

Step 3: Check which pair adds to 7:

$$1 + 12 = 13 \quad \times$$

$$2 + 6 = 8 \quad \times$$

$$3 + 4 = 7 \quad \checkmark$$

Step 4: Write factors: $(x + 3)(x + 4)$

Example 2.2: Factorize $x^2 - 5x - 14$

Step 1: Identify $b = -5$, $c = -14$

Step 2: Find number pairs that multiply to -14:

(1, -14), (2, -7), (-1, 14), (-2, 7)

Step 3: Check which pair adds to -5:

$$1 + (-14) = -13 \quad \times$$

$$2 + (-7) = -5 \quad \checkmark$$

Step 4: Write factors: $(x + 2)(x - 7)$

Memory Aid: "SUM and PRODUCT"

- SUM tells you what to ADD
- PRODUCT tells you what to MULTIPLY

Case 3: Trinomials with $a \neq 1$

Form: $ax^2 + bx + c$

Method: Splitting the Middle Term

Step-by-Step Algorithm:

1. Multiply $a \times c$
2. Find two numbers that:
 - Add to b
 - Multiply to $a \times c$
3. Split the middle term using these numbers
4. Factor by grouping

Visual: The AC Method Flowchart

$$ax^2 + bx + c$$

↓ [Multiply $a \times c$]

Find numbers: $m + n = b$, $m \times n = a \times c$

↓ [Split]

$$ax^2 + mx + nx + c$$

↓ [Group]

$$(ax^2 + mx) + (nx + c)$$

↓ [Factor groups]
 $x(ax + m) + k(nx + c) \rightarrow$ Common factor!
 ↓
 $(px + q)(rx + s)$

Example 3.1: Factorize $3x^2 + 10x - 8$

Step 1: $a \times c = 3 \times (-8) = -24$

Step 2: Find numbers:

Need: sum = 10, product = -24

Numbers: -2 and 12 (because $-2 + 12 = 10$, $-2 \times 12 = -24$)

Step 3: Split: $3x^2 - 2x + 12x - 8$

Step 4: Group: $(3x^2 - 2x) + (12x - 8)$

Step 5: Factor groups: $x(3x - 2) + 4(3x - 2)$

Step 6: Common factor: $(x + 4)(3x - 2)$

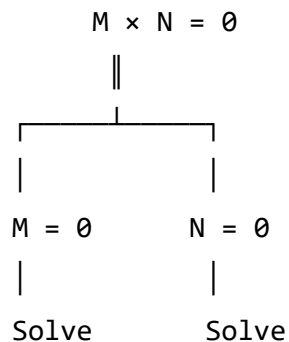
Verification: Expand $(x + 4)(3x - 2) = 3x^2 - 2x + 12x - 8 = 3x^2 + 10x - 8 \checkmark$

C. SOLVING QUADRATIC EQUATIONS BY FACTORIZATION

The Zero Product Property

Fundamental Principle: If $M \times N = 0$, then either $M = 0$ or $N = 0$ (or both)

Visual: The Zero Product Rule



Why This Works:

- The only way a product can be zero is if at least one factor is zero
- This transforms a quadratic equation into two simple linear equations

Solving Procedure

Step 1: Write in standard form: $ax^2 + bx + c = 0$ **Step 2:** Factorize the quadratic expression

Step 3: Set each factor equal to zero **Step 4:** Solve each linear equation

DETAILED WORKED EXAMPLES

Example 9.1: Simple Trinomial

Solve: $x^2 - 8x + 15 = 0$

Step-by-Step Solution:

Step 1: Already in standard form

Step 2: Factorize:

Numbers that add to -8, multiply to 15: -3 and -5

$$(x - 3)(x - 5) = 0$$

Step 3: Set each factor to zero:

$$x - 3 = 0 \quad \text{OR} \quad x - 5 = 0$$

Step 4: Solve:

$$x = 3 \quad \text{OR} \quad x = 5$$

Graphical Interpretation: The parabola $y = x^2 - 8x + 15$ crosses the x-axis at $x = 3$ and $x = 5$

✓ **Answer:** $x = 3$ or $x = 5$

Example 9.2: Equation Requiring Rearrangement

Solve: $6x^2 - 5x = 4$

Step-by-Step Solution:

Step 1: Standard form: $6x^2 - 5x - 4 = 0$

Step 2: Factorize using AC method:

$$a \times c = 6 \times (-4) = -24$$

Numbers: sum = -5, product = -24 $\rightarrow$ 3 and -8

$$\text{Split: } 6x^2 + 3x - 8x - 4 = 0$$

$$\text{Group: } (6x^2 + 3x) + (-8x - 4)$$

$$\text{Factor groups: } 3x(2x + 1) - 4(2x + 1)$$

$$\text{Common factor: } (3x - 4)(2x + 1) = 0$$

Step 3: Set factors to zero:

$$3x - 4 = 0 \quad \text{OR} \quad 2x + 1 = 0$$

Step 4: Solve:

$$x = 4/3 \quad \text{OR} \quad x = -1/2$$

Check Solutions:

- For $x = \frac{4}{3}$: $6(\frac{16}{9}) - 5(\frac{4}{3}) = \frac{96}{9} - \frac{20}{3} = \frac{32}{3} - \frac{20}{3} = \frac{12}{3} = 4 \checkmark$
- For $x = -\frac{1}{2}$: $6(\frac{1}{4}) - 5(-\frac{1}{2}) = \frac{6}{4} + \frac{5}{2} = 1.5 + 2.5 = 4 \checkmark$

✓ **Answer:** $x = \frac{4}{3}$ or $x = -\frac{1}{2}$

Example 9.3: Difference of Squares

Solve: $4x^2 - 25 = 0$

Step-by-Step Solution:

Step 1: Already in standard form

Step 2: Recognize difference of squares:

$$4x^2 - 25 = (2x)^2 - 5^2 = (2x - 5)(2x + 5) = 0$$

Step 3: Set factors to zero:

$$2x - 5 = 0 \quad \text{OR} \quad 2x + 5 = 0$$

Step 4: Solve:

$$x = 5/2 \quad \text{OR} \quad x = -5/2$$

✓ **Answer:** $x = \frac{5}{2}$ or $x = -\frac{5}{2}$

Example 9.4: Common Factor First

Solve: $3x^2 - 12x = 0$

Step-by-Step Solution:

Step 1: Already in standard form

Step 2: Factor out common factor:

$$3x(x - 4) = 0$$

Step 3: Set factors to zero:

$$3x = 0 \quad \text{OR} \quad x - 4 = 0$$

Step 4: Solve:

$$x = 0 \quad \text{OR} \quad x = 4$$

Important: Always look for common factors first—it simplifies the problem!

✓ **Answer:** $x = 0$ or $x = 4$

3. COMMON PITFALLS AND TIPS

Pitfall 1: Forgetting the Standard Form **Tip:** Always rearrange to $ax^2 + bx + c = 0$ before factoring

Pitfall 2: Incorrect Sign Handling **Tip:** Double-check that your numbers:

- ADD to b
- MULTIPLY to $a \times c$

Pitfall 3: Not Checking Solutions **Tip:** Always verify by substituting back into the original equation

Pitfall 4: Overlooking Common Factors **Tip:** Always factor out the greatest common factor first

Pitfall 5: Assuming All Quadratics Factor **Tip:** Some quadratics don't factor nicely—we'll learn other methods later!

4. REVISION QUESTIONS

1. What distinguishes a quadratic equation from a linear equation? Provide examples of each.
 2. Explain the Zero Product Property and why it's essential for solving quadratic equations.
 3. Describe the geometric interpretation of the difference of two squares.
 4. Walk through the "splitting the middle term" method step by step with an example.
 5. Why can't $x^2 + 9 = 0$ be solved by factorization in real numbers? What does this tell us about its solutions?
-

5. ASSESSMENT

EVALUATION (IN-CLASS)

1. Factorize $x^2 - 121$
2. Factorize $y^2 + 10y + 21$
3. Factorize $x^2 + 2x - 15$
4. Solve for k : $k^2 - 9k = 0$ (Hint: common factor)
5. Solve for x : $x^2 - 11x + 28 = 0$

ASSIGNMENT (TAKE-HOME)

1. Factorize $5x^2 - 13x - 6$
2. Solve for y : $y(y + 3) = 10$ (Hint: expand and rearrange)
3. Solve for m : $3m^2 + 19m = 14$
4. Solve for p : $4p^2 - 25 = 0$ (Hint: difference of two squares)
5. **Word Problem:** The product of two consecutive positive integers is 56.
 - Let the integers be n and $n + 1$

- Write the quadratic equation
 - Solve by factorization to find the integers
-

6. REAL-WORLD PROBLEM SOLVING

Application: Consecutive Integer Problem

Problem: The product of two consecutive positive integers is 56. Find the integers.

Step-by-Step Solution:

Step 1: Define variables:

Let the integers be n and $n+1$

Step 2: Set up equation:

$$n(n + 1) = 56$$

Step 3: Expand and rearrange:

$$n^2 + n = 56$$

$$n^2 + n - 56 = 0$$

Step 4: Factorize:

Numbers that add to 1, multiply to -56: -7 and 8

$$(n - 7)(n + 8) = 0$$

Step 5: Solve:

$$n - 7 = 0 \rightarrow n = 7$$

$$n + 8 = 0 \rightarrow n = -8$$

Step 6: Interpret:

We need positive integers, so $n = 7$

The integers are 7 and 8

Verification: $7 \times 8 = 56$ ✓

✓ **Answer:** The integers are 7 and 8

Memory Poem:

Quadratic equations, degree two,
 $ax^2 + bx + c$, that's the clue!
Factorize to solve with ease,
Using methods such as these:

Difference of squares, a special case,
Makes solving problems a real race!
Trinomials need number pairs,
That satisfy both sums and shares!

Zero product property's the key,
Set each factor to zero, you'll see!
Solutions one, or sometimes two,
Now quadratic skills belong to you!

Answers to Assessment

In-Class:

1. $(x - 11)(x + 11)$
2. $(y + 3)(y + 7)$
3. $(x + 5)(x - 3)$
4. $k = 0$ or $k = 9$
5. $x = 4$ or $x = 7$

Take-Home:

1. $(5x + 2)(x - 3)$
2. $y = 2$ or $y = -5$
3. $m = \frac{2}{3}$ or $m = -7$
4. $p = \frac{5}{2}$ or $p = -\frac{5}{2}$
5. Integers are 7 and 8

WEEK 10: QUADRATIC EQUATIONS II

Concept: From Roots to Graphs - The Complete Picture of Quadratic Equations

1. INTRODUCTION: BEYOND FACTORIZATION

Building on Previous Knowledge

Last Week (Factorization):

Equation: $x^2 - 5x + 6 = 0$

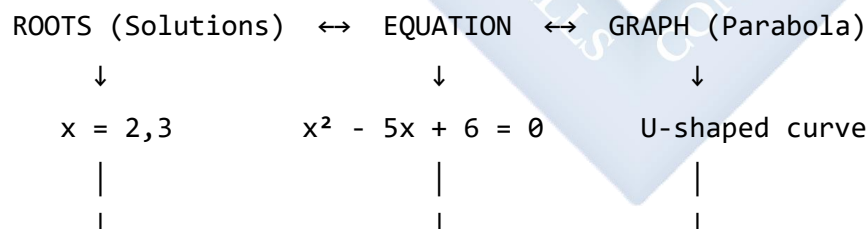
Factorization: $(x - 2)(x - 3) = 0$

Solutions: $x = 2$ or $x = 3$

This Week (Advanced Concepts):

- **Forming equations** from known solutions
- **Graphical interpretation** of quadratics
- **Real-world applications** in physics, business, and engineering

Visual: The Quadratic Equation Ecosystem



All representations are connected!

Real-World Significance

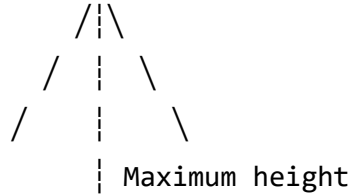
Diagram 1: Quadratic Equations in Action

PHYSICS & ENGINEERING



Projectile Motion:

$$h = -5t^2 + 20t$$

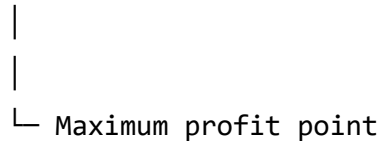


BUSINESS & ECONOMICS



Profit Optimization:

$$P = -2x^2 + 100x - 500$$



ARCHITECTURE & DESIGN



Bridge arches:

$$y = ax^2 + bx + c$$



COMPUTER GRAPHICS



3D modeling:

Parabolic surfaces



Key Insight: Understanding the complete picture of quadratic equations unlocks their power to model and solve real-world problems.

2. CONTENT

A. FORMATION OF QUADRATIC EQUATIONS FROM GIVEN ROOTS

The Fundamental Connection

If we know the roots (solutions) α and β , we can reconstruct the equation.

Visual: The Root-to-Equation Process

Roots: α and β

↓ [Reverse thinking]

Factors: $(x - \alpha)$ and $(x - \beta)$

↓ [Multiply]

Equation: $(x - \alpha)(x - \beta) = 0$

↓ [Expand]

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

Mathematical Derivation:

Given: Roots are α and β

Then: $(x - \alpha)(x - \beta) = 0$

Expand: $x^2 - \beta x - \alpha x + \alpha\beta = 0$

Simplify: $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

✓ Golden Formula:

$$x^2 - (\text{Sum of roots})x + (\text{Product of roots}) = 0$$

Where:

- **Sum of roots** = $\alpha + \beta$
- **Product of roots** = $\alpha\beta$

Memory Aid: "SP Equation"

- **Sum** → Coefficient of x
 - **Product** → Constant term
-

DETAILED WORKED EXAMPLES

Example 10.1: Integer Roots

Find the quadratic equation whose roots are 4 and -3.

Step-by-Step Solution:

Step 1: Identify roots

$$\alpha = 4, \beta = -3$$

Step 2: Calculate sum and product

$$\text{Sum} = \alpha + \beta = 4 + (-3) = 1$$

$$\text{Product} = \alpha \times \beta = 4 \times (-3) = -12$$

Step 3: Apply formula

$$x^2 - (\text{Sum})x + (\text{Product}) = 0$$

$$x^2 - (1)x + (-12) = 0$$

$$x^2 - x - 12 = 0$$

Verification:

$$\text{Solve } x^2 - x - 12 = 0$$

$$\text{Factorization: } (x - 4)(x + 3) = 0$$

$$\text{Roots: } x = 4, x = -3 \checkmark$$

$$\checkmark \text{ Answer: } x^2 - x - 12 = 0$$

Example 10.2: Fractional Roots

Find the quadratic equation whose roots are $\frac{1}{2}$ and $\frac{3}{2}$.

Step-by-Step Solution:

Step 1: Identify roots

$$\alpha = 1/2, \beta = 3/2$$

Step 2: Calculate sum and product

$$\text{Sum} = 1/2 + 3/2 = 4/2 = 2$$

$$\text{Product} = (1/2) \times (3/2) = 3/4$$

Step 3: Apply formula

$$x^2 - (2)x + (3/4) = 0$$

$$x^2 - 2x + 3/4 = 0$$

Step 4: Clear fractions (multiply by 4)

$$4x^2 - 8x + 3 = 0$$

Verification:

Solve $4x^2 - 8x + 3 = 0$

Factorization: $(2x - 1)(2x - 3) = 0$

Roots: $x = 1/2, x = 3/2$ ✓

✓ **Answer:** $4x^2 - 8x + 3 = 0$

Example 10.3: Complex Roots

Find the quadratic equation whose roots are $2 + \sqrt{3}$ and $2 - \sqrt{3}$.

Step-by-Step Solution:

Step 1: Identify roots

$$\alpha = 2 + \sqrt{3}, \beta = 2 - \sqrt{3}$$

Step 2: Calculate sum and product

$$\text{Sum} = (2 + \sqrt{3}) + (2 - \sqrt{3}) = 4$$

$$\text{Product} = (2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$$

Step 3: Apply formula

$$x^2 - (4)x + (1) = 0$$

$$x^2 - 4x + 1 = 0$$

Mathematical Insight: When roots are conjugate pairs ($a \pm b\sqrt{c}$), the sum and product are always rational!

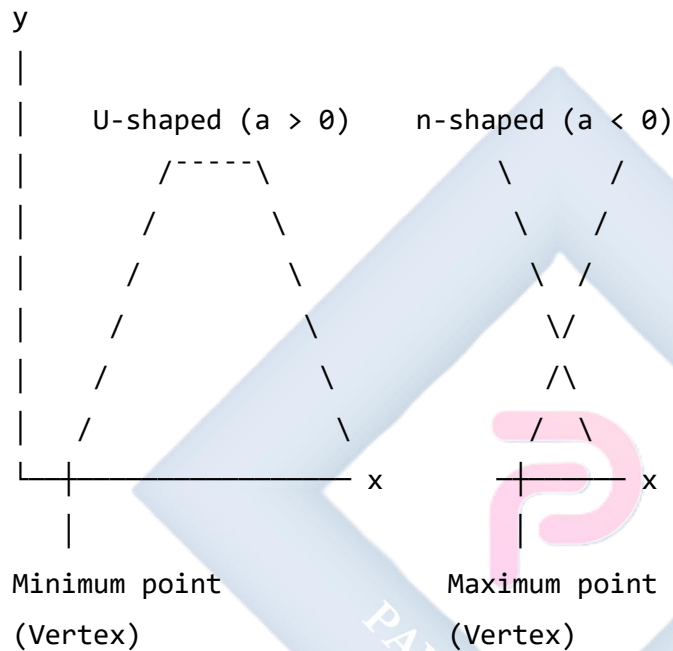
✓ **Answer:** $x^2 - 4x + 1 = 0$

B. DRAWING QUADRATIC GRAPHS

Understanding the Parabola

Definition: A **parabola** is the U-shaped or n-shaped curve formed by graphing a quadratic equation.

Visual: Parabola Characteristics



Key Features:

- **Vertex:** The turning point (minimum or maximum)
- **Axis of Symmetry:** Vertical line through the vertex
- **Roots:** x-intercepts (where $y = 0$)
- **y-intercept:** Where $x = 0$

Step-by-Step Graphing Procedure

Step 1: Create a Table of Values Choose x-values around the expected vertex

Step 2: Calculate Corresponding y-values Substitute each x into the equation

Step 3: Choose Appropriate Scales Ensure the graph fits nicely on your paper

Step 4: Plot Points Carefully Mark each (x, y) coordinate accurately

Step 5: Draw a Smooth Curve Connect points with a continuous, curved line

COMPREHENSIVE GRAPHING EXAMPLE

Example 10.4: Complete Graphing Process

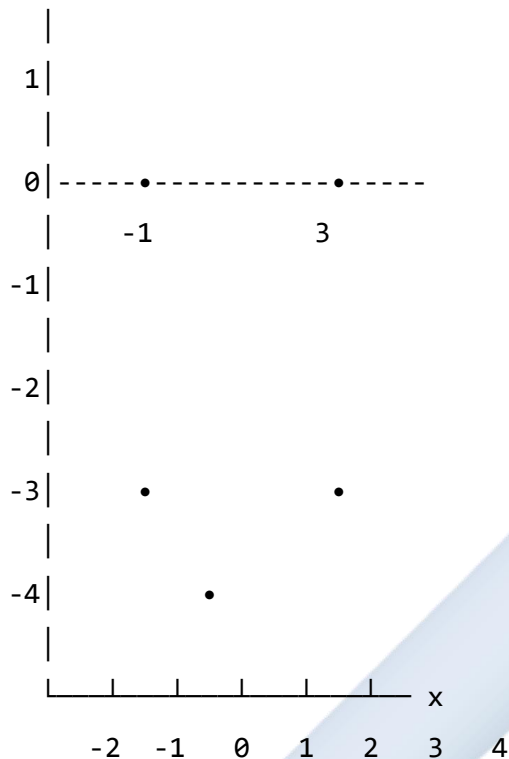
Draw the graph of $y = x^2 - 2x - 3$ for $-2 \leq x \leq 4$

Step 1: Table of Values

x	x^2	-2x	-3	$y = x^2 - 2x - 3$	Point
-2	4	4	-3	5	(-2, 5)
-1	1	2	-3	0	(-1, 0)
0	0	0	-3	-3	(0, -3)
1	1	-2	-3	-4	(1, -4)
2	4	-4	-3	-3	(2, -3)
3	9	-6	-3	0	(3, 0)
4	16	-8	-3	5	(4, 5)

Step 2: Plot Points and Draw Curve

y
|
5 | •
|
4 |
|
3 |
|
2 |



Step 3: Identify Key Features

- **Roots:** $x = -1$ and $x = 3$ (where curve crosses x-axis)
- **Vertex:** $(1, -4)$ - minimum point
- **Axis of Symmetry:** $x = 1$ (vertical line through vertex)
- **y-intercept:** $(0, -3)$

C. EXTRACTING INFORMATION FROM GRAPHS

Three Types of Graphical Solutions

1. Finding Roots of the Original Equation

Equation: $x^2 - 2x - 3 = 0$

On graph: Find where $y = 0$ (x-intercepts)

Solution: $x = -1, x = 3$

2. Finding Minimum/Maximum Values

For $y = x^2 - 2x - 3$:

Vertex at (1, -4)

Minimum value: $y = -4$

3. Solving Related Equations

To solve: $x^2 - 2x - 1 = 0$

Rewrite: $x^2 - 2x - 3 = 2$

Find where original curve intersects $y = 2$

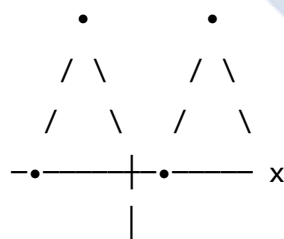
Approximate solutions: $x \approx -0.4$, $x \approx 2.4$

Visual: Graphical Solution Methods

Original: $y = x^2 - 2x - 3$

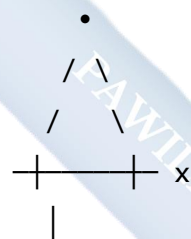
ROOTS:

Where curve hits
x-axis ($y=0$)



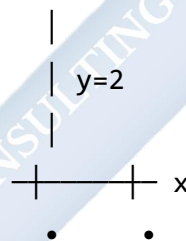
MINIMUM:

Lowest point
(1, -4)



RELATED EQ:

Shift horizontal line
to solve new equations



D. REAL-WORLD APPLICATIONS

Example 10.5: Garden Design Problem

A rectangular garden has a length 4 m longer than its width. If the area is 45 m^2 , find the dimensions.

Step-by-Step Solution:

Step 1: Define variables

Let width = w

Then length = $w + 4$

Step 2: Set up equation

Area = length $\times$ width

$$45 = (w + 4) \times w$$

$$45 = w^2 + 4w$$

Step 3: Rearrange to standard form

$$w^2 + 4w - 45 = 0$$

Step 4: Solve by factorization

Find numbers that add to 4, multiply to -45: 9 and -5

$$(w + 9)(w - 5) = 0$$

Step 5: Interpret solutions

$$w = -9 \text{ or } w = 5$$

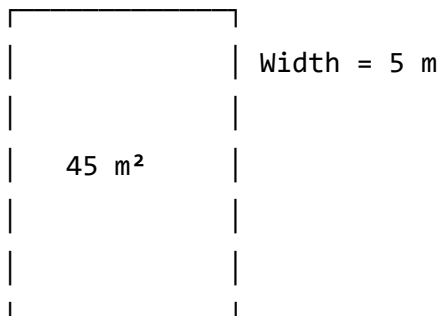
Width cannot be negative $\Rightarrow w = 5$

$$\text{Length} = w + 4 = 9$$

Verification: $5 \times 9 = 45 \text{ m}^2 \checkmark$

Visual Representation:

Garden Layout:



$$\text{Length} = 9 \text{ m}$$

✓ **Answer:** Width = 5 m, Length = 9 m

Example 10.6: Projectile Motion

A ball is thrown upward with initial velocity. Its height h (in meters) after t seconds is given by $h = -5t^2 + 20t$. Find:

- Maximum height reached
- Time when ball hits the ground

Graphical Approach:

Step 1: Recognize this is a quadratic: $h = -5t^2 + 20t$

Step 2: Graph shows inverted parabola ($a = -5 < 0$)

Step 3: Maximum height at vertex

Step 4: Ground level when $h = 0$

Algebraic Solution:

For $h = -5t^2 + 20t$:

Vertex at $t = -b/(2a) = -20/(2 \times -5) = 2$ seconds

Maximum height = $-5(2)^2 + 20(2) = -20 + 40 = 20$ meters

Time when $h = 0$:

$$-5t^2 + 20t = 0$$

$$-5t(t - 4) = 0$$

$t = 0$ (start) or $t = 4$ seconds

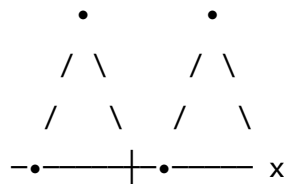
✓ **Answer:** Maximum height = 20 m at $t = 2$ s; Hits ground at $t = 4$ s

3. GRAPHICAL INTERPRETATION MASTERY

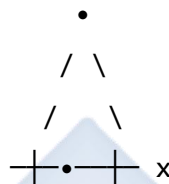
Different Cases of Quadratic Graphs

Diagram 2: The Six Possible Graph Scenarios

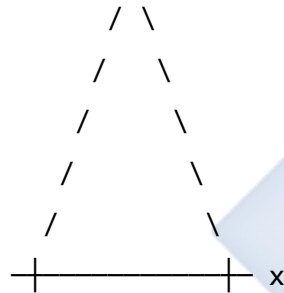
Case 1: Two distinct real roots



Case 2: One repeated root



Case 3: No real roots



Case 4: Maximum, two roots



Case 5: Maximum, one root



Case 6: Maximum, no roots



Key Insights:

- **Discriminant ($b^2 - 4ac$)** determines the number of roots
- **Positive discriminant:** Two distinct roots
- **Zero discriminant:** One repeated root
- **Negative discriminant:** No real roots

4. REVISION QUESTIONS

1. Explain the relationship between the roots of a quadratic equation and its coefficients.
 2. Describe the step-by-step process for drawing an accurate quadratic graph.
 3. How can you use a quadratic graph to solve equations other than the original?
 4. What real-world situations can be modeled by quadratic equations? Provide specific examples.
 5. How does the sign of the coefficient 'a' affect the shape of a quadratic graph?
-

5. ASSESSMENT

EVALUATION (IN-CLASS)

1. Find the quadratic equation whose roots are 5 and 2.
2. Find the quadratic equation whose roots are 0 and -7.
3. Find the sum and product of roots of $2x^2 - 6x + 5 = 0$.
4. From the graph of $y = x^2 - 2x - 3$, find the axis of symmetry.
5. What is the minimum value of y in $y = x^2 - 2x - 3$?

ASSIGNMENT (TAKE-HOME)

1. Find the quadratic equation whose roots are $\frac{2}{3}$ and $-\frac{1}{2}$.
2. Draw the graph of $y = 8 - 2x - x^2$ for $-5 \leq x \leq 3$.
3. From your graph in Q2, find:
 - The roots of $8 - 2x - x^2 = 0$
 - The roots of $8 - 2x - x^2 = 5$
 - The maximum value of y
4. The product of two consecutive odd integers is 63. Form a quadratic equation and solve for the integers. (Hint: Let them be n and n + 2)

EXTENSION (CHALLENGE PROBLEMS)

1. The sum of the roots of a quadratic equation is 5 and the product is -14. Find the equation.
 2. A quadratic graph passes through points (1, 0), (3, 0), and (0, 6). Find its equation.
 3. The height of a projectile is given by $h = -4t^2 + 16t$. Find:
 - Maximum height reached
 - Time when height is 12 meters
-

6. STUDY TIPS AND COMMON PITFALLS

Tip 1: Always Verify Roots

- Substitute back into the original equation
- Check if they satisfy the equation

Tip 2: Choose Appropriate x-values

- Include points around the vertex
- Ensure you capture the curve's shape

Tip 3: Handle Fractions Carefully

- Multiply through to clear denominators
- Keep work neat and organized

Tip 4: Interpret Real-World Context

- Reject solutions that don't make sense (negative lengths, etc.)
- Include appropriate units in final answers

Common Pitfalls:

- Forgetting the negative sign in $x^2 - (sum)x + (product) = 0$
- Incorrectly plotting points from the table of values
- Drawing straight lines between points instead of smooth curves

- Misidentifying maximum vs. minimum points

Memory Poem:

From roots to graphs, the story's told,
Of equations brave and stories bold.
Sum and product, a magical pair,
Create new equations from thin air!

Parabolas curve, both up and down,
In every city, in every town.
Maximum profit, projectile's flight,
Quadratic equations shine so bright!

So draw your graphs with careful hand,
Across the coordinate land.
Find the roots, the vertex too,
The power of quadratics belongs to you!

Answers to Assessment

In-Class:

1. $x^2 - 7x + 10 = 0$
2. $x^2 + 7x = 0$
3. Sum = 3, Product = 2.5
4. $x = 1$
5. $y = -4$

Take-Home:

1. $6x^2 - x - 2 = 0$
2. Graph should show inverted parabola with vertex at $(-1, 9)$
3. a) $x = -4, 2$; b) $x \approx -3, 1$; c) Maximum $y = 9$

4. Integers are 7 and 9, or -9 and -7

Extension:

1. $x^2 - 5x - 14 = 0$
2. $y = 2x^2 - 8x + 6$
3. a) Maximum height = 16 m; b) $t = 1$ s or 3 s

WEEK 11: LOGICAL REASONING I

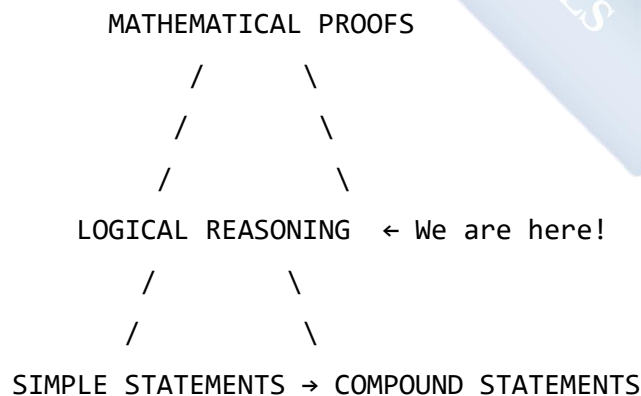
Concept: The Foundation of Mathematical Thinking and Proof

1. INTRODUCTION: THE BEDROCK OF MATHEMATICS

Beyond Numbers and Formulas

Mathematics is fundamentally about **structured thinking**, **precise reasoning**, and **logical proof**. Every mathematical argument, theorem, or solution depends on logical principles to determine validity.

Visual: The Logic Pyramid



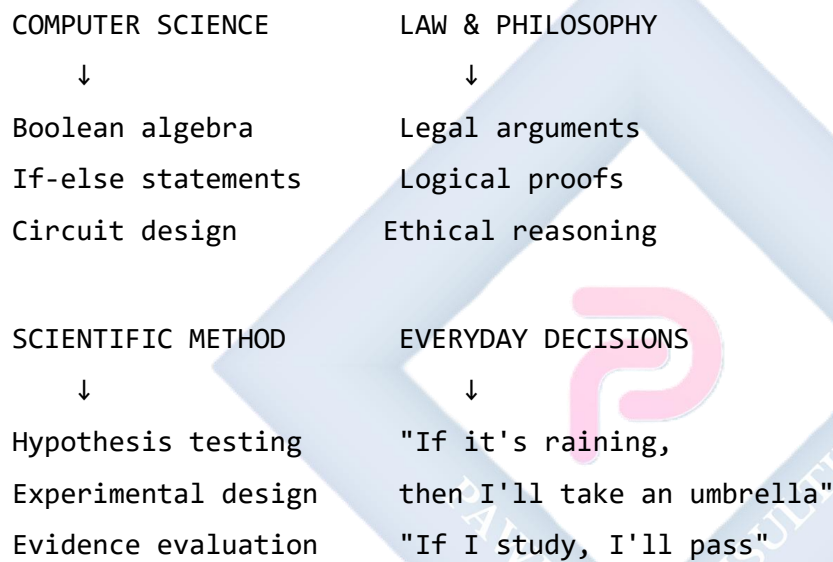
Why Logical Reasoning Matters:

- Distinguishes between valid and invalid arguments

- Forms the basis of mathematical proofs
- Essential for computer programming and algorithms
- Critical for legal reasoning and scientific method
- Helps detect fallacies in everyday thinking

Real-World Applications

Diagram 1: Logic in Everyday Life



Key Insight: Logic is the **operating system** of rational thought, whether in mathematics, science, or daily decision-making.

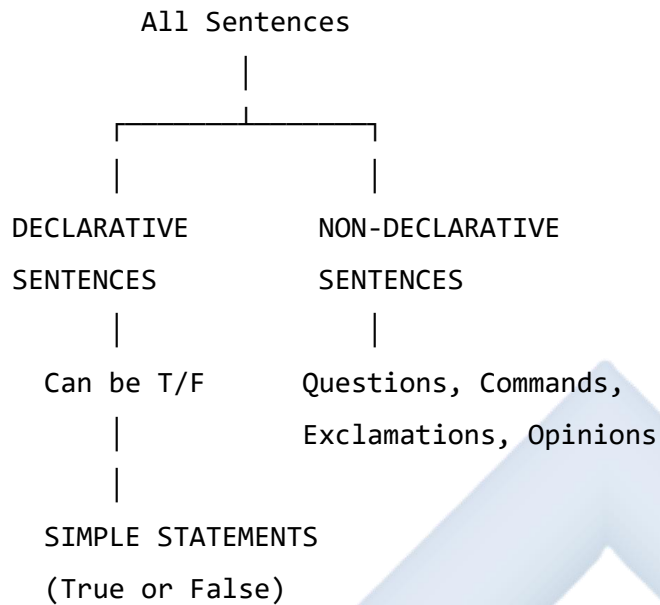
2. CONTENT

A. SIMPLE STATEMENTS (PROPOSITIONS)

Definition and Characteristics

A **simple statement** (or **proposition**) is a declarative sentence that can be clearly classified as either **True (T)** or **False (F)** — but not both, and not neither.

Visual: Statement Classification



Mathematical Notation: We use lowercase letters: **p, q, r, s** to represent statements

Examples of Simple Statements

Symbol	Statement	Truth Value	Reasoning
p	"Lagos is a city in Nigeria."	True (T)	Verifiable geographical fact
q	"The human head is made of wood."	False (F)	Anatomically incorrect
r	" $2 + 5 = 7$ "	True (T)	Mathematical truth
s	" $10 < 3$ "	False (F)	Mathematical falsehood

What is NOT a Simple Statement?

Visual: The "Not Statements" Gallery

QUESTIONS:

"What is your name?"



Asks something

COMMANDS:

"Pass the salt."



Tells to act

OPINIONS:

"This food is delicious."



Subjective feeling

No truth value

No truth value

No objective truth

OPEN SENTENCES:

" $x + 3 = 8$ "

↓

Depends on x

Not T/F until

x is known

EXCLAMATIONS:

"Wow, amazing!"

↓

Emotional expression

Not T/F

Detailed Analysis:

Type	Example	Why Not a Statement
Question	"What time is it?"	Asks rather than declares
Command	"Close the door."	Instructs action
Opinion	"Mathematics is beautiful."	Subjective, not objectively verifiable
Open Sentence	" $x > 5$ "	Truth depends on variable value
Exclamation	"Great job!"	Expresses emotion

Critical Thinking Exercise

Analyze These Sentences:

1. "The square root of 16 is 4." → **Statement** (True)
2. "Please solve this problem." → **Not statement** (Command)
3. "This sentence is false." → **Paradox** (Neither T nor F consistently)
4. "Water boils at 100°C at sea level." → **Statement** (True)
5. "That movie was boring." → **Not statement** (Opinion)

B. TRUE AND FALSE STATEMENTS

Characteristics of Valid Statements

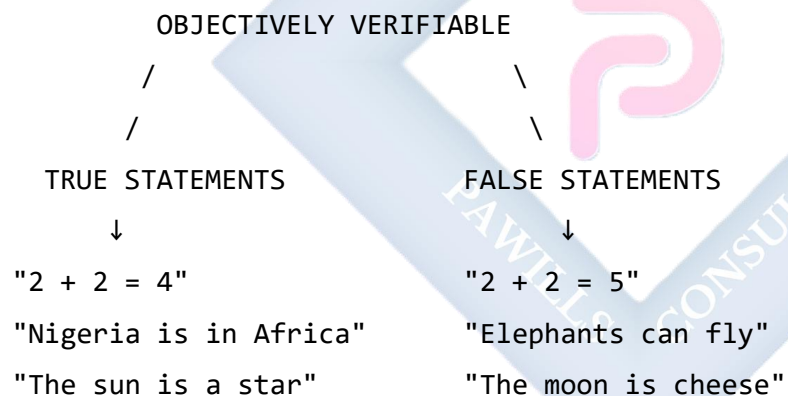
True Statements Must Be:

- Objectively verifiable
- Factually correct
- Context-independent

False Statements Are:

- Objectively disprovable
- Factually incorrect
- Still valid statements (because they have a truth value)

Visual: The Truth Spectrum



More Examples

True Statements:

- p: "Nigeria gained independence in 1960."
- q: "The sum of angles in a triangle is 180°."
- r: "Water freezes at 0°C at standard pressure."

False Statements:

- s: "There are 8 days in a week."
- t: "Humans can breathe underwater."
- u: " $5 \times 6 = 31$ "

Important: A false statement is still a **valid statement** because it has a definite truth value (False).

C. NEGATION OF SIMPLE STATEMENTS

Concept and Symbolism

The **negation** of a statement expresses its logical opposite — it denies the original statement.

Symbol: $\sim$

Read as: "not"

The negation of p is written as $\sim p$ and read as "not p."

Visual: The Negation Machine

INPUT: p		OUTPUT: $\sim p$
"It is raining"	→	"It is not raining"
" $2 + 2 = 4$ "	→	" $2 + 2 \neq 4$ "
" $x < 5$ "	→	" $x \geq 5$ "

Forming Correct Negations

Method 1: Add "It is not the case that..."

- p: "Monday is the first day of the week."
- $\sim p$: "It is not the case that Monday is the first day of the week."

Method 2: Insert "not" grammatically

- p: "All cats are black."

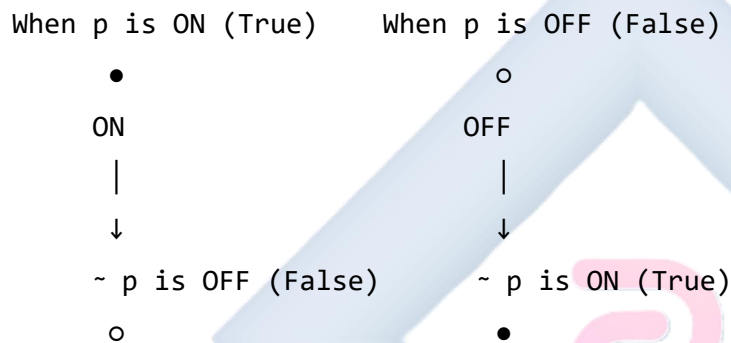
- $\sim p$: "Not all cats are black." or "Some cats are not black."

Method 3: Reverse mathematical relations

- $p: "x > 5" \rightarrow \sim p: "x \leq 5"$
- $q: "a = b" \rightarrow \sim q: "a \neq b"$

Truth Table for Negation

Visual: The Negation Switch



Formal Truth Table:

p	$\sim p$	Verbal Example
T	F	p : "It is raining" (True) $\sim p$: "It is not raining" (False)
F	T	p : " $2 + 2 = 5$ " (False) $\sim p$: " $2 + 2 \neq 5$ " (True)

Key Rule: Negation always **flips** the truth value.

DETAILED WORKED EXAMPLES

Example 11.1: Identifying Statements

Which of the following are simple statements? For those that are, state their truth value.

- "The year 2024 is a leap year."
- "Solve the equation."

c) "Red is a beautiful colour."

d) " $5 - 1 = 3$ "

e) " $x > 10$ "

Step-by-Step Analysis:

a) "The year 2024 is a leap year."

→ DECLARATIVE, VERIFIABLE

→ STATEMENT, Truth Value: TRUE ($2024 \div 4 = 506$, no remainder)

b) "Solve the equation."

→ COMMAND, INSTRUCTS ACTION

→ NOT A STATEMENT

c) "Red is a beautiful colour."

→ OPINION, SUBJECTIVE

→ NOT A STATEMENT

d) " $5 - 1 = 3$ "

→ DECLARATIVE, MATHEMATICALLY VERIFIABLE

→ STATEMENT, Truth Value: FALSE ($5 - 1 = 4 \neq 3$)

e) " $x > 10$ "

→ OPEN SENTENCE, DEPENDS ON x

→ NOT A STATEMENT (until x is specified)

✓ **Answer:** Only (a) and (d) are statements.

Example 11.2: Writing Negations

Write the negation of each statement:

- a) p: "Monday is the first day of the week."
b) q: "A triangle does not have 4 sides."
c) r: " $10 \geq 7$ "

Solution:

- a) p: "Monday is the first day of the week."
~ p: "Monday is not the first day of the week."
OR: "It is not the case that Monday is the first day of the week."
- b) q: "A triangle does not have 4 sides."
~ q: "A triangle has 4 sides."
(Note: We remove the "not" rather than adding another one)
- c) r: " $10 \geq 7$ " (10 is greater than or equal to 7)
~ r: " $10 < 7$ " (10 is less than 7)
(Mathematical opposite of " $\geq$ " is "<")

Critical Thinking: The negation must completely reverse the meaning, not just add "not" carelessly.

Example 11.3: Truth Values of Negations

Let:

- p: "A square has four equal sides." (True)
- q: "2 is an odd number." (False)

Find the truth values of: a) ~ p b) ~ q

Solution:

- a) p is True $\rightarrow$ ~ p must be False
~ p: "A square does not have four equal sides." $\rightarrow$ FALSE

b) q is False $\rightarrow \sim q$ must be True
 $\sim q$: "2 is not an odd number." $\rightarrow$ TRUE

Summary Table:

Statement	Truth Value	Negation	Negation Truth Value
p : A square has four equal sides.	T	$\sim p$: A square does not have four equal sides.	F
q : 2 is an odd number.	F	$\sim q$: 2 is not an odd number.	T

Practice Exercise: Tricky Negations

Original: "All cats are black."

Common Mistake: "No cats are black." (Wrong!)

Correct Negation: "Not all cats are black." or "Some cats are not black."

Why this is correct:

- If "All cats are black" is False, it doesn't mean "No cats are black"
- It only means "There exists at least one cat that is not black"

Visual Understanding:

"All cats are black"

●●●●●●●●●●

(All black)

Negation: "Not all cats are black"

●●●●○●●●●●

(Most black, but one isn't)

D. THE IMPORTANCE OF LOGICAL REASONING

Real-World Applications

Diagram 2: Logic Across Disciplines

MATHEMATICS



Proofs and theorems

Valid arguments

COMPUTER SCIENCE



Boolean logic

If-then programming

Circuit design

LAW & PHILOSOPHY



Legal reasoning

Argument analysis

Ethical proofs

EVERYDAY THINKING



Critical thinking

Detecting fallacies

Decision making

SCIENCE



Hypothesis testing

Experimental design

Evidence evaluation

MEDICINE



Diagnostic logic

Symptom analysis

Treatment decisions

Common Logical Fallacies in Everyday Life

Ad Hominem: Attacking the person instead of the argument

- "Your idea is wrong because you're young."

False Cause: Assuming causation from correlation

- "I wore red and won, so red is lucky."

Hasty Generalization: Drawing conclusions from insufficient evidence

- "I met one rude person from that city, so everyone there is rude."

Appeal to Emotion: Using emotions rather than logic

- "You should agree with me because it will make me happy."

3. REVISION QUESTIONS

1. What two conditions must a sentence satisfy to be considered a "simple statement"?
 2. Provide one example each of: a command, an opinion, and an open sentence.
 3. Explain the symbol and meaning of negation in logic.
 4. If statement p is False, what must be the truth value of $\sim p$? Why?
 5. Create a truth table for negation and explain what it demonstrates.
-

4. ASSESSMENT

EVALUATION (IN-CLASS)

1. Which of the following are simple statements? a) "Good morning!" b) "5 is a prime number." c) "He is very smart."
2. Write the negation of p : "The sun rises in the West."
3. What is the truth value of the negation you wrote in Q2?
4. Let q be " $4 + 4 = 8$." State the truth value of q and $\sim q$.
5. Write the negation of r : " $x < 5$."

ASSIGNMENT (TAKE-HOME)

1. Write three true simple statements and three false simple statements from different categories (mathematics, geography, science).
2. Write the negation of each: a) p : "All birds can fly." b) q : "A rhombus is not a square." c) r : "12 is a multiple of 5." d) s : "A student is present."
3. State the truth value of each negation from Q2.
4. Let p be " $x = 3$ " and q be " $x^2 = 9$."

- If $x = 3$, what are the truth values of p and q ?
 - If $x = -3$, what are the truth values of p and q ?
5. **Critical Thinking:** Explain why "This statement is false" creates a logical paradox and cannot be a simple statement.

EXTENSION / ENRICHMENT

Visualization Activity:

- Create "Logic Cards" with statements on one side and their truth values on the back
- Practice identifying statements vs. non-statements in newspaper headlines

Group Discussion:

- "Is mathematics discovered or invented?" - Analyze the arguments logically
- "Can artificial intelligence truly think?" - Apply logical reasoning to this question

Real-World Investigation:

- Find three examples of logical fallacies in advertisements
- Analyze a political argument and identify any logical errors

5. COMMON PITFALLS AND TIPS

Pitfall 1: Confusing opinions with statements

Tip: Ask "Can this be objectively verified as true or false?"

Pitfall 2: Incorrect negation formation

Tip: Always check if the negation completely reverses the original meaning

Pitfall 3: Overlooking mathematical opposites

Tip: Remember: $< \rightarrow \geq$, $> \rightarrow \leq$, $= \rightarrow \neq$

Pitfall 4: Treating open sentences as statements

Tip: Look for variables - if present, it's not a simple statement

Memory Poem:

Simple statements, true or false,
That's the logic we must solve.
Declarative and clear they stand,
With truth values close at hand.

Negation flips what is declared,
The opposite is what is shared.
If true becomes a falsehood new,
If false becomes a statement true.

In math and life and computer's core,
It's logic that we all adore.
So think with care and reason well,
As logic's stories we now tell!

Answers to Assessment

In-Class:

1. a) Not statement (greeting), b) Statement (True), c) Not statement (opinion)
2. $\sim p$: "The sun does not rise in the West." or "It is not the case that the sun rises in the West."
3. True (since p is false)
4. q : True, $\sim q$: False
5. $\sim r$: " $x \geq 5$ "

Take-Home:

1. Examples: True - "Paris is in France", " $7 \times 8 = 56$ ", "Water is H_2O "; False - "Rome is in Asia", " $9 - 4 = 6$ ", "Fish breathe air"

2. a) "Not all birds can fly" or "Some birds cannot fly"; b) "A rhombus is a square"; c) "12 is not a multiple of 5"; d) "A student is not present"
3. a) True, b) False, c) True, d) Depends on context
4. When $x=3$: $p=\text{True}$, $q=\text{True}$; When $x=-3$: $p=\text{False}$, $q=\text{True}$
5. If it's true, then it says it's false (contradiction). If it's false, then it says it's true (contradiction). Therefore, it cannot have a consistent truth value.

WEEK 12: LOGICAL REASONING II

Concept: Building Complex Arguments from Simple Statements Using Logical Connectives

1. INTRODUCTION: FROM SIMPLE TO COMPOUND REASONING

Building on Week 11 Foundations

Last Week (Simple Statements):

p : "It is raining." (True/False)

q : "I will take an umbrella." (True/False)

Individual statements with definite truth values

This Week (Compound Statements):

$p \rightarrow q$: "If it is raining, then I will take an umbrella."

A complex relationship connecting simple statements

Visual: The Logic Building Process

SIMPLE STATEMENTS $\rightarrow$ LOGICAL CONNECTIVES $\rightarrow$ COMPOUND STATEMENTS

p, q, r

$\wedge, \vee, \rightarrow, \leftrightarrow$

$p \wedge q, p \vee q, p \rightarrow q$

$\downarrow$

$\downarrow$

$\downarrow$

"It is raining"

"AND, OR, IF"

"If it rains, I take umbrella"

"I take umbrella"

"It's raining AND I take umbrella"

Real-World Significance:

- Computer programming: if-else conditions
- Legal contracts: "If party A fails to deliver, then party B may terminate"
- Scientific reasoning: "If hypothesis is true, then we expect these results"
- Everyday decisions: "If it's weekend, then I'll sleep late"

The Power of Logical Connections

Diagram 1: Logic in Everyday Decision Making

WEATHER DECISIONS



If raining → take umbrella
If sunny → go to park

ACADEMIC PLANNING



If exam → then study
If weekend → then relax

COMPUTER PROGRAMMING



If user_input = "yes"
Then proceed
Else ask again

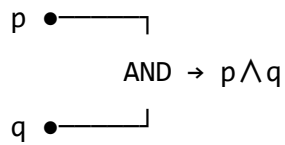
MEDICAL DIAGNOSIS



If fever AND cough
Then suspect infection
Else check other symptoms

Key Insight: Compound statements allow us to model complex real-world conditions and relationships.

Visual: The AND Gate



TRUTH TABLE:

p	q	$p \wedge q$	
●	●	●	(Both ON $\rightarrow$ ON)
●	○	○	(One OFF $\rightarrow$ OFF)
○	●	○	(One OFF $\rightarrow$ OFF)
○	○	○	(Both OFF $\rightarrow$ OFF)

Formal Truth Table:

p	q	$p \wedge q$	Explanation
T	T	T	Both true $\rightarrow$ Statement true
T	F	F	One false $\rightarrow$ Statement false
F	T	F	One false $\rightarrow$ Statement false
F	F	F	Both false $\rightarrow$ Statement false

Real-World Example:

p : "You have a ticket" (True)

q : "The show is today" (True)

$p \wedge q$: "You have a ticket AND the show is today" $\rightarrow$ True

But if either is false, the whole statement becomes false.

Memory Aid: "AND demands perfection - both must be true!"

2. DISJUNCTION ("OR") - The Inclusive Operator

Symbol: $\vee$

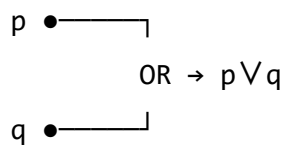
Read as: "p or q"

Form: $p \vee q$

Rule: False only when **both** p and q are False

(This is **INCLUSIVE OR** - one, the other, or both)

Visual: The OR Gate



TRUTH TABLE:

p	q	$p \vee q$	
●	●	●	(Both ON $\rightarrow$ ON)
●	○	●	(One ON $\rightarrow$ ON)
○	●	●	(One ON $\rightarrow$ ON)
○	○	○	(Both OFF $\rightarrow$ OFF)

Formal Truth Table:

p	q	$p \vee q$	Explanation
T	T	T	Both true $\rightarrow$ Statement true
T	F	T	One true $\rightarrow$ Statement true
F	T	T	One true $\rightarrow$ Statement true
F	F	F	Both false $\rightarrow$ Statement false

Real-World Example:

p: "I will have tea" (True/False)

q: "I will have coffee" (True/False)

$p \vee q$: "I will have tea OR coffee" $\rightarrow$ True if I have either or both

Only false if I have neither tea nor coffee.

Important Note: This is **inclusive OR** (mathematical sense), not exclusive OR (either/or but not both).

Memory Aid: "OR is generous - just one true makes it true!"

3. *CONDITIONAL ("IF... THEN...") - The Implication Operator*

Symbol: $\rightarrow$

Read as: "If p, then q" or "p implies q"

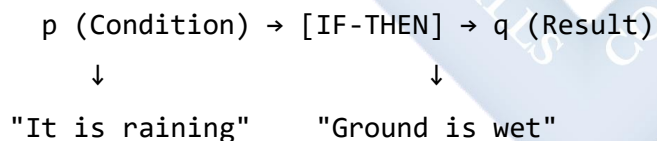
Form: $p \rightarrow q$

Components:

- **p:** Antecedent (condition)
- **q:** Consequent (result)

Rule: False only when p is True and q is False

Visual: The IF-THEN Relationship



TRUTH TABLE:

p	q	$p \rightarrow q$	
•	•	•	(Rain + Wet $\rightarrow$ Promise kept)
•	○	○	(Rain + Dry $\rightarrow$ Promise broken)
○	•	•	(No rain + Wet $\rightarrow$ Promise not broken)
○	○	•	(No rain + Dry $\rightarrow$ Promise not broken)

Formal Truth Table:

p	q	$p \rightarrow q$	Explanation
T	T	T	Condition true, result true $\rightarrow$ Promise kept
T	F	F	Condition true, result false $\rightarrow$ Promise broken
F	T	T	Condition false $\rightarrow$ Promise not applicable
F	F	T	Condition false $\rightarrow$ Promise not applicable

Real-World Example:

p: "It is raining" (Condition)

q: "The ground is wet" (Result)

$p \rightarrow q$: "If it is raining, then the ground is wet"

The ONLY time this is false: It's raining but ground is dry.

Counterintuitive but Important: When the condition (p) is false, the entire statement is considered **true** by convention.

Memory Aid: "IF-THEN only fails when the promise is broken!"

4. BICONDITIONAL ("IF AND ONLY IF") - The Equality Operator

Symbol: $\leftrightarrow$

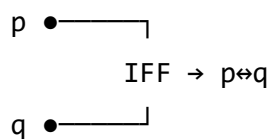
Read as: "p if and only if q" (abbreviated "iff")

Form: $p \leftrightarrow q$

Meaning: p and q must have **the same truth value**

Equivalent to: $(p \rightarrow q) \wedge (q \rightarrow p)$

Visual: The IFF Relationship



TRUTH TABLE:

p	q	$p \leftrightarrow q$
•	•	• (Both same $\rightarrow$ True)
•	○	○ (Different $\rightarrow$ False)
○	•	○ (Different $\rightarrow$ False)
○	○	• (Both same $\rightarrow$ True)

Formal Truth Table:

p	q	$p \leftrightarrow q$	Explanation
T	T	T	Both true $\rightarrow$ Equivalent
T	F	F	Different values $\rightarrow$ Not equivalent
F	T	F	Different values $\rightarrow$ Not equivalent
F	F	T	Both false $\rightarrow$ Equivalent

Real-World Example:

p: "A figure is a square"

q: "A figure has four equal sides and four right angles"

$p \leftrightarrow q$: "A figure is a square IF AND ONLY IF it has four equal sides and four right angles"

Both describe exactly the same condition.

Memory Aid: "IFF requires perfect matching - both true or both false!"

C. COMPREHENSIVE WORKED EXAMPLES

Example 12.1: Translation Between English and Symbols

Given:

- p: "The food is hot"
- q: "The drink is cold"

Translate to Symbolic Form:

English Statement	Symbolic Form	Step-by-Step Breakdown
"The food is hot and the drink is cold"	$p \wedge q$	Simple conjunction
"The food is not hot or the drink is cold"	$\sim p \vee q$	Negation + disjunction
"If the food is hot, then the drink is not cold"	$p \rightarrow (\sim q)$	Conditional with negation

Visual Translation Process:

English: "If the food is hot, then the drink is not cold"

```
| | | | |  
| | | | └ "not cold" = ~ q  
| | | └ "then" indicates →  
| | └ "if" starts conditional  
| └ "food is hot" = p  
└ Entire structure: p → (~ q)
```

Example 12.2: Evaluating Truth Values

Given:

- $p: \text{"Lagos is in Ghana"} \rightarrow \text{False (Lagos is in Nigeria)}$
- $q: \text{"}2 \times 3 = 6\text{"} \rightarrow \text{True}$

Evaluate: "If Lagos is in Ghana, then $2 \times 3 = 6$ "

Step-by-Step Solution:

Step 1: Identify the form: $p \rightarrow q$

Step 2: Substitute truth values: $F \rightarrow T$

Step 3: Consult truth table for conditional:

p	q	$p \rightarrow q$
F	T	T

Step 4: Therefore, the statement is TRUE

Explanation: Even though the condition (p) is false, the conditional statement as a whole is considered true because there's no case where the condition is true but the result is false.

Counterintuitive Insight: In logic, a false premise can lead to a true conditional statement. This preserves the consistency of mathematical reasoning.

Example 12.3: Truth Table Construction

Construct truth table for: $p \wedge \sim q$

Step-by-Step Construction:

Step 1: List all possible truth combinations for p and q

Step 2: Add column for $\sim q$ (negation of q)

Step 3: Add column for $p \wedge \sim q$ (conjunction of p and $\sim q$)

p	q	$\sim q$	$p \wedge \sim q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

Visual Table Construction:

Start:		Step 2:			Step 3:			
p	q	p	q	$\sim q$	p	q	$\sim q$	$p \wedge \sim q$
T	T	T	T	F	T	T	F	F
T	F	T	F	T	T	F	T	T
F	T	F	T	F	F	T	F	F
F	F	F	F	T	F	F	T	F

Interpretation: The compound statement $p \wedge \sim q$ is true only when p is true AND q is false.

D. ADVANCED CONCEPTS

Logical Equivalences and Special Cases

Tautology: A statement that is **always true** regardless of truth values

- Example: $p \vee \sim p$ ("It is raining or it is not raining")
- Truth table: Always T

Contradiction: A statement that is **always false** regardless of truth values

- Example: $p \wedge \sim p$ ("It is raining and it is not raining")
- Truth table: Always F

Visual: Tautology vs Contradiction

TAUTOLOGY (Always True)

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T
↑		

Always true!

CONTRADICTION (Always False)

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F
↑		

Always false!

3. REAL-WORLD APPLICATIONS AND PATTERNS

Application 1: Computer Programming

```
IF (user_age >= 18) AND (has_id = True) THEN
    grant_access = True
ELSE
    grant_access = False
END IF
```

Application 2: Legal Reasoning

```
IF (contract_signed = True) AND (payment_made = True) THEN
    obligation_fulfilled = True
ELSE
    obligation_fulfilled = False
```

Application 3: Scientific Hypothesis Testing

```
IF (hypothesis = True) THEN (predicted_result = True)
Actual: (predicted_result = False)
CONCLUSION: Therefore, (hypothesis = False)
```

4. COMMON PITFALLS AND TIPS

Pitfall 1: Confusing inclusive vs exclusive OR **Tip:** Mathematical "OR" ($\vee$) is inclusive - remember "one or both"

Pitfall 2: Misunderstanding conditional statements **Tip:** $p \rightarrow q$ is only false when p is true and q is false

Pitfall 3: Incorrect negation placement **Tip:** Use parentheses to group: $p \rightarrow \sim q$ means "if p then not q"

Pitfall 4: Forgetting truth table patterns **Tip:** Create memory aids:

- AND: T only with T,T
- OR: F only with F,F
- $\rightarrow$: F only with T,F
- $\leftrightarrow$: T only with same values

Study Strategy: Practice building truth tables systematically:

1. List all variable combinations
2. Compute inner operations first

3. Work outward through parentheses
 4. Verify with real-world examples
-

5. REVISION QUESTIONS

1. What distinguishes a compound statement from a simple statement?
 2. Explain the four logical operators with their symbols and truth conditions.
 3. Why is $p \rightarrow q$ considered true when p is false?
 4. How is the biconditional operator different from the conditional operator?
 5. Provide real-world examples of each logical operator in action.
 6. What are tautologies and contradictions? Give examples.
 7. How do logical operators relate to computer programming?
-

6. ASSESSMENT

EVALUATION (IN-CLASS)

1. Let p = "It is sunny" and q = "It is hot." Write "It is not sunny and it is hot" in symbols.
2. Let $p = T$ and $q = T$. What is the truth value of $p \rightarrow (\sim q)$?
3. Let $p = F$ and $q = F$. What is the truth value of $p \leftrightarrow q$?
4. Construct the truth table for $p \vee \sim q$.
5. Identify the antecedent in: "If $x = 2$, then $x^2 = 4$."

ASSIGNMENT (TAKE-HOME)

1. Let p : "You are in SS1" and q : "You study Maths." Translate into English: a) $p \wedge q$ b) $\sim p \vee q$
c) $q \rightarrow p$
2. Let p : "An elephant can fly" (F), q : "A square has 4 sides" (T). Find truth values of: a) $p \wedge q$ b) $p \vee q$ c) $p \rightarrow q$ d) $\sim q \rightarrow p$
3. Construct truth table for $(p \vee q) \rightarrow p$.

4. Construct truth table for $(p \rightarrow q) \wedge (q \rightarrow p)$ and verify it matches $p \leftrightarrow q$.
5. A statement that is always True is a **Tautology** (e.g., $p \vee \sim p$). A statement that is always False is a **Contradiction** (e.g., $p \wedge \sim p$). Construct truth table for $p \wedge \sim p$ and state whether it's a tautology or contradiction.

EXTENSION (CHALLENGE PROBLEMS)

1. Create a truth table for $(p \rightarrow q) \vee (q \rightarrow p)$. What pattern do you notice?
2. Using truth tables, prove that $\sim (p \wedge q)$ is equivalent to $(\sim p \vee \sim q)$.
3. Design a logical expression for a security system that grants access if:
 - The user has a valid ID AND knows the password, OR
 - The user has admin privileges AND the system is in maintenance mode

7. VISUAL SUMMARY AND MEMORY AIDS

Diagram 2: Logical Operators Quick Reference

CONJUNCTION (AND) $\wedge$

T only with T,T

"Both must be true"

DISJUNCTION (OR) $\vee$

F only with F,F

"At least one true"

CONDITIONAL ($\rightarrow$)

F only with T,F

"Promise broken"

BICONDITIONAL ($\leftrightarrow$)

T only with same values

"Perfect match"

Truth Table Patterns:

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Memory Poem:

Compound statements, logical might,
Connect with AND, OR, day and night.
IF-THEN promises we must keep,
IFF makes equivalence deep.

Truth tables show how values flow,
With T and F in perfect row.
AND wants both to be so true,
OR is happy with just a few.

IF-THEN fails with T and F,
IFF matches, that's the test.
In math and code and thoughtful art,
It's logic playing its smart part!

Answers to Assessment

In-Class:

1. $\sim p \wedge q$
2. F (since p is T, $\sim q$ is F, so $T \rightarrow F$ is F)
3. T (since both F, so same values)

p	q	$\sim q$	$p \vee \sim q$
T	T	F	T
T	F	T	T
F	T	F	F
F	F	T	T

5. "x = 2"

Take-Home:

1. a) "You are in SS1 and you study Maths" b) "You are not in SS1 or you study Maths" c) "If you study Maths, then you are in SS1"

2. a) $F \wedge T = F$ b) $T \vee T = T$

c) $T \rightarrow T = T$ d) $F \rightarrow F = T$? Wait: $\sim q$ means "A square does not have 4 sides" = F, so $F \rightarrow F = T$

p	q	$p \vee q$	$(p \vee q) \rightarrow p$
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	T

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p)$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

They match!

5.

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

This is a CONTRADICTION (always false)

WEEK 13: REVISION, EXAMINATION & CLOSING

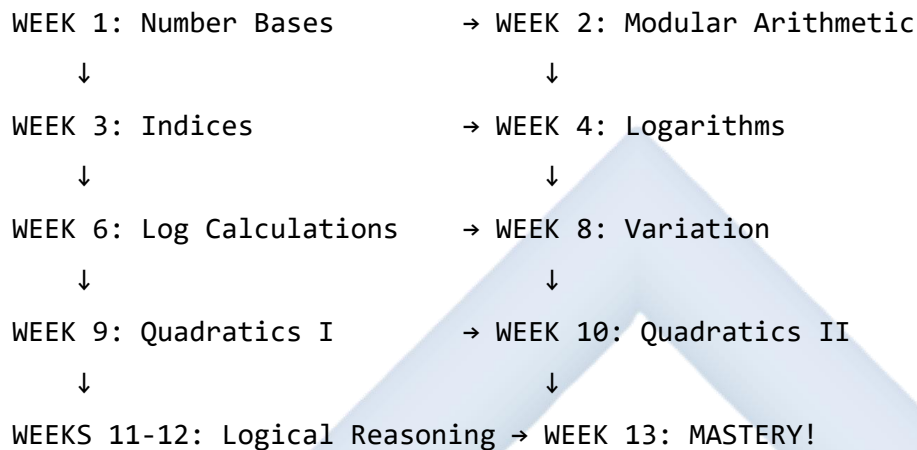
Concept: Mastering Mathematical Connections and Building Exam Confidence

1. INTRODUCTION: THE MATHEMATICAL JOURNEY REVIEW

From Foundation to Mastery

Visual: Our Mathematical Journey Map

TERM 1 ROADMAP:



Key Insight: Mathematics builds cumulatively - each topic connects to and reinforces the others. Understanding these connections is the key to true mastery.

2. COMPREHENSIVE REVISION STRATEGY

The Three-Level Revision Approach

Level 1: Conceptual Understanding (Know the "Why")

- Definitions, principles, and relationships
- Why formulas work the way they do

Level 2: Procedural Fluency (Know the "How")

- Step-by-step methods and algorithms
- Problem-solving techniques

Level 3: Application Mastery (Know the "When")

- Real-world applications
- Recognizing which tools to use when

3. TOPIC-BY-TOPIC REVISION GUIDE

WEEK 1 – NUMBER BASE SYSTEMS

Core Concept: Different ways of representing numbers

Visual: Base Conversion Relationships

Binary (Base 2) ↔ Decimal (Base 10) ↔ Other Bases

$$1101_2 = 13_{10} = 15_8$$

Key Formulas:

- To Decimal: Expand using place values
- From Decimal: Repeated division/multiplication

✓ **Self-Check Challenge:**

1. Convert 1101_2 to Base 10 → 13_{10}
2. Express 75_{10} in Base 5 → 300_5
3. Add $243_5 + 132_5$ → 430_5
4. Explain Base 8 → Uses digits 0-7, each place = power of 8

Common Pitfall: Forgetting that "10" in any base equals that base value

🕒 WEEK 2 – MODULAR ARITHMETIC

Core Concept: Mathematics of cycles and remainders

Visual: The Modular Clock

Mod 5 System: $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 0 \rightarrow 1 \dots$

Equivalent Equivalent

Key Insight: $a \equiv b \pmod{m}$ means $(a - b)$ is divisible by m

✓ Self-Check Challenge:

1. $17 \pmod{5} \rightarrow 2$
2. Wednesday + 100 days $\rightarrow 100 \pmod{7} = 2 \rightarrow$ **Friday**
3. Explain $7 \equiv 2 \pmod{5} \rightarrow 7-2=5$, **divisible by 5**
4. $(8 + 15) \pmod{6} \rightarrow 23 \pmod{6} = 5$

Memory Aid: "Modular arithmetic = clock mathematics"

⚡ WEEK 3 – INDICES (EXPONENTS)

Core Concept: Efficient representation of repeated multiplication

Visual: The 7 Laws of Indices

1. Multiplication: $a^m \times a^n = a^{m+n}$
2. Division: $a^m \div a^n = a^{m-n}$
3. Power of Power: $(a^m)^n = a^{m \cdot n}$
4. Zero Power: $a^0 = 1$
5. Negative: $a^{-n} = 1/a^n$
6. Fractional: $a^{1/n} = \sqrt[n]{a}$
7. Combined: $a^{m/n} = (\sqrt[n]{a})^m$

✓ Self-Check Challenge:

1. $2^3 \times 2^5 \rightarrow 2^8 = 256$
2. $(3^2)^4 \rightarrow 3^8 = 6561$
3. Solve $9^x = 27 \rightarrow x = 1.5$
4. $1/8$ as power of 2 $\rightarrow 2^{-3}$

Pro Tip: When bases differ, convert to same base first

WEEK 4 – INTRODUCTION TO LOGARITHMS

Core Concept: The inverse of exponentiation

Visual: The Exponent-Logarithm Relationship

Exponential: $10^3 = 1000$

Logarithmic: $\log_{10} 1000 = 3$



Key Formulas:

- $\log_a b = c$ means $a^c = b$
- Characteristic = integer part from power of 10
- Mantissa = decimal part from log tables

✓ **Self-Check Challenge:**

1. $3^4 = 81$ in log form $\rightarrow \log_3 81 = 4$
 2. Characteristic of $\log_{10}(0.045) \rightarrow 2$ (from 4.5×10^{-2})
 3. $\log_{10}(12.5)$ using tables $\rightarrow \approx 1.0969$
 4. Three basic log laws $\rightarrow \log(ab) = \log a + \log b$, $\log(a/b) = \log a - \log b$, $\log a^b = b \log a$
-

WEEK 6 – LOGARITHMIC CALCULATIONS

Core Concept: Using logs as a "human calculator"

Visual: The Logarithmic Transformation

Complex Operations $\rightarrow$ Simple Operations

Multiplication $\rightarrow$ Addition

Division $\rightarrow$ Subtraction

Powers	→	Multiplication
Roots	→	Division

✔ Self-Check Challenge:

1. 32×2.5 using logs $\rightarrow \log 32 + \log 2.5 \approx 1.5051 + 0.3979 = 1.9030 \rightarrow \text{antilog} \approx 80$
2. Cube root using logs $\rightarrow$ **divide log by 3, then antilog**
3. Explain $\log(ab) = \log a + \log b \rightarrow$ **because $a^m \times a^n = a^{m+n}$**

Real-World Connection: This is how calculations were done before calculators!

WEEK 8 – VARIATION

Core Concept: Mathematical relationships between changing quantities

Visual: Variation Types

Direct: $y = kx$	(both increase together)
Inverse: $y = k/x$	(one increases, other decreases)
Joint: $y = kxz$	(depends on multiple variables)
Combined: $y = kx^2/z$	(mix of direct and inverse)

✔ Self-Check Challenge:

1. $y \propto x^2$, $y=8$ when $x=2$, find y when $x=5 \rightarrow y = 50$
2. $y \propto 1/x^2$, x doubles $\rightarrow y$ becomes $\frac{1}{4}$
3. Make k subject in $y = k/x^2 \rightarrow k = yx^2$

Memory Aid: "Direct = same direction, Inverse = opposite direction"

WEEK 9 – QUADRATICS I (FACTORIZATION)

Core Concept: Solving second-degree equations by factoring

Visual: The Zero Product Principle

If $A \times B = 0$, then $A=0$ OR $B=0$

↓

$$(x - 2)(x - 3) = 0$$

↓

$$x = 2 \text{ OR } x = 3$$

✔ **Self-Check Challenge:**

1. Factorize $3x^2 - 7x + 2 \rightarrow (3x - 1)(x - 2)$
2. Solve $x^2 + 5x = 14 \rightarrow x^2 + 5x - 14 = 0 \rightarrow (x + 7)(x - 2) = 0 \rightarrow x = -7, 2$
3. Check solution $\rightarrow$ **substitute back into original equation**

Pro Tip: Always check your factors by expanding back

WEEK 10 – QUADRATICS II (ROOTS & GRAPHS)

Core Concept: Complete understanding of quadratic behavior

Visual: From Roots to Graph

Roots: α and β

↓

$$\text{Equation: } x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

↓

Graph: Parabola with vertex, axis of symmetry

Key Formulas:

- Sum of roots = $-b/a$
- Product of roots = c/a
- Vertex x-coordinate = $-b/2a$

✔ **Self-Check Challenge:**

1. Roots 2 and 3 $\rightarrow x^2 - 5x + 6 = 0$
2. Quadratic formula $\rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

3. Graph of $y = x^2 + 3x + 2 \rightarrow$ **U-shaped, roots at -1 and -2, vertex at (-1.5, -0.25)**
-

WEEKS 11-12 – LOGICAL REASONING

Core Concept: Systematic thinking and argument analysis

Visual: Logical Operator Summary

Conjunction (AND $\wedge$): T only with T,T

Disjunction (OR $\vee$): F only with F,F

Conditional ($\rightarrow$): F only with T,F

Biconditional ($\leftrightarrow$): T only with same values

✔ Self-Check Challenge:

1. Identify statements $\rightarrow$ **only declarative sentences with truth values**
2. Negation of "The sun rises in the east" $\rightarrow$ **"The sun does not rise in the east"**
3. $p \vee q$ is False when $\rightarrow$ **both p and q are False**
4. $p \rightarrow q$ is False when $\rightarrow$ **p is True and q is False**
5. Truth table for $(p \vee q) \rightarrow p \rightarrow$ **shows it's not always true**

Critical Thinking: Logic helps detect fallacies in everyday arguments

4. EXAMINATION PREPARATION MASTERY

The 4-Pillar Success Strategy

Pillar 1: Conceptual Clarity

Before memorizing formulas, understand:

- Why the formula works
- When to apply it
- What the variables represent

Pillar 2: Procedural Practice

Systematic problem-solving:

1. Understand the problem
2. Choose the right method
3. Execute carefully
4. Verify your answer

Pillar 3: Time Management

Exam strategy:

- Scan entire paper first
- Start with confident questions
- Allocate time proportionally
- Leave time for review

Pillar 4: Error Analysis

Learn from mistakes:

- Keep an error log
- Identify patterns in errors
- Practice similar problems

Exam-Specific Tips

Multiple Choice Questions:

- Eliminate obviously wrong answers first
- Watch for "none of the above" and "all of the above"
- Check units and dimensions

Show-Your-Work Problems:

- Write steps clearly and logically
- Include units where appropriate

- Box your final answer
- Even if stuck, show partial work for partial credit

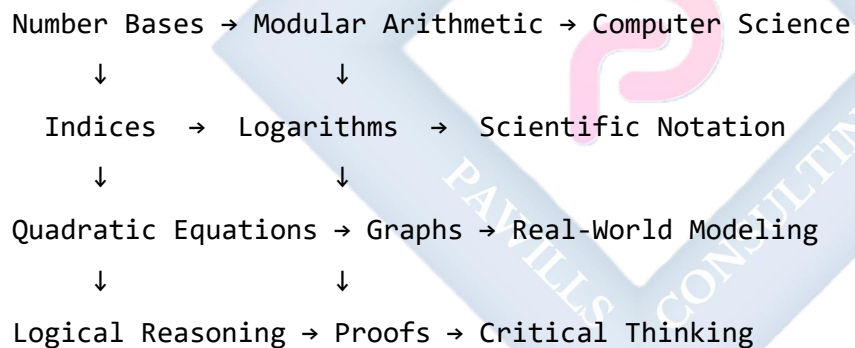
Word Problems:

- Underline key information
- Define variables clearly
- Write down what you're looking for
- Check if answer makes sense in context

5. INTERCONNECTIONS REVIEW

How Topics Connect

Diagram: The Mathematical Web



Key Connections:

- **Logarithms** are the inverse of **indices**
- **Quadratic graphs** visualize **quadratic equations**
- **Logical reasoning** underpins **mathematical proofs**
- **Modular arithmetic** relates to **computer number systems**

6. MINDset FOR SUCCESS

Building Mathematical Confidence

Growth Mindset Statements:

- "Mistakes are learning opportunities"
- "I can improve with practice"
- "Challenge strengthens understanding"
- "Every problem has a solution path"

Exam Mindset:

- "I am prepared and capable"
- "I will read questions carefully"
- "I will manage my time wisely"
- "I will check my work thoroughly"

7. CLOSING REFLECTION & LEGACY

Your Mathematical Journey

What You've Gained:

- **Analytical thinking** skills that transfer to all subjects
- **Problem-solving** strategies for academic and real-life challenges
- **Logical reasoning** abilities for clear, critical thinking
- **Mathematical confidence** that builds with each success

Visual: The Mathematical Mindset

From: "I can't do math"

↓

To: "I haven't figured it out yet"

↓

To: "I understand the pattern"

↓

To: "I can explain it to others"

Final Words of Wisdom

"Mathematics is not about speed; it's about depth of understanding.

It's not about memorization; it's about pattern recognition.

It's not about getting the right answer; it's about the journey of reasoning that leads you there.

The skills you've developed this term—logical thinking, systematic problem-solving, attention to detail—will serve you far beyond the mathematics classroom. They are the foundation of scientific discovery, technological innovation, and informed citizenship.

As you prepare for your examination, remember that you are not just recalling facts—you are demonstrating your ability to think, to reason, and to solve problems. That is the true value of mathematics education."

8. FINAL PREPARATION CHECKLIST

✓ Last-Minute Review

The Night Before:

Review formula sheet and key definitions

Get adequate sleep (7-8 hours)

Prepare materials: calculator, pencils, ruler

Exam Morning:

Eat a healthy breakfast

Arrive early and relaxed

Review mental strategies

During Exam:

Read instructions carefully

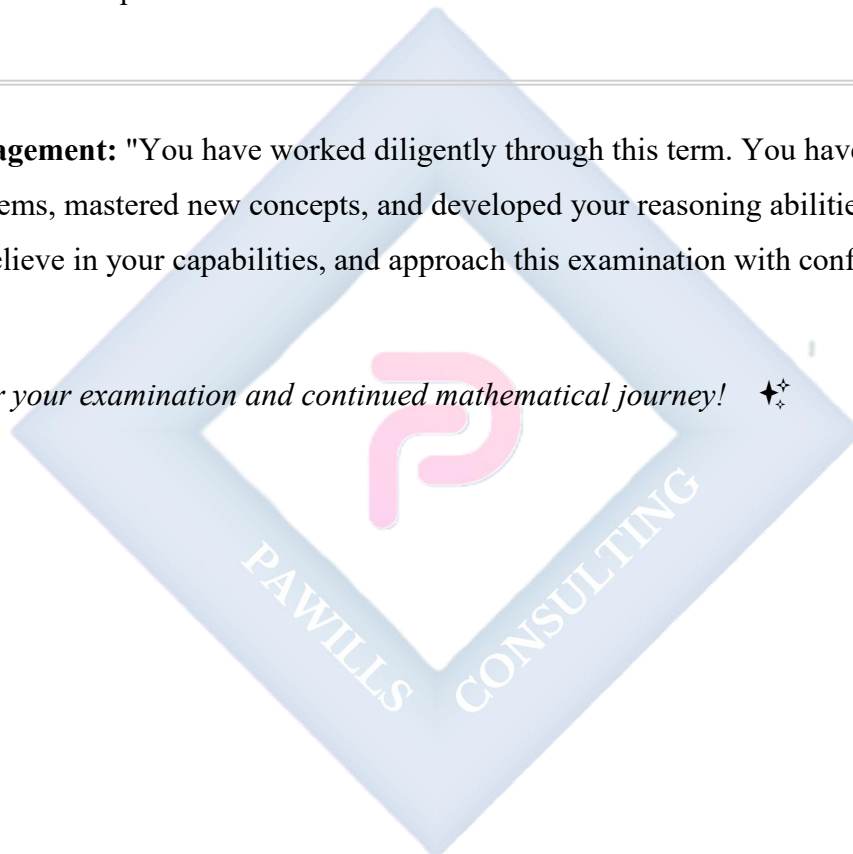
Manage time effectively

Show all work clearly

Review answers if time permits

Final Encouragement: "You have worked diligently through this term. You have solved complex problems, mastered new concepts, and developed your reasoning abilities. Trust in your preparation, believe in your capabilities, and approach this examination with confidence. You are ready!"

Best wishes for your examination and continued mathematical journey! ✨



SS1 MATHEMATICS LESSON NOTES

SECOND TERM

WEEK 1: CONSTRUCTIONS I

1. INTRODUCTION

Welcome to the Second Term! We will begin with **Geometric Constructions**. This is a foundational skill in geometry that involves drawing accurate geometric figures using only two simple tools:

- An **unmarked straightedge** (a ruler, used only for drawing straight lines).
- A **pair of compasses** (used for drawing circles and arcs).

A protractor may be used only to **check** your work at the end, not to create the angle. The goal is to create perfect geometric shapes based on pure principles.

Why Pure Constructions?

Geometric constructions date back to ancient Greece, where mathematicians like Euclid established that perfect geometric figures could be created using only these two tools. This demonstrates the elegance and precision of mathematical reasoning.

2. CONTENT

A. BISECTION OF ANGLES

"**To bisect**" means to cut something into two equal halves.

Procedure: To Bisect any Angle $\angle ABC$

Step 1: Place the compass point at the vertex of the angle (B).

Step 2: Draw an arc that cuts both arms of the angle (at points P and Q).

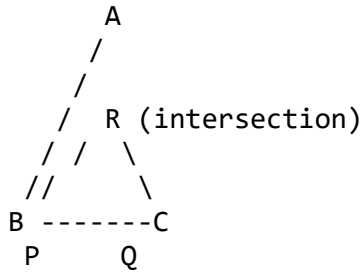
Step 3: Place the compass point at P and draw an arc in the middle of the angle.

Step 4: Without changing the compass radius, place the point at Q and draw another arc to intersect the first arc (at point R).

Step 5: Use your straightedge to draw a line from the vertex B through the intersection point R .

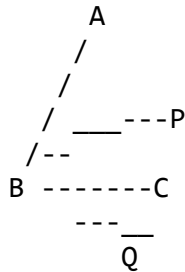
This line BR has now perfectly bisected $\angle ABC$. ($\angle ABR = \angle CBR$).

Illustration:

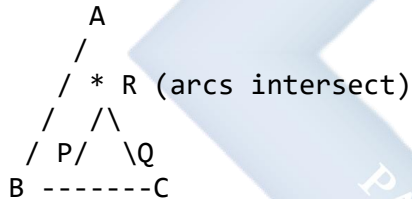


Detailed Diagram:

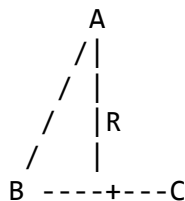
Step 1-2: Draw arc from B



Step 3-4: Draw arcs from P and Q



Step 5: Draw line BR



B. CONSTRUCTION OF ANGLES 60° AND 90°

Construction of 60° (The "Equilateral Triangle" Method)

Step 1: Draw a straight line and mark a point *P* on it.

Step 2: Place your compass point on *P* and draw a wide arc (a semi-circle) that cuts the line at point *Q*.

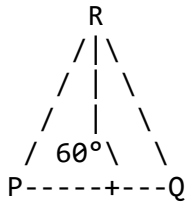
Step 3: Without changing the compass radius, move the compass point to Q .

Step 4: Draw a new arc that intersects the first arc at a point R .

Step 5: Draw a straight line from P through R .

The angle $\angle RPQ$ is exactly 60° . (This works because you have essentially drawn an equilateral triangle).

Illustration:



Why This Works:

When you draw arcs of equal radius from P and Q , the triangle PQR has three equal sides ($PQ = QR = PR$), making it equilateral. All angles in an equilateral triangle are 60° .

Construction of 90° (The "Perpendicular" Method)

Step 1: Draw a straight line and mark a point P in the middle.

Step 2: Place your compass point on P and draw an arc that cuts the line on both sides of P (at points X and Y).

Step 3: Increase the radius of your compasses.

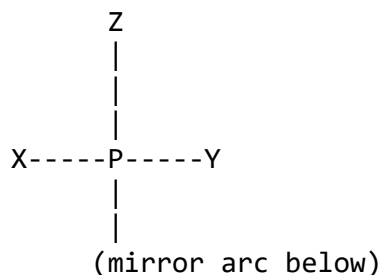
Step 4: Place the point on X and draw a large arc above the line.

Step 5: Without changing the radius, place the point on Y and draw another arc to intersect the one from Step 4 (at point Z).

Step 6: Draw a straight line from P through Z .

The angle $\angle ZPY$ is exactly 90° .

Illustration:

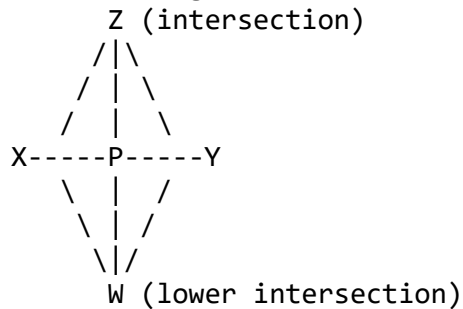


Detailed Step-by-Step:

Step 2: Equal arcs from P

___X___P___Y___

Step 4-5: Large arcs from X and Y



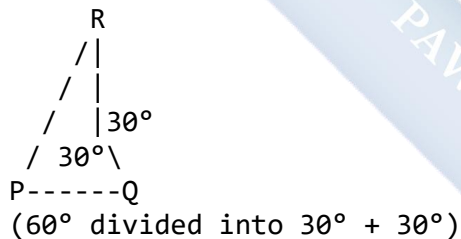
C. CONSTRUCTION OF 30° AND 45°

These angles are made by **bisecting** the angles we just built.

Construction of 30° :

1. First, construct a 60° angle.
2. Then, follow the steps for angle bisection (from section A) to bisect the 60° angle.
3. The result is two 30° angles.

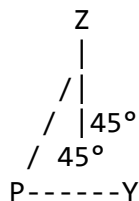
Illustration:



Construction of 45° :

1. First, construct a 90° angle.
2. Then, follow the steps for angle bisection to bisect the 90° angle.
3. The result is two 45° angles.

Illustration:



D. CONSTRUCTION OF TRIANGLES

To construct a unique triangle, you need at least **three pieces of information**:

- **SSS** (Side-Side-Side) – All three sides known
- **SAS** (Side-Angle-Side) – Two sides and the included angle
- **ASA** (Angle-Side-Angle) – Two angles and the included side
- **RHS** (Right angle-Hypotenuse-Side) – For right-angled triangles

General Steps for Triangle Construction:

1. **Start with the base:** Draw the longest or most convenient side first
 2. **Use the second piece of information:** Construct an angle or measure another side
 3. **Complete the triangle:** Find the third vertex using the remaining information
-

3. ROBUST WORKED EXAMPLES

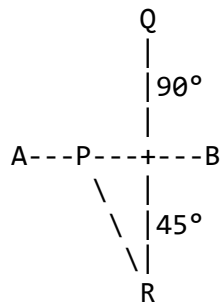
Example 1.1: Construct an angle of 135°

Thought Process: 135° is not a base angle. But $135^\circ = 90^\circ + 45^\circ$. And 45° is half of 90° . So, we can construct 90° and then bisect the other 90° angle on the straight line.

Solution:

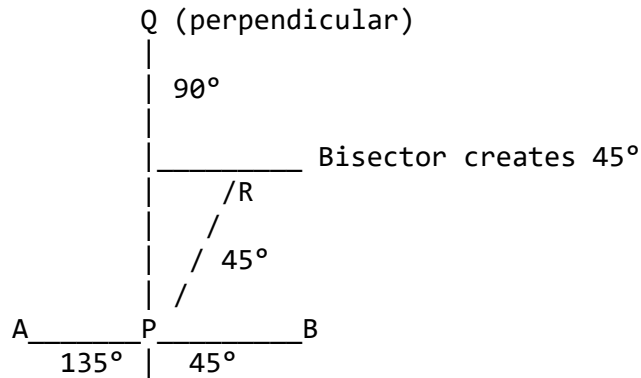
1. Draw a straight line AB and mark a point P on it.
2. Construct a 90° angle at P . Call the new line PQ .
3. The angle $\angle QPA$ is 90° . The angle $\angle QPB$ is also 90° .
4. Bisect the angle $\angle QPB$. This will create a 45° angle. Call the bisector PR .
5. The total angle $\angle RPA$ is $90^\circ + 45^\circ = 135^\circ$.

Illustration:



$$\angle RPA = 90^\circ + 45^\circ = 135^\circ$$

Alternative Visualization:

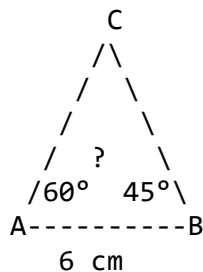


Example 1.2: Construct $\triangle ABC$ where $AB = 6$ cm, $\angle CAB = 60^\circ$, and $\angle CBA = 45^\circ$

Solution:

1. Draw a straight line and mark a point A .
2. Use your ruler to measure 6 cm from A and mark point B . You now have the base AB .
3. **At point A :** Construct a 60° angle. Draw a long line AX through this angle.
4. **At point B :** Construct a 90° angle (lightly). Then, bisect this 90° angle to create a 45° angle. Draw a long line BY through this angle.
5. The point where line AX and line BY intersect is point C .
6. $\triangle ABC$ is the required triangle.

Illustration:



Step-by-Step Visual:

Step 2:

A _____ B (6 cm)

Step 3-4:

X (construction line)

/

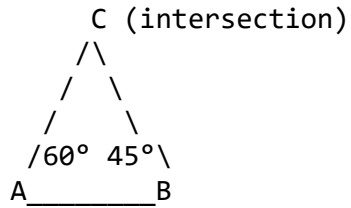
60°

A/ _____ B

45°\

Y (construction line)

Step 5:



4. REVISION QUESTIONS (WEEK 1)

1. What are the only two tools permitted in pure geometric construction?
 2. What does "to bisect" mean?
 3. How is a 60° angle related to an equilateral triangle?
 4. How would you construct an angle of 15° ? (Hint: $60^\circ \rightarrow 30^\circ \rightarrow 15^\circ$)
 5. How would you construct an angle of 22.5° ?
-

5. ASSESSMENT (WEEK 1)

EVALUATION (IN-CLASS)

1. Describe the steps to bisect a given line segment XY .
 2. Describe the steps to construct a 90° angle at a point P on a line.
 3. What is the first step in constructing a 30° angle?
 4. If you construct a 60° angle and a 90° angle that share a common vertex and arm, what is the value of the angle between them?
 5. What two base angles are added to get 105° ? ($60^\circ + 45^\circ$)
-

ASSIGNMENT (TAKE-HOME)

1. Using only a ruler and compasses:
 - Construct a 90° angle. Bisect it to create a 45° angle.
 - Construct a 60° angle. Bisect it to create a 30° angle.
 2. Construct $\triangle PQR$ such that $PQ = 7$ cm, $QR = 8$ cm, and $PR = 6$ cm. (This is SSS construction).
 3. Construct $\triangle XYZ$ such that $XY = 8$ cm, $\angle ZXY = 30^\circ$, and $\angle ZYX = 60^\circ$.
 - a. Measure the length of XZ .
 - b. What is the value of $\angle XZY$? (It should be 90°).
-

WEEK 2: CONSTRUCTIONS II

1. INTRODUCTION

This week, we build on our construction skills. We learn how to **copy an angle**, which is a vital skill for constructing quadrilaterals. We will also be introduced to **Locus**, which is a fundamental concept in geometry that answers the question, "Where are all the points that follow this rule?"

Historical Context:

The concept of locus problems dates back to ancient Greek mathematicians who sought to understand the paths traced by moving points under specific conditions. This concept is fundamental in modern applications like GPS tracking, robotics, and computer graphics.

2. CONTENT

A. CONSTRUCTION OF EQUAL ANGLES (COPYING AN ANGLE)

Sometimes you are given a random angle and asked to copy it, not just construct a 60° or 90° angle.

Procedure: To Copy $\angle ABC$ to a new point P

Step 1: Draw a line PQ . This will be the base of your new angle.

Step 2 (At the original angle): Place your compass point at B and draw an arc that cuts both arms of $\angle ABC$ (at M and N).

Step 3 (At the new angle): Without changing the radius, place your compass point at P and draw the same arc, cutting line PQ at a point X .

Step 4 (Measure the "opening"): Go back to the original angle. Set your compass radius to be the exact distance between M and N .

Step 5 (Mark the "opening"): Go to your new angle. Place the compass point on X . Draw an arc that intersects the arc from Step 3 at a point Y .

Step 6: Draw a straight line from P through Y .

$\angle YPQ$ is now an exact copy of $\angle ABC$.

Illustration:

Original Angle:

A
/
/

New Angle:

Y
/
/

$\angle M$
 $B-----N--C$

$\angle$
 $P-----X--Q$

Detailed Step-by-Step:

Step 2: Arc from B

A
 $\angle$
 $\angle M$ (arc cuts here)
 $\angle$
 $B-----N--C$

Step 3: Same arc from P

$P-----X--Q$

Step 4-5: Measure MN, mark from X
 Y (intersection)

$\angle$
 $\angle$
 $\angle$
 $P-----X--Q$

Result: $\angle YPQ = \angle ABC$

B. CONSTRUCTION OF QUADRILATERALS

A quadrilateral (like a parallelogram or kite) is not rigid. To construct one, you usually need **5 pieces of information**. The method is to break the shape into two triangles.

Example: Construct a Parallelogram ABCD with $AB = 6$ cm, $BC = 4$ cm, and $\angle ABC = 60^\circ$

Steps:

1. Draw the line $AB = 6$ cm.
2. At point B , construct a 60° angle.
3. Measure 4 cm along this new line and mark point C .
4. Now you have two options to find D :

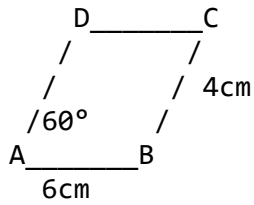
Method 1 (Compass):

- Set compass to 4 cm. Draw an arc from A .
- Set compass to 6 cm. Draw an arc from C .
- The intersection is D .

Method 2 (Parallel lines):

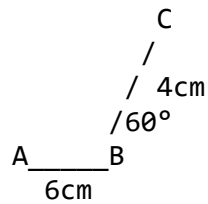
- Copy the 60° angle from B to A (using alternate interior angles rule for parallel lines).
- 5. Join A to D and C to D .

Illustration:

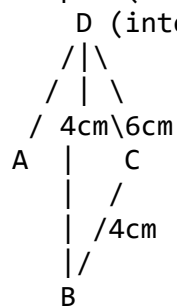


Construction Process:

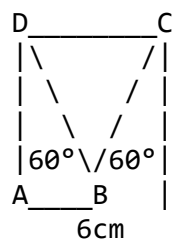
Step 1-3:



Step 4 (Method 1 - Compass Arcs):



Final:



Properties of a Parallelogram (for verification):

- Opposite sides are equal and parallel
- Opposite angles are equal
- Adjacent angles are supplementary (add to 180°)
- Diagonals bisect each other

C. LOCUS OF POINTS

Definition: A **locus** (plural: **loci**) is the set of all points that satisfy a given geometric rule or condition. It is the "path" a point traces as it moves according to that rule.

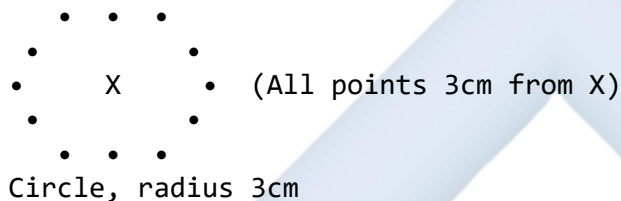
THE FOUR FUNDAMENTAL LOCI:

1. Locus of points at a fixed distance from a single point

Rule: "Find all points P that are 3 cm away from point X ."

Locus: A circle with center X and radius 3 cm.

Illustration:



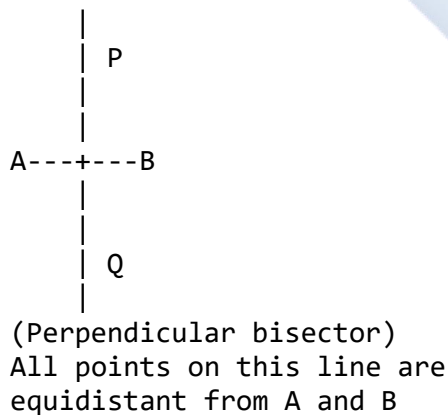
Real-World Example: The coverage area of a cell phone tower.

2. Locus of points equidistant from two fixed points

Rule: "Find all points P that are the same distance from A as they are from B ."

Locus: The perpendicular bisector of the line segment AB .

Illustration:



Real-World Example: The boundary line between two countries or properties where every point is equally distant from two capital cities or landmarks.

3. Locus of points at a fixed distance from a single line

Rule: "Find all points P that are 2 cm away from a line AB ."

Locus: A pair of parallel lines, one on each side of AB , both 2 cm away.

Illustration:

_____ (2cm above)

===== Line AB

_____ (2cm below)

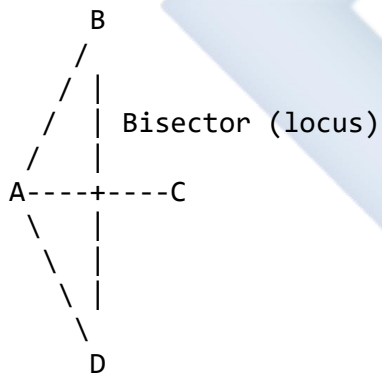
Real-World Example: The boundaries of a road or railway track of fixed width.

4. Locus of points equidistant from two intersecting lines

Rule: "Find all points P that are the same distance from line AB as they are from line AC ."

Locus: The angle bisector of the angle $\angle BAC$. (There are actually two, one for the interior angle and one for the exterior).

Illustration:



(Both interior and exterior bisectors)

Real-World Example: The path followed by a ball that bounces off two walls at equal angles.

3. ROBUST WORKED EXAMPLES

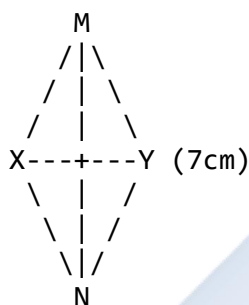
Example 2.1: Constructing a Locus

Draw a line segment $XY = 7$ cm. Construct the locus of all points P such that P is equidistant from X and Y.

Solution:

1. Draw the line $XY = 7$ cm.
2. The rule "equidistant from two points" is Locus #2. The locus is the perpendicular bisector.
3. Place your compass at X . Set the radius to be more than half of XY (e.g., 5 cm). Draw a large arc above and below the line.
4. Without changing the radius, place the compass at Y . Draw another large arc to intersect the first one at two points (call them M and N).
5. Draw a straight line connecting M and N . This line is the required locus.

Any point on this line is exactly the same distance from X and Y .

Illustration:

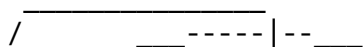
Line MN is the perpendicular bisector and the required locus

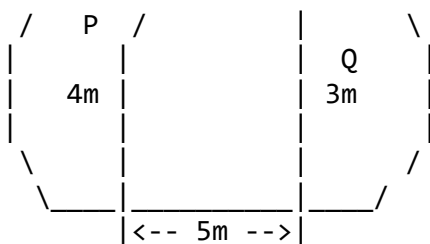
Example 2.2: Locus Application

A goat is tied to a post P with a rope 4 meters long. It is also tied to another post Q with a rope 3 meters long. P and Q are 5 meters apart. Construct and shade the region where the goat can graze.

Solution: (Scale: 1m = 1cm)

1. Draw the line $PQ = 5$ cm.
2. **Locus 1:** The goat must be within 4m of P . This is Locus #1. Construct a circle with center P and radius 4 cm. Shade the inside of this circle.
3. **Locus 2:** The goat must be within 3m of Q . This is Locus #1 again. Construct a circle with center Q and radius 3 cm. Shade the inside of this circle.
4. **Final Answer:** The goat can only graze where both conditions are true. The required region is the intersection (overlap) of the two shaded circles.

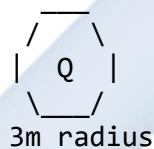
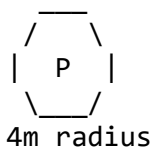
Illustration:



Shaded region = Overlap (lens shape)
where goat can reach both posts

Cross-Section View:

Circle from P (4m): Circle from Q (3m):



Combined (goat's grazing area):



4. REVISION QUESTIONS (WEEK 2)

1. What is a "locus"?
2. What is the locus of a point equidistant from two points A and B ?
3. What is the locus of a point equidistant from two intersecting lines PQ and PR ?
4. Describe the locus of the tip of the minute hand of a clock.
5. What are the key steps to "copy" an angle?

5. ASSESSMENT (WEEK 2)

EVALUATION (IN-CLASS)

1. Describe, in words, the locus of a point P that moves so that its distance from a fixed point O is always 5 cm.
2. Describe the locus of points equidistant from two parallel lines L_1 and L_2 that are 6 cm apart.
3. What two tools are used to construct the perpendicular bisector of a line?

4. When constructing a quadrilateral, how many pieces of information do you generally need?
 5. What is the first step in finding the locus of points equidistant from two intersecting lines?
-

ASSIGNMENT (TAKE-HOME)

1. Using only a ruler and compasses:
 - Draw an angle $\angle ABC = 75^\circ$ (by constructing $60^\circ + 15^\circ$). Construct a separate line XY and copy the 75° angle to point X .
 2. Construct a rhombus $PQRS$ with side lengths of 5 cm and one angle $\angle PQR = 60^\circ$.
 3. Draw a line $AB = 6$ cm.
 - a. Construct L_1 , the locus of all points 4 cm away from the line AB .
 - b. Construct L_2 , the locus of all points equidistant from A and B .
 - c. Mark the points X and Y where L_1 and L_2 intersect.
 4. Draw two intersecting lines WX and WY forming an angle of 45° .
 - a. Construct L_1 , the locus of points equidistant from WX and WY .
 - b. Construct L_2 , the locus of points 3 cm from the point W .
 - c. Mark the point P that satisfies both conditions.
-

WEEK 3: PROOFS OF THEOREMS I

1. INTRODUCTION

In geometry, we have **axioms** (or **postulates**), which are statements we accept as true without proof (e.g., "the shortest distance between two points is a straight line"). A **Theorem** is a statement that is not obvious, but which we can prove to be true by using logic and axioms.

This week, we will learn to write a formal deductive proof for two of the most important theorems in all of geometry.

Structure of a Proof:

1. **Given:** What information are you starting with?
 2. **To Prove:** What are you trying to show?
 3. **Construction (if needed):** A line or shape you add to the diagram to help.
 4. **Proof:** The step-by-step logical argument, with a reason for every statement.
 5. **Conclusion:** A final statement, often "**Q.E.D.**" (*quod erat demonstrandum*, Latin for "what was to be shown").
-

2. CONTENT

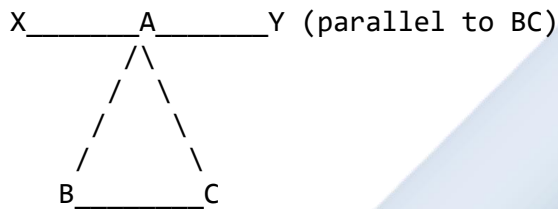
A. THEOREM 1: THE SUM OF ANGLES IN A TRIANGLE IS 180°

Given: A triangle ABC .

To Prove: $\angle A + \angle B + \angle C = 180^\circ$.

Construction: Draw a line XY through the vertex A , such that XY is parallel to the base BC .

Illustration:



Proof:

Statement	Reason
$\angle XAB = \angle ABC$ (Let's call this $\angle B$)	Alternate interior angles are equal ($XY \parallel BC$)
$\angle YAC = \angle ACB$ (Let's call this $\angle C$)	Alternate interior angles are equal ($XY \parallel BC$)
$\angle XAB + \angle BAC + \angle YAC = 180^\circ$	Angles on a straight line (line XY) add up to 180° .
Substitute (1) and (2) into (3):	
$\angle B + \angle BAC + \angle C = 180^\circ$	Substitution
$\therefore \angle A + \angle B + \angle C = 180^\circ$	
Q.E.D.	

Visual Proof Diagram:

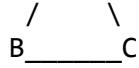
Step-by-Step Visualization:

1. Draw triangle ABC :



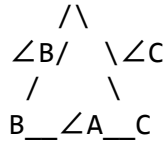
2. Add parallel line XY through A :





3. Mark alternate angles:

X ___ [$\angle B$] A [$\angle C$] ___ Y



4. On line XY: $\angle B + \angle A + \angle C = 180^\circ$

B. THEOREM 2: THE EXTERIOR ANGLE OF A TRIANGLE

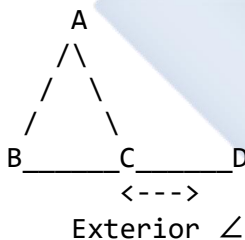
The **exterior angle** is the angle formed by extending one of the sides.

Theorem: The exterior angle of a triangle is equal to the sum of the two opposite interior angles.

Given: A triangle ABC , with the side BC extended to a point D .

To Prove: The exterior angle $\angle ACD = \angle BAC + \angle ABC$. (or $\angle_{\text{ext}} = \angle A + \angle B$)

Illustration:

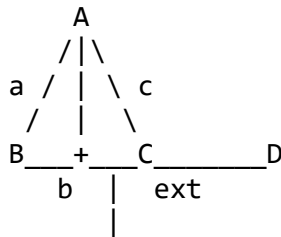


Proof:

Statement	Reason
$\angle BAC + \angle ABC + \angle BCA = 180^\circ$	Sum of angles in $\triangle ABC$ is 180° (Theorem 1).
$\angle BCA + \angle ACD = 180^\circ$	Angles on a straight line (line BD) add up to 180° .
Set (1) equal to (2):	
$\angle BAC + \angle ABC + \angle BCA = \angle BCA + \angle ACD$	Both sides equal 180° .
Subtract $\angle BCA$ from both sides:	
$\angle BAC + \angle ABC = \angle ACD$	
Q.E.D.	

Visual Proof Diagram:

Annotated Diagram:



Interior angles: a, b, c
 $a + b + c = 180^\circ$ (Theorem 1)

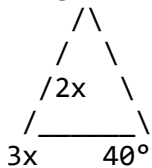
On line BCD: $c + \text{ext} = 180^\circ$

Therefore: $a + b + c = c + \text{ext}$
Simplify: $a + b = \text{ext}$

3. ROBUST WORKED EXAMPLES (APPLICATIONS)

Example 3.1: Find the value of x in the diagram

Triangle with angles:



Solution:

Reason: Sum of angles in a triangle is 180° .

$$2x + 3x + 40^\circ = 180^\circ$$

$$5x + 40^\circ = 180^\circ$$

$$5x = 180^\circ - 40^\circ$$

$$5x = 140^\circ$$

$$x = 140^\circ / 5$$

Answer: $x = 28^\circ$

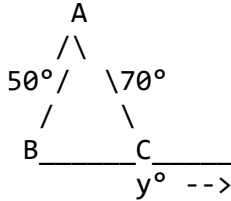
Verification:

- First angle: $2(28^\circ) = 56^\circ$
- Second angle: $3(28^\circ) = 84^\circ$
- Third angle: 40°

- Sum: $56^\circ + 84^\circ + 40^\circ = 180^\circ \checkmark$

Example 3.2: Find the value of y in the diagram

Triangle with exterior angle:



Solution:

Reason: Exterior angle = sum of opposite interior angles.

$$y = 50^\circ + 70^\circ$$

Answer: $y = 120^\circ$

Alternative method:

- Find the third interior angle C : $C = 180^\circ - 50^\circ - 70^\circ = 60^\circ$
- Then find y on the straight line: $y = 180^\circ - 60^\circ = 120^\circ \checkmark$

Example 3.3: Multi-Step Problem

In triangle PQR, $\angle P = 2x$, $\angle Q = 3x$, and the exterior angle at R is 140° . Find the value of x and all interior angles.

Solution:

Step 1: Use the exterior angle theorem.

Exterior angle at $R = \angle P + \angle Q$

$$140^\circ = 2x + 3x$$

$$140^\circ = 5x$$

$$x = 28^\circ$$

Step 2: Find the interior angles.

- $\angle P = 2(28^\circ) = 56^\circ$
- $\angle Q = 3(28^\circ) = 84^\circ$
- $\angle R = 180^\circ - 140^\circ = 40^\circ$ (supplementary to exterior angle)

Verification: $56^\circ + 84^\circ + 40^\circ = 180^\circ \checkmark$

4. REVISION QUESTIONS (WEEK 3)

1. What is the difference between an axiom and a theorem?
 2. What does "Q.E.D." stand for?
 3. In the proof of Theorem 1, what construction line did we draw?
 4. What geometric rule did we use to relate the angles in the triangle to the angles on the straight line in Theorem 1?
 5. What two theorems (or axioms) can be used to prove the Exterior Angle Theorem?
-

5. ASSESSMENT (WEEK 3)

EVALUATION (IN-CLASS)

1. State the theorem about the sum of angles in a triangle.
 2. State the theorem about the exterior angle of a triangle.
 3. One angle in a triangle is 90° . The other two are x and $2x$. Find x .
 4. The exterior angle of a triangle is 110° . One of the opposite interior angles is 40° . What is the other?
 5. Can a triangle have angles 50° , 60° , and 80° ? Why or why not?
-

ASSIGNMENT (TAKE-HOME)

1. Find the value of x and y in the diagram below:
(Diagram showing triangle with angles marked as algebraic expressions)
 2. Find the value of a , b , and c in the diagram below:
(Diagram showing intersecting lines and triangles)
 3. The angles of a triangle are in the ratio 2: 3: 4. Find the size of each angle. (Let the angles be $2x$, $3x$, $4x$).
 4. Write out the full proof that the sum of angles in a triangle is 180° . (Given, To Prove, Construction, Proof).
-

WEEK 4: PROOFS OF THEOREMS II (RIDERS)

1. INTRODUCTION

A **rider** is a problem in geometry that requires you to be a "detective." You must use a chain of logical deductions, based on the theorems you know, to find a missing angle or prove a relationship. This week is about applying all the foundational theorems to solve complex problems.

Think of riders as:

- Geometric puzzles
- Logic challenges
- Step-by-step detective work
- Applications of all your "theorem tools"

2. CONTENT (THE THEOREM "TOOLBOX")

To solve riders, you must know these rules by heart.

ANGLES ON A STRAIGHT LINE:

Add up to 180° .

$$\begin{array}{c} a^\circ \quad b^\circ \\ \hline a + b = 180^\circ \end{array}$$

ANGLES AT A POINT:

Add up to 360° .

$$\begin{array}{c} a^\circ \\ | \\ d^\circ - | - b^\circ \\ | \\ c^\circ \end{array}$$

$$a + b + c + d = 360^\circ$$

VERTICALLY OPPOSITE ANGLES:

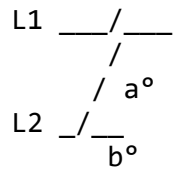
Are equal.

$$\begin{array}{c} a^\circ \backslash \quad / b^\circ \\ \quad \vee \quad \wedge \\ c^\circ / \quad \backslash d^\circ \end{array}$$

$$a = c, b = d$$

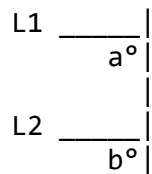
PARALLEL LINE THEOREMS (Given $L_1 \parallel L_2$):

1. Alternate Interior Angles: Are equal. (Z-angle)



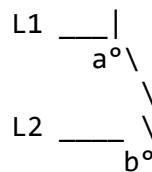
$a = b$ (Z-pattern)

2. Corresponding Angles: Are equal. (F-angle)



$a = b$ (F-pattern)

3. Co-interior (Allied) Angles: Add up to 180° . (C-angle)

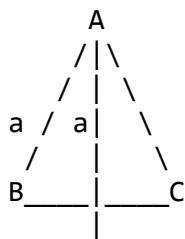


$a + b = 180^\circ$ (C-pattern)

TRIANGLE THEOREMS:

- Angles add to 180° .
- Exterior angle = sum of two opposite interior angles.
- Isosceles Triangle:** Base angles (angles opposite equal sides) are equal.

Isosceles Triangle:



Equal sides $\rightarrow$ Equal base angles

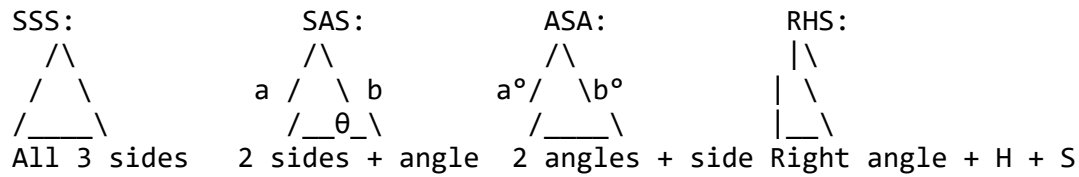
CONGRUENT TRIANGLES:

Two triangles are identical in shape and size if one of these 4 conditions is met:

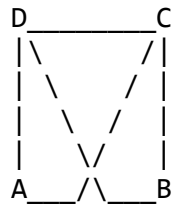
- SSS** (Side-Side-Side)

2. **SAS** (Side-Angle-Side): The angle must be **between** the two sides.
3. **ASA** (Angle-Side-Angle): The side must be **between** the two angles.
4. **RHS** (Right-angle-Hypotenuse-Side)

Visual Guide:



PROPERTIES OF PARALLELOGRAMS:



- Opposite sides are equal and parallel
- Opposite angles are equal
- Diagonals bisect each other
- Adjacent angles are supplementary (add to 180°)

POLYGONS:

Sum of interior angles = $(n - 2) \times 180^\circ$ (where n is the number of sides).

Sum of exterior angles = 360° (always!)

Visual Explanation:

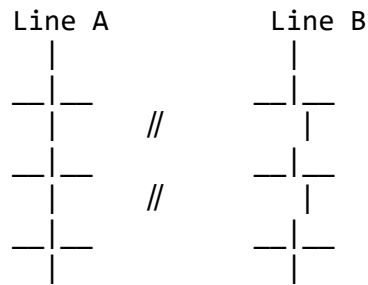
Triangle ($n=3$):	Square ($n=4$):
Sum = $(3-2) \times 180 = 180^\circ$	Sum = $(4-2) \times 180 = 360^\circ$



INTERCEPT THEOREM:

If three or more parallel lines cut two transversals (lines A and B), then they divide the transversals in the same ratio.

Simple case: If the parallel lines cut off equal intercepts on line A , they will also cut off equal intercepts on line B .

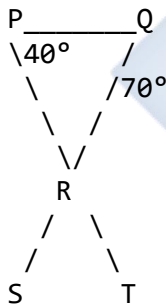


Equal spacing $\rightarrow$ Equal spacing

3. ROBUST WORKED EXAMPLES (RIDERS)

Example 4.1: Parallel Lines and Triangles

In the diagram, $PQ \parallel RS$. Find the value of $\angle QRT$.



Thought Process: We need to find $\angle QRT$. This angle is inside $\triangle QRT$. We know one angle ($\angle QTR = 70^\circ$). If we can find $\angle TQR$, we can solve the problem.

Solution:

Step 1: $\angle PQT = \angle TQR = 40^\circ$

Reason: Alternate Interior Angles ($PQ \parallel RS$).

Step 2: Now consider $\triangle QRT$.

$$\angle TQR + \angle QTR + \angle QRT = 180^\circ$$

Reason: Sum of angles in a triangle.

Step 3: $40^\circ + 70^\circ + \angle QRT = 180^\circ$

$$110^\circ + \angle QRT = 180^\circ$$

$$\angle QRT = 180^\circ - 110^\circ = 70^\circ$$

Answer: 70°

Example 4.2: Polygon Angles

A regular polygon has 9 sides (a nonagon). Find:

- **a)** The sum of its interior angles.
- **b)** The size of one interior angle.
- **c)** The size of one exterior angle.

Solution:

a) Sum of interior angles $= (n - 2) \times 180^\circ$

$$= (9 - 2) \times 180^\circ$$

$$= 7 \times 180^\circ = \mathbf{1260^\circ}$$

b) One interior angle (since it's regular):

$$\text{Angle} = \frac{\text{Sum}}{n} = \frac{1260^\circ}{9} = \mathbf{140^\circ}$$

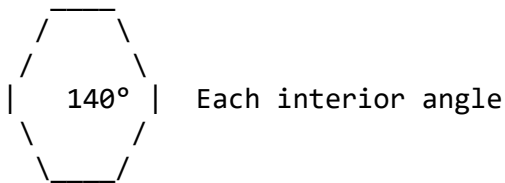
c) One exterior angle (Method 1):

$$\text{Exterior Angle} = \frac{360^\circ}{n} = \frac{360^\circ}{9} = \mathbf{40^\circ}$$

(Check: Interior + Exterior = $140^\circ + 40^\circ = 180^\circ$. Correct.)

Visual Representation:

Regular Nonagon (9 sides):



Exterior angles: 40° each

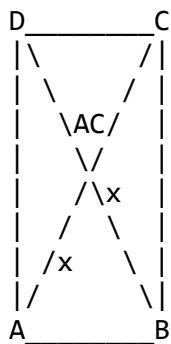
Sum of all exterior: $9 \times 40^\circ = 360^\circ$ ✓

Example 4.3: Deductive Proof with Parallelograms

Given: Parallelogram $ABCD$. The diagonal AC bisects $\angle DAB$.

To Prove: $ABCD$ is a rhombus (i.e., all four sides are equal).

Illustration:



Proof:

Statement	Reason
$\angle DAC = \angle CAB$ (call this x)	Given (AC bisects $\angle DAB$)
$AB \parallel DC$	Property of parallelogram
$\therefore \angle CAB = \angle DCA = x$	Alternate Interior Angles
$\therefore \angle DAC = \angle DCA = x$	Both are equal to x
Consider $\triangle ADC$:	
Since $\angle DAC = \angle DCA$, $\triangle ADC$ is an isosceles triangle.	Base angles are equal.
$\therefore$ Side $AD =$ Side DC	Sides opposite equal angles are equal.
But $AD = BC$ and $DC = AB$	Properties of a parallelogram.
$\therefore AD = DC = AB = BC$	Substitution
$\therefore ABCD$ is a rhombus.	
Q.E.D.	

4. REVISION QUESTIONS (WEEK 4)

1. List the three main parallel line angle theorems.
2. What are the four conditions for proving triangles are congruent?
3. What is the formula for the sum of the interior angles of an n -sided polygon?
4. What is the sum of all the exterior angles of any polygon?
5. What is the difference between a parallelogram and a rhombus?

5. ASSESSMENT (WEEK 4)

EVALUATION (IN-CLASS)

1. Find the value of x .

(Diagram with parallel lines and angles marked with expressions involving x)

2. Find the interior angle of a regular 5-sided polygon (a pentagon).
3. State the "SAS" condition for congruence. Why is the "A" in the middle?
4. In parallelogram $ABCD$, $\angle A = 100^\circ$. What is $\angle B$?
5. What does the Intercept Theorem state?

ASSIGNMENT (TAKE-HOME)

1. A polygon has an interior angle sum of 900° . How many sides does it have?
2. The exterior angle of a regular polygon is 30° . How many sides does it have?
3. In the diagram below, AB is a straight line. Find the values of x , y , and z .

(Diagram showing intersecting lines with angles marked)

4. **Proof:** Given an isosceles triangle PQR with $PQ = PR$. M is the midpoint of QR . Prove that the triangles $\triangle PQM$ and $\triangle PRM$ are congruent. (Which condition do you use?)

WEEK 5: MIDTERM EXAMINATION

This week is for your **midterm examination**, which will cover all content from Weeks 1-4 (Constructions, Theorems, Proofs, and Riders).

Examination Coverage:

- ✓ Geometric constructions (angles, triangles, quadrilaterals)
- ✓ Locus problems
- ✓ Angle theorems and proofs
- ✓ Rider problems
- ✓ Polygon properties

Preparation Tips:

1. Review all worked examples from Weeks 1-4
2. Practice construction techniques with compass and straightedge
3. Memorize key theorems and their proofs
4. Work through past assessment questions
5. Ensure you can clearly explain your reasoning in proofs

WEEK 6: TRIGONOMETRIC RATIOS

1. INTRODUCTION

This week, we begin the study of **Trigonometry**. The word comes from the Greek *trigonon* ("triangle") and *metron* ("measure"). It is the branch of mathematics that studies the relationship between the side lengths and angles of triangles.

We will start with the **right-angled triangle**, which is the foundation of all trigonometry.

Historical Note:

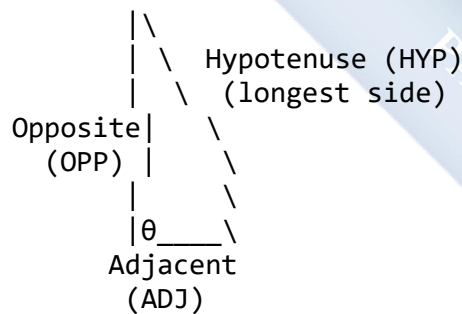
Trigonometry has ancient origins, developed by Greek, Indian, and Islamic mathematicians for astronomy, navigation, and architecture. Today, it's essential in engineering, physics, computer graphics, and GPS technology.

2. CONTENT

A. THE RIGHT-ANGLED TRIANGLE

A **right-angled triangle** has one angle that is exactly 90° .

The sides have special names relative to a chosen angle θ (theta).



The Three Sides:

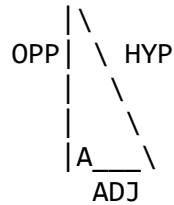
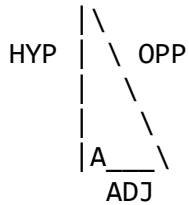
1. **Hypotenuse (HYP):** The longest side, always opposite the 90° angle.
2. **Opposite (OPP):** The side directly opposite the angle θ .
3. **Adjacent (ADJ):** The side next to the angle θ (that is not the hypotenuse).

Important Note: If we change the angle θ to the other acute angle, the Opposite and Adjacent sides will swap!

Example:

For angle A:
C

For angle C:
C

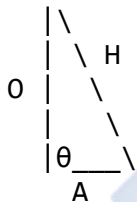


B. THE THREE TRIGONOMETRIC RATIOS

We define the three basic ratios as follows. The mnemonic **SOH CAH TOA** is essential to remember them.

SINE (sin)

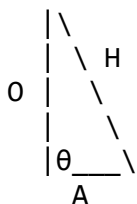
$$\text{SOH} \Rightarrow \sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$



$$\sin \theta = O/H$$

COSINE (cos)

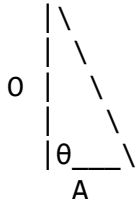
$$\text{CAH} \Rightarrow \cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$



$$\cos \theta = A/H$$

TANGENT (tan)

$$\text{TOA} \Rightarrow \tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$



$$\tan \theta = O/A$$

Complete SOH CAH TOA Visual:

SOH CAH TOA MEMORY AID		
S O H	→	Sin = Opp/Hyp
C A H	→	Cos = Adj/Hyp
T O A	→	Tan = Opp/Adj

C. SOLVING PROBLEMS

We can use these ratios to find missing sides or angles.

Finding a missing SIDE: You need 1 angle and 1 side.

Steps:

1. Label the sides (OPP, ADJ, HYP) relative to the given angle.
2. Choose the correct ratio (SOH, CAH, or TOA) that uses the side you have and the side you want.
3. Write the equation, substitute values, and solve.

Finding a missing ANGLE: You need 2 sides.

Steps:

1. Label the sides relative to the unknown angle θ .
2. Choose the correct ratio that uses the two sides you have.
3. Write the equation.
4. Use the "inverse" function on your calculator ($\sin^{-1}$, $\cos^{-1}$, or $\tan^{-1}$) to find the angle.

Decision Tree for Choosing Ratios:

What do you have?

- Have: H and O → Want: angle → Use $\sin^{-1}$
- Have: H and A → Want: angle → Use $\cos^{-1}$
- Have: O and A → Want: angle → Use $\tan^{-1}$

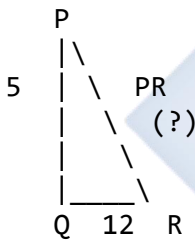
- Have: angle & H → Want: O → Use $\sin$
- Have: angle & H → Want: A → Use $\cos$
- Have: angle & O → Want: A → Use $\tan$
- Have: angle & A → Want: O → Use $\tan$

3. ROBUST WORKED EXAMPLES

Example 6.1: Finding Ratios

In $\triangle PQR$, $\angle Q = 90^\circ$, $PQ = 5$ cm, $QR = 12$ cm. Find PR , $\sin P$, $\cos P$, and $\tan R$.

Diagram:



Solution:

Part 1: Find PR (Hypotenuse)

Use Pythagoras.

$$PR^2 = PQ^2 + QR^2 = 5^2 + 12^2 = 25 + 144 = 169$$

$$PR = \sqrt{169} = 13 \text{ cm.}$$

Part 2: Find ratios for $\angle P$

From P 's perspective:

- Opp = 12 (side QR)
- Adj = 5 (side PQ)
- Hyp = 13 (side PR)

$$\sin P = \frac{\text{Opp}}{\text{Hyp}} = \frac{12}{13}$$

$$\cos P = \frac{\text{Adj}}{\text{Hyp}} = \frac{5}{13}$$

Part 3: Find tan R

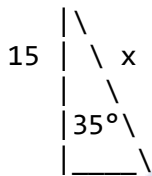
From R's perspective:

- Opp = 5 (side PQ)
- Adj = 12 (side QR)
- Hyp = 13 (side PR)

$$\tan R = \frac{\text{Opp}}{\text{Adj}} = \frac{5}{12}$$

Example 6.2: Finding a Missing Side

Find the value of x in the diagram.



Solution:

Step 1: Label

Relative to 35°:

- x is **Adjacent**
- 15 is **Hypotenuse**

Step 2: Choose

We have A and H. Use **CAH** (Cosine).

Step 3: Equation

$$\cos(35^\circ) = \frac{\text{Adj}}{\text{Hyp}} = \frac{x}{15}$$

Step 4: Solve

$$x = 15 \times \cos(35^\circ)$$

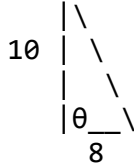
Calculator: $\cos(35^\circ) \approx 0.8192$

$$x = 15 \times 0.8192 = 12.288$$

Answer: $x \approx 12.29$ cm

Example 6.3: Finding a Missing Angle

Find the angle θ in the diagram.



Solution:

Step 1: Label

Relative to θ :

- 10 is **Opposite**
- 8 is **Adjacent**

Step 2: Choose

We have O and A. Use **TOA** (Tangent).

Step 3: Equation

$$\tan(\theta) = \frac{\text{Opp}}{\text{Adj}} = \frac{10}{8} = 1.25$$

Step 4: Solve

$$\theta = \tan^{-1}(1.25)$$

Calculator: $\theta \approx 51.34^\circ$

Answer: $\theta \approx 51.3^\circ$

4. REVISION QUESTIONS (WEEK 6)

1. What does the mnemonic "SOH CAH TOA" stand for?
 2. If $\sin\theta = 0.5$, what does this tell you about the relationship between the opposite side and the hypotenuse?
 3. If $\tan\theta = 1$, what does this tell you about the opposite and adjacent sides?
 4. What is the "inverse" function used for? (e.g., $\sin^{-1}$).
 5. In $\triangle XYZ$, $\angle Y = 90^\circ$. What is the "adjacent" side relative to $\angle X$? What is the "adjacent" side relative to $\angle Z$?
-

5. ASSESSMENT (WEEK 6)

EVALUATION (IN-CLASS)

1. Draw a right-angled triangle and label the Hypotenuse, Opposite, and Adjacent sides relative to an angle α .
 2. State the three trigonometric ratios in terms of Opp, Adj, and Hyp.
 3. If $\cos\theta = \frac{4}{5}$, what is $\sin\theta$ and $\tan\theta$? (Hint: Draw the triangle. Use Pythagoras to find the 3rd side).
 4. In $\triangle ABC$, $\angle C = 90^\circ$, $\angle B = 50^\circ$, and $AC = 7$ cm. Which ratio would you use to find the length of BC ?
 5. What button(s) would you press on a calculator to solve $\tan\theta = 2.5$?
-

ASSIGNMENT (TAKE-HOME)

1. In $\triangle ABC$, $\angle C = 90^\circ$.
 - a. If $AB = 10$ cm and $\angle A = 40^\circ$, find the length of BC .
 - b. If $AC = 12$ cm and $BC = 5$ cm, find the size of $\angle A$.
 2. A 12-meter ladder leans against a wall, making an angle of 70° with the ground. How high up the wall does the ladder reach?
 3. A rectangle is 15 cm long and 8 cm wide. Find the angle that the diagonal makes with the 15 cm side.
 4. If $\sin x = \frac{3}{5}$ and x is an acute angle, find $(\cos x)^2 + (\sin x)^2$ without finding x .
-

WEEK 7: MIDTERM BREAK

This week is for your **midterm break**.

Suggested Activities:

- Review all topics covered so far
 - Practice construction techniques
 - Work on trig problems
 - Rest and recharge!
-

WEEK 8: TRIGONOMETRIC RATIOS OF SPECIAL ANGLES

1. INTRODUCTION

While our calculators can find the ratios for any angle, there are three "special" angles (30° , 45° , 60°) whose ratios are so common that we are expected to know them by heart. These ratios are **exact** (they use surds like $\sqrt{2}$) and allow us to solve problems without a calculator.

Why These Angles Are Special:

These angles appear frequently in:

- Regular polygons (equilateral triangles, squares)
- Engineering and architecture
- Physics problems
- Computer graphics and game design

2. CONTENT

A. DERIVATION OF RATIOS FOR 45°

Step 1: Draw an isosceles right-angled triangle. Let the two equal sides be **1** unit.

Step 2: By Pythagoras, the hypotenuse h is: $h^2 = 1^2 + 1^2 = 2 \Rightarrow h = \sqrt{2}$.

Step 3: The angles are 45° , 45° , 90° .

Step 4: Use SOH CAH TOA for 45° :

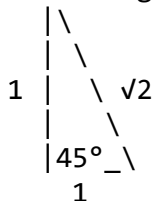
$$\sin 45^\circ = \frac{\text{Opp}}{\text{Hyp}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\cos 45^\circ = \frac{\text{Adj}}{\text{Hyp}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\tan 45^\circ = \frac{\text{Opp}}{\text{Adj}} = \frac{1}{1} = 1$$

Illustration:

45-45-90 Triangle:



All measurements:

- Equal legs: 1, 1
- Hypotenuse: $\sqrt{2}$
- Angles: 45° , 45° , 90°

B. DERIVATION OF RATIOS FOR 30° AND 60°

Step 1: Draw an equilateral triangle with all sides **2** units. All angles are 60° .

Step 2: Bisect the triangle vertically. This creates two 30-60-90 right-angled triangles.

Step 3: The sides of this new triangle are:

- Hypotenuse = 2
- Base = 1

Step 4: By Pythagoras, the height h is: $2^2 = 1^2 + h^2 \Rightarrow 4 = 1 + h^2 \Rightarrow h^2 = 3 \Rightarrow h = \sqrt{3}$.

Step 5: Use SOH CAH TOA:

For 60° : Opp = $\sqrt{3}$, Adj = 1, Hyp = 2

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\tan 60^\circ = \frac{\sqrt{3}}{1} = \sqrt{3}$$

For 30° : Opp = 1, Adj = $\sqrt{3}$, Hyp = 2

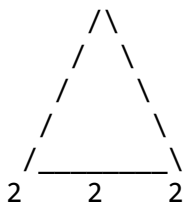
$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

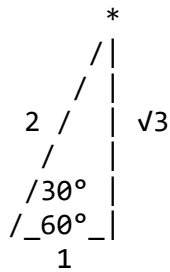
$$\tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

Illustration:

Equilateral Triangle (side = 2):



Bisected to create 30-60-90:



30-60-90 Triangle:

- Short leg (opp 30°): 1
- Long leg (opp 60°): $\sqrt{3}$
- Hypotenuse: 2
- Angles: 30° , 60° , 90°

C. SUMMARY TABLE (MEMORIZE THIS)

Angle (θ)	$\sin \theta$	$\cos \theta$	$\tan \theta$
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$ or $\frac{\sqrt{3}}{3}$
45°	$\frac{1}{\sqrt{2}}$ or $\frac{\sqrt{2}}{2}$	$\frac{1}{\sqrt{2}}$ or $\frac{\sqrt{2}}{2}$	1
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

Memory Aid Pattern:

	30°	45°	60°	
sin:	$1/2$	$\sqrt{2}/2$	$\sqrt{3}/2$	(increasing)
cos:	$\sqrt{3}/2$	$\sqrt{2}/2$	$1/2$	(decreasing - mirror of sin)
tan:	$1/\sqrt{3}$	1	$\sqrt{3}$	(increasing)

Notice: $\sin 30^\circ = \cos 60^\circ = 1/2$
 $\sin 60^\circ = \cos 30^\circ = \sqrt{3}/2$

3. ROBUST WORKED EXAMPLES

Example 8.1: Evaluating Expressions

Without using a calculator, find the exact value of $\sin 60^\circ \cos 30^\circ + \tan 45^\circ$.

Solution:

Step 1: Substitute the values from the table:

$$\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + (1)$$

Step 2: Simplify:

$$\left(\frac{\sqrt{3} \times \sqrt{3}}{2 \times 2}\right) + 1$$
$$\left(\frac{3}{4}\right) + 1$$

Answer: $1\frac{3}{4}$ or $\frac{7}{4}$ or 1.75

Example 8.2: Solving Equations

Find x (an acute angle) if $2\sin x = \sqrt{3}$.

Solution:

Step 1: Isolate the trig ratio: $\sin x = \frac{\sqrt{3}}{2}$

Step 2: Ask: "Which special angle has a sine of $\frac{\sqrt{3}}{2}$?"

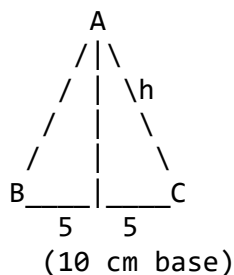
Step 3: Look at the table: $\sin 60^\circ = \frac{\sqrt{3}}{2}$.

Answer: $x = 60^\circ$

Example 8.3: Application in Geometry

An equilateral triangle has sides of length 10 cm. Find its exact perpendicular height.

Diagram:



Solution:

Step 1: Draw the triangle and bisect it. You get a 30-60-90 triangle.

The Hypotenuse = 10 cm. The Base = 5 cm. The Height = h .

Step 2: We can use Pythagoras (as in the derivation) OR trigonometry.

Using Trig: We need to find h , which is Opposite the 60° angle. We have the Hypotenuse (10). Use SOH.

$$\sin 60^\circ = \frac{\text{Opp}}{\text{Hyp}} = \frac{h}{10}$$

Step 3: We know $\sin 60^\circ = \frac{\sqrt{3}}{2}$.

$$\frac{\sqrt{3}}{2} = \frac{h}{10}$$

$$2h = 10\sqrt{3}$$

$$h = 5\sqrt{3} \text{ cm}$$

Answer: The exact height is $5\sqrt{3}$ cm (or approximately 8.66 cm).

4. REVISION QUESTIONS (WEEK 8)

1. How do you derive the ratios for 45° ? (What triangle do you use?)
 2. How do you derive the ratios for 30° and 60° ? (What triangle do you use?)
 3. What is the value of $\sin 30^\circ$? What is $\cos 60^\circ$? What do you notice?
 4. What is the value of $\tan 45^\circ$? Why does this make sense?
 5. What does "exact value" mean in this context?
-

5. ASSESSMENT (WEEK 8)

EVALUATION (IN-CLASS)

Do not use a calculator for this section.

1. What is the exact value of $\tan 30^\circ$?
 2. What is the exact value of $\cos 45^\circ$?
 3. Evaluate: $\sin 45^\circ \times \cos 45^\circ$.
 4. Evaluate: $\tan 60^\circ \div \sin 60^\circ$. (Hint: $1/\cos 60^\circ$)
 5. Solve for x : $\tan x = 1$.
-

ASSIGNMENT (TAKE-HOME)

Do not use a calculator for this section. Show all your steps.

1. Evaluate: $2\sin 30^\circ \cos 30^\circ$.
2. Evaluate: $\frac{\tan 60^\circ - \tan 30^\circ}{1 + \tan 60^\circ \tan 30^\circ}$.
3. A square $ABCD$ has a diagonal AC of length 10 cm. Find the exact length of its side AB . (Hint: The diagonal splits the square into two 45-45-90 triangles).
4. Solve for θ (acute): $4\cos^2 \theta = 3$. (Hint: Isolate $\cos \theta$).

WEEK 9: APPLICATION OF TRIGONOMETRIC RATIOS

1. INTRODUCTION

This week, we apply our SOH CAH TOA knowledge to solve real-world problems. This involves two new, crucial concepts: **Angle of Elevation** and **Angle of Depression**. We will also expand our understanding of trig ratios beyond 90° to cover all four quadrants (0° to 360°).

Real-World Applications:

Trigonometry is used in:

- Aviation and navigation
- Architecture and construction
- Surveying and mapping
- Physics and engineering
- Computer graphics and animation

2. CONTENT

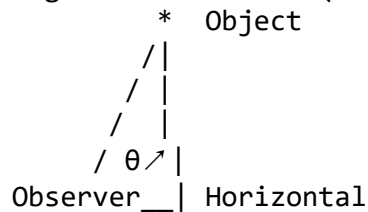
A. ANGLES OF ELEVATION AND DEPRESSION

Angle of Elevation: The angle between the horizontal line of sight and the line of sight **UP** to an object.

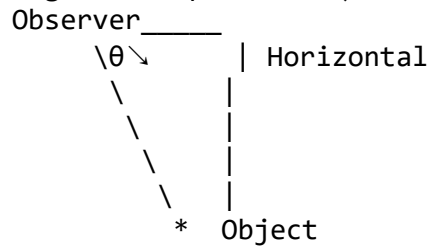
Angle of Depression: The angle between the horizontal line of sight and the line of sight **DOWN** to an object.

Illustration:

Angle of Elevation (looking up):

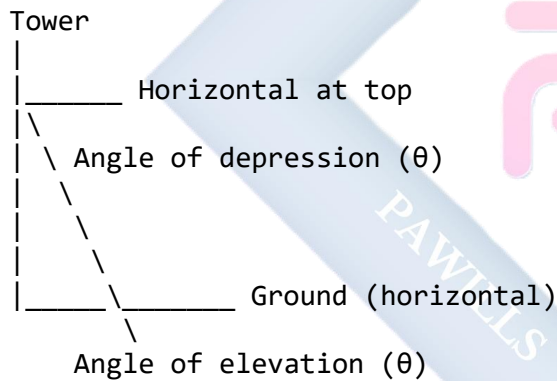


Angle of Depression (looking down):



Key Fact: Due to parallel lines (the two horizontals) and a transversal, the **Angle of Elevation from the ground to an object is always equal to the Angle of Depression from the object back to the ground**. They are alternate interior angles.

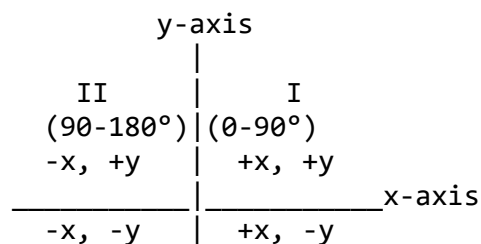
Detailed Diagram:



θ (elevation) = θ (depression)
[Alternate angles, parallel horizontals]

B. SINE, COSINE, TANGENT OF ANGLES 0° TO 360°

We use the Cartesian (x, y) plane to define angles, split into four quadrants.



III (180°-270°)	IV (270°-360°)
--------------------	-------------------

Quadrant I (0°-90°): x is +, y is +. (Acute angles) **Quadrant II (90°-180°):** x is -, y is +. (Obtuse angles) **Quadrant III (180°-270°):** x is -, y is -. (Reflex angles) **Quadrant IV (270°-360°):** x is +, y is -. (Reflex angles)

ASTC Rule:

We use the mnemonic "**All Students Take Cash**" (or ASTC) to remember which ratios are positive in which quadrant, starting from Quadrant I.

S (Sin+)	A (All+)
T (Tan+)	C (Cos+)

- **A** - Quadrant I (0°-90°): **All** ratios (sin, cos, tan) are positive.
- **S** - Quadrant II (90°-180°): Only **Sine** is positive. (cos and tan are negative).
- **T** - Quadrant III (180°-270°): Only **Tangent** is positive. (sin and cos are negative).
- **C** - Quadrant IV (270°-360°): Only **Cosine** is positive. (sin and tan are negative).

Finding Equivalent Angles:

We always relate the angle back to the acute angle it makes with the horizontal x-axis.

Quadrant II (e.g., 150°): $150^\circ = 180^\circ - 30^\circ$. Use 30° as the reference.

- $\sin 150^\circ = + \sin 30^\circ = \frac{1}{2}$
- $\cos 150^\circ = - \cos 30^\circ = -\frac{\sqrt{3}}{2}$

Quadrant III (e.g., 225°): $225^\circ = 180^\circ + 45^\circ$. Use 45° as the reference.

- $\sin 225^\circ = - \sin 45^\circ = -\frac{\sqrt{2}}{2}$
- $\tan 225^\circ = + \tan 45^\circ = 1$

Quadrant IV (e.g., 300°): $300^\circ = 360^\circ - 60^\circ$. Use 60° as the reference.

- $\cos 300^\circ = + \cos 60^\circ = \frac{1}{2}$
- $\tan 300^\circ = - \tan 60^\circ = -\sqrt{3}$

Visual Guide:

Reference Angle Formulas:

Quadrant II (90° - 180°):

Reference = 180° - angle

Example: $150^\circ \rightarrow \text{ref} = 30^\circ$

Quadrant III (180° - 270°):

Reference = angle - 180°

Example: $225^\circ \rightarrow \text{ref} = 45^\circ$

Quadrant IV (270° - 360°):

Reference = 360° - angle

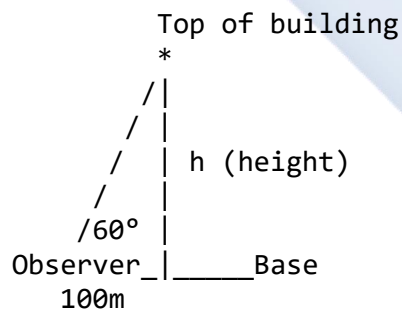
Example: $300^\circ \rightarrow \text{ref} = 60^\circ$

3. ROBUST WORKED EXAMPLES

Example 9.1: Angle of Elevation

A man stands 100 meters from the base of a tall building. He measures the angle of elevation to the top of the building as 60° . What is the height of the building?

Diagram:



Solution:

Step 1: Draw the triangle. The man is at one corner. The building is the Opposite side. The ground is the Adjacent side.

Step 2: Identify known values:

- Angle $\theta = 60^\circ$
- Adj = 100 m
- Opp = h (height)

Step 3: We have O and A. Use **TOA** (Tangent).

$$\tan 60^\circ = \frac{\text{Opp}}{\text{Adj}} = \frac{h}{100}$$

Step 4: We know $\tan 60^\circ = \sqrt{3}$.

$$\sqrt{3} = \frac{h}{100}$$

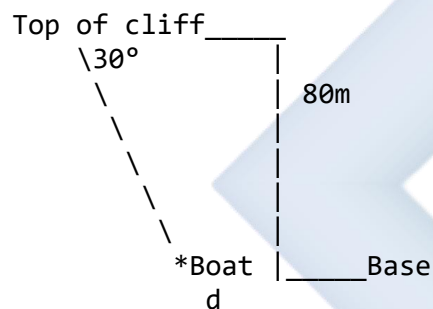
$$h = 100\sqrt{3} \text{ meters}$$

Answer: The exact height is $100\sqrt{3}$ m (or approx. 173.2 m).

Example 9.2: Angle of Depression

From the top of a cliff 80 meters high, the angle of depression to a boat at sea is 30° . How far is the boat from the base of the cliff?

Diagram:



Solution:

Step 1: Draw the diagram. The cliff is vertical. The horizontal "line of sight" from the top is parallel to the sea.

Step 2: The angle down from the horizontal is 30° .

The angle of elevation from the boat up to the cliff top is also 30° (alternate angles).

Step 3: Now we have a right-angled triangle with the boat.

- Angle $\theta = 30^\circ$
- Opp = 80 m (height of cliff)
- Adj = d (distance)

Step 4: We have O and A. Use **TOA** (Tangent).

$$\tan 30^\circ = \frac{\text{Opp}}{\text{Adj}} = \frac{80}{d}$$

Step 5: We know $\tan 30^\circ = \frac{1}{\sqrt{3}}$.

$$\frac{1}{\sqrt{3}} = \frac{80}{d}$$

$$1 \times d = 80 \times \sqrt{3}$$

$$d = 80\sqrt{3} \text{ m}$$

Answer: The boat is $80\sqrt{3}$ m away (or approx. 138.6 m).

Example 9.3: ASTC

Without a calculator, find the exact value of $\sin 120^\circ$ and $\cos 240^\circ$.

Solution:

Part 1: $\sin 120^\circ$

Step 1: 120° is in Quadrant II (ASTC). **Sine is Positive.**

Step 2: Reference angle = $180^\circ - 120^\circ = 60^\circ$.

Step 3: $\sin 120^\circ = + \sin 60^\circ = \frac{\sqrt{3}}{2}$.

Part 2: $\cos 240^\circ$

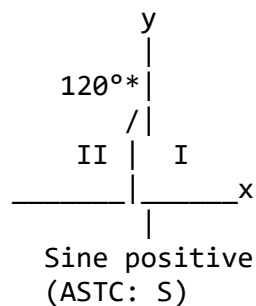
Step 1: 240° is in Quadrant III (ASTC). **Cosine is Negative.**

Step 2: Reference angle = $240^\circ - 180^\circ = 60^\circ$.

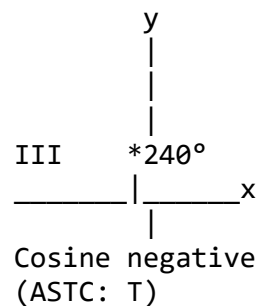
Step 3: $\cos 240^\circ = - \cos 60^\circ = -\frac{1}{2}$.

Visualization:

120° location:



240° location:



4. REVISION QUESTIONS (WEEK 9)

1. What is the difference between an angle of elevation and an angle of depression?
 2. If the angle of depression from a tower to a car is 40° , what is the angle of elevation from the car to the tower?
 3. What does the mnemonic "ASTC" stand for?
 4. In which two quadrants is the sine ratio positive?
 5. In which two quadrants is the tangent ratio negative?
 6. What is the "reference angle" for 315° ?
-

5. ASSESSMENT (WEEK 9)

EVALUATION (IN-CLASS)

1. A vertical pole is 10m high. What is the angle of elevation to the top of the pole from a point 10m away on the ground? (Use special angles).
 2. From a window, the angle of depression to an object on the ground is 45° . The object is 20m from the base of the house. How high is the window?
 3. Is $\cos 100^\circ$ positive or negative?
 4. Is $\tan 200^\circ$ positive or negative?
 5. What is the reference angle for 130° ?
-

ASSIGNMENT (TAKE-HOME)

1. The shadow of a mast is 30m long. If the angle of elevation of the sun is 30° , find the exact height of the mast.
2. From the top of a lighthouse 120m high, the angle of depression of two boats *A* and *B* (in a straight line on the same side of the lighthouse) are 30° and 60° respectively.
 - a. Find the distance of boat *A* from the base of the lighthouse.
 - b. Find the distance of boat *B* from the base of the lighthouse.
 - c. Find the distance between the two boats.
3. Without using a calculator, find the exact value of:
 - a. $\tan 315^\circ$
 - b. $\sin 210^\circ$
 - c. $\cos 135^\circ$
4. If $\sin \theta = \frac{3}{5}$ and θ is in Quadrant II, find $\cos \theta$ and $\tan \theta$. (Hint: Draw a triangle, find the adjacent side (it will be 4). Then apply the ASTC rule for Quadrant II).

WEEK 10: TRIGONOMETRIC RATIOS & GRAPHS

1. INTRODUCTION

This week, we visualize the trigonometric ratios as functions. We will see how $\sin(x)$, $\cos(x)$, and $\tan(x)$ behave as the angle x changes from 0° to 360° . The **unit circle** is the key to understanding this.

Why Study Trig Graphs?

Trigonometric graphs model:

- Sound waves and music
- Light waves and colors
- Ocean tides and weather patterns
- Electrical currents and signals
- Cyclical phenomena in nature

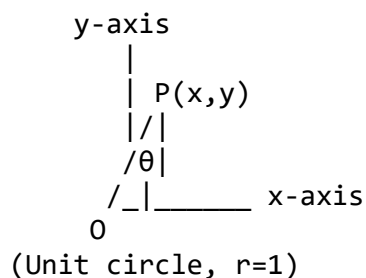
2. CONTENT

A. TRIGONOMETRIC RATIOS WITH RESPECT TO THE UNIT CIRCLE

The **Unit Circle** is a circle with its center at the origin $(0, 0)$ and a radius $r = 1$.

Let $P(x, y)$ be any point on the circle, forming an angle θ with the positive x-axis.

Illustration:



For point P on the circle:

- x-coordinate = $\cos \theta$
- y-coordinate = $\sin \theta$
- radius = 1

By SOH CAH TOA, we have:

$$\cos\theta = \frac{\text{Adj}}{\text{Hyp}} = \frac{x}{r} = \frac{x}{1} \Rightarrow x = \cos\theta$$

$$\sin\theta = \frac{\text{Opp}}{\text{Hyp}} = \frac{y}{r} = \frac{y}{1} \Rightarrow y = \sin\theta$$

$$\tan\theta = \frac{\text{Opp}}{\text{Adj}} = \frac{y}{x}$$

This gives us a new, powerful definition:

Cosine is the x-coordinate, and Sine is the y-coordinate of a point on the unit circle.

Key Points on the Unit Circle:

At 0°: The point is (1,0).

- $\cos 0^\circ = x = 1$
- $\sin 0^\circ = y = 0$
- $\tan 0^\circ = \frac{y}{x} = \frac{0}{1} = 0$

At 90°: The point is (0,1).

- $\cos 90^\circ = x = 0$
- $\sin 90^\circ = y = 1$
- $\tan 90^\circ = \frac{y}{x} = \frac{1}{0} = \text{Undefined}$

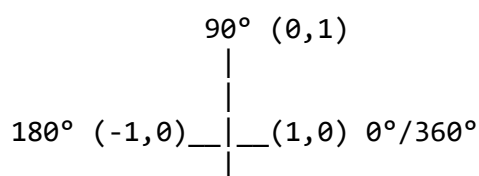
At 180°: The point is (−1,0).

- $\cos 180^\circ = x = -1$
- $\sin 180^\circ = y = 0$
- $\tan 180^\circ = \frac{y}{x} = \frac{0}{-1} = 0$

At 270°: The point is (0, −1).

- $\cos 270^\circ = x = 0$
- $\sin 270^\circ = y = -1$
- $\tan 270^\circ = \frac{y}{x} = \frac{-1}{0} = \text{Undefined}$

Complete Unit Circle:



$$270^\circ (0, -1)$$

Key angles marked:

$$0^\circ: (1, 0)$$

$$30^\circ: (\sqrt{3}/2, 1/2)$$

$$45^\circ: (\sqrt{2}/2, \sqrt{2}/2)$$

$$60^\circ: (1/2, \sqrt{3}/2)$$

$$90^\circ: (0, 1)$$

etc.

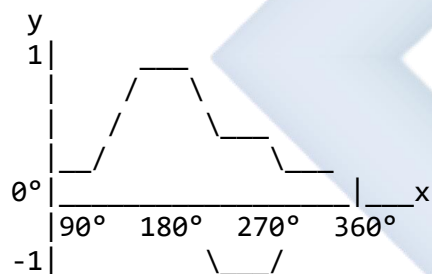
B. GRAPH OF SINE ($y = \sin x$)

This graph plots the height (y-coordinate) of the point P as it moves around the unit circle.

Table of Values:

x	0°	90°	180°	270°	360°
$\sin x$	0	1	0	-1	0

Graph:



Shape: Smooth wave

Starts at: 0

Peak at 90° : 1

Returns to 0 at 180°

Trough at 270° : -1

Back to 0 at 360°

Properties:

- **Period:** It repeats every 360° .
- **Amplitude:** The maximum height from the center is 1.
- **Range:** $-1 \leq \sin x \leq 1$

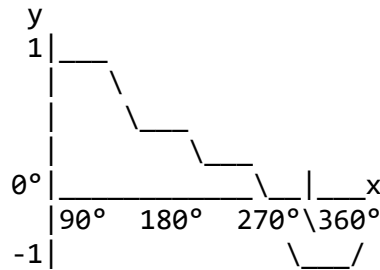
C. GRAPH OF COSINE ($y = \cos x$)

This graph plots the horizontal position (x-coordinate) of the point P .

Table of Values:

x	0°	90°	180°	270°	360°
$\cos x$	1	0	-1	0	1

Graph:



Shape: Smooth wave (shifted sine)

Starts at: 1

Zero at 90° : 0

Minimum at 180° : -1

Zero at 270° : 0

Back to 1 at 360°

Properties:

- **Period:** 360° .
- **Amplitude:** 1.
- **Range:** $-1 \leq \cos x \leq 1$
- **Note:** It's the same shape as the sine wave, just shifted by 90° .

D. GRAPH OF TANGENT ($y = \tan x$)

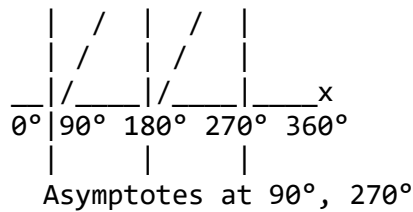
This graph plots the ratio $\frac{y}{x}$.

Behavior:

- At 0° , $\tan 0^\circ = 0$.
- As $x \rightarrow 90^\circ$, $y \rightarrow 1$ and $x \rightarrow 0$, $\tan x \rightarrow \frac{1}{0} \rightarrow +\infty$.
- At 90° , it is **undefined**. This creates a **vertical asymptote**.
- At 180° , $\tan 180^\circ = 0$.
- At 270° , it is **undefined** again (another asymptote).

Graph:





Shape: Repeating "S" curves
 Asymptotes where undefined
 Period: 180° (not 360° !)

Properties:

- **Period:** It repeats every 180° .
- **Amplitude:** Undefined (it goes to infinity).
- **Range:** All real numbers ($-\infty$ to $+\infty$)
- **Asymptotes:** At 90° , 270° (where $\cos x = 0$)

3. ROBUST WORKED EXAMPLES

Example 10.1: Reading the Sine Graph

Using the graph of $y = \sin x$, find the values of x (between 0° and 360°) for which $\sin x = 0.5$.

Solution:

Step 1: Draw the graph of $y = \sin x$.

Step 2: Draw the horizontal line $y = 0.5$.

Step 3: Find where the line cuts the curve.

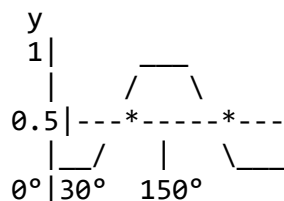
It cuts at two points.

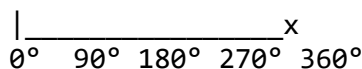
Step 4: From our knowledge of special angles, the first one is $x = 30^\circ$.

Step 5: The second one is in Quadrant II. The graph is symmetrical. The second root is $180^\circ - 30^\circ = 150^\circ$.

Answer: $x = 30^\circ, 150^\circ$.

Visual:





Example 10.2: Reading the Cosine Graph

Using the graph of $y = \cos x$, find the values of x for which $\cos x = -0.5$.

Solution:

Step 1: Draw the graph of $y = \cos x$.

Step 2: Draw the horizontal line $y = -0.5$.

Step 3: It cuts the curve at two points, both in the "negative" zones (Quadrants II and III).

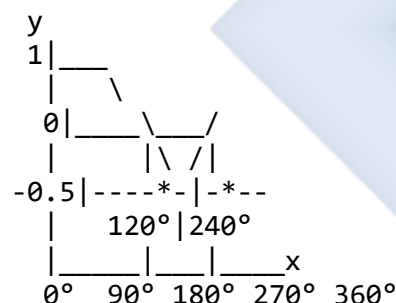
Step 4: First, find the reference angle (where $\cos \theta = +0.5$). From special angles, this is $\theta = 60^\circ$.

Step 5: The Quadrant II solution is $180^\circ - 60^\circ = 120^\circ$.

Step 6: The Quadrant III solution is $180^\circ + 60^\circ = 240^\circ$.

Answer: $x = 120^\circ, 240^\circ$.

Visual:



4. REVISION QUESTIONS (WEEK 10)

1. On the unit circle, which coordinate (x or y) represents sine? Which represents cosine?
2. Why is $\tan 90^\circ$ undefined?
3. What are the "amplitude" and "period" of a wave graph?
4. What is the period of $y = \sin x$? What is the period of $y = \tan x$?
5. What is the maximum value of $y = \cos x$? What is the minimum value?

5. ASSESSMENT (WEEK 10)

EVALUATION (IN-CLASS)

1. What are the x and y coordinates of the point on the unit circle at 180° ?
 2. What is the exact value of $\sin 270^\circ$?
 3. What is the exact value of $\cos 0^\circ$?
 4. Looking at the graph of $y = \sin x$, where is the graph at its minimum?
 5. What is an "asymptote"? Which of the three trig graphs has one?
-

ASSIGNMENT (TAKE-HOME)

1. Draw a table of values for $y = \cos x$ using the angles $0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ, 225^\circ, 270^\circ, 315^\circ$, and 360° .
 2. Use your table from Q1 to plot an accurate graph of $y = \cos x$ on graph paper. (Use a scale of $2\text{cm} = 45^\circ$ on the x -axis and $4\text{cm} = 1$ unit on the y -axis).
 3. From your graph, find the approximate values of x for which $\cos x = 0.8$.
 4. What is the main difference between the graph of $y = \sin x$ and $y = \cos x$?
-

WEEK 11: REVISION

1. INTRODUCTION

This week is for a comprehensive review of all topics from the term. We will consolidate our knowledge of constructions, theorems, and trigonometry in preparation for the final examination.

Revision Strategy:

- Review all worked examples
 - Practice key skills repeatedly
 - Make summary notes
 - Attempt past questions
 - Identify weak areas for extra practice
-

2. TOPIC SUMMARIES

Weeks 1-2 (Constructions & Loci):

Skills:

- Bisection of lines/angles
- Construction of 30° , 45° , 60° , 90°
- Copying an angle
- Constructing triangles/quads

Loci: The 4 fundamental loci:

1. Fixed distance from a point $\rightarrow$ Circle
2. Equidistant from two points $\rightarrow$ Perpendicular bisector
3. Fixed distance from a line $\rightarrow$ Parallel lines
4. Equidistant from two lines $\rightarrow$ Angle bisector

Key: Use only compasses and a straightedge.

Weeks 3-4 (Theorems & Riders):

Proofs: Know the formal proofs for:

- "Angles in $\triangle = 180^\circ$ "
- "Exterior Angle = Sum of Opp. Interior"

Toolbox: You must know all the rules:

- Parallel lines (Alt, Corr, Co-int)
- Polygons: $(n - 2) \times 180^\circ$
- Congruence (SSS, SAS, ASA, RHS)
- Parallelogram properties

Riders: Use the toolbox to solve problems step-by-step, giving a reason for each step.

Weeks 6-10 (Trigonometry):**SOH CAH TOA:**

- $\sin = \frac{O}{H}$
- $\cos = \frac{A}{H}$
- $\tan = \frac{O}{A}$

Used to find sides and angles.

Special Angles: Know the table for 30° , 45° , 60° by heart.

Angle	sin	cos	tan
30°	1/2	$\sqrt{3}/2$	$1/\sqrt{3}$
45°	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60°	$\sqrt{3}/2$	1/2	$\sqrt{3}$

Applications:

- Angles of Elevation (looking up)
- Angles of Depression (looking down)

ASTC: The 4 quadrants:

- Quadrant I: **All** positive
- Quadrant II: **Sin** positive
- Quadrant III: **Tan** positive
- Quadrant IV: **Cos** positive

Graphs: Know the shapes of:

- $y = \sin x$ (wave, period 360°)
- $y = \cos x$ (wave, period 360°)
- $y = \tan x$ (repeating curve, period 180°, asymptotes)

3. COMPREHENSIVE REVISION ASSIGNMENT (WEEK 11)

(This serves as the final assignment, covering all topics)

SECTION A: CONSTRUCTION

Question 1:

- Using a ruler and compasses only, construct a triangle ABC with $AB = 8$ cm, $\angle A = 60^\circ$, and $\angle B = 45^\circ$.
- Construct the locus L_1 of points equidistant from A and B .
- Construct the locus L_2 of points equidistant from AB and AC .
- Let the loci intersect at P . Measure the length of AP .

Guidance:

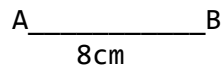
- Start by drawing the base $AB = 8$ cm
- At point A , construct 60° angle
- At point B , construct 45° angle
- Mark intersection point as C

- For L_1 : Construct perpendicular bisector of AB
- For L_2 : Construct angle bisector of $\angle BAC$
- Mark point P where L_1 and L_2 meet

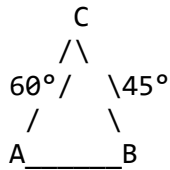
Expected Solution Structure:

Construction Steps:

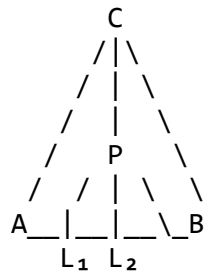
1. Draw AB = 8cm



2. Construct 60° at A and 45° at B



3. Add loci L_1 and L_2

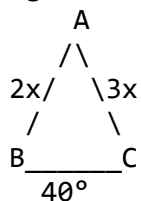


SECTION B: THEOREMS & RIDERS

Question 2:

- a. Find the value of x in the diagram below. (State your reasons).

Diagram:



Solution:

Step	Statement	Reason
1	$2x + 3x + 40^\circ = 180^\circ$	Sum of angles in a triangle

Step	Statement	Reason
2	$5x + 40^\circ = 180^\circ$	Simplification
3	$5x = 140^\circ$	Subtract 40° from both sides
4	$x = 28^\circ$	Divide both sides by 5

Verification:

- Angle A = $2(28^\circ) = 56^\circ$
- Angle B = $3(28^\circ) = 84^\circ$
- Angle C = 40°
- Sum = $56^\circ + 84^\circ + 40^\circ = 180^\circ$ ✓

Question 2b:

A regular polygon has 10 sides. Find:

- (i) its exterior angle
- (ii) its interior angle

Solution:

(i) Exterior angle:

$$\begin{aligned} \text{Formula: Exterior angle} &= \frac{360^\circ}{n} \\ &= \frac{360^\circ}{10} = 36^\circ \end{aligned}$$

(ii) Interior angle:

Method 1 (Using sum of interior angles):

$$\text{Sum of interior angles} = (n - 2) \times 180^\circ = (10 - 2) \times 180^\circ = 8 \times 180^\circ = 1440^\circ$$

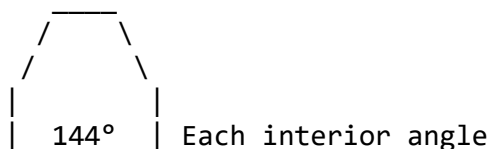
$$\text{One interior angle} = \frac{1440^\circ}{10} = 144^\circ$$

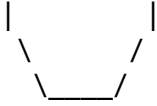
Method 2 (Using supplementary angles):

$$\text{Interior angle} = 180^\circ - \text{Exterior angle} = 180^\circ - 36^\circ = 144^\circ$$

Visual:

Regular Decagon (10 sides):





Each exterior: 36°

Sum of exterior: $10 \times 36^\circ = 360^\circ \checkmark$

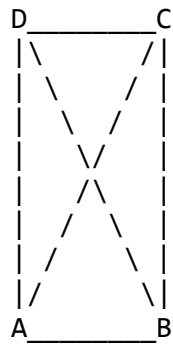
Question 2c: PROOF

Prove that in a parallelogram, opposite angles are equal.

Given: Parallelogram $ABCD$.

To Prove: $\angle A = \angle C$ and $\angle B = \angle D$.

Diagram:



Proof:

Statement	Reason
$AB \parallel DC$	Property of parallelogram
$\angle A + \angle D = 180^\circ$	Co-interior angles (AD is transversal)
$AD \parallel BC$	Property of parallelogram
$\angle D + \angle C = 180^\circ$	Co-interior angles (DC is transversal)
From statements (2) and (4):	
$\angle A + \angle D = \angle D + \angle C$	Both equal 180°
Subtract $\angle D$ from both sides:	
$\therefore \angle A = \angle C$	
Similarly, $\angle B = \angle D$	Using the same method
Q.E.D.	

SECTION C: TRIGONOMETRY

Question 3:

a. Without a calculator, find the exact value of: $\frac{\sin 30^\circ \times \tan 60^\circ}{\cos 45^\circ}$.

Solution:

Step 1: Substitute values from the special angles table:

$$\frac{\sin 30^\circ \times \tan 60^\circ}{\cos 45^\circ} = \frac{\frac{1}{2} \times \sqrt{3}}{\frac{\sqrt{2}}{2}}$$

Step 2: Simplify the numerator:

$$= \frac{\frac{\sqrt{3}}{2}}{\frac{\sqrt{2}}{2}}$$

Step 3: Divide fractions (multiply by reciprocal):

$$= \frac{\sqrt{3}}{2} \times \frac{2}{\sqrt{2}}$$

Step 4: Cancel the 2s:

$$= \frac{\sqrt{3}}{\sqrt{2}}$$

Step 5: Rationalize the denominator:

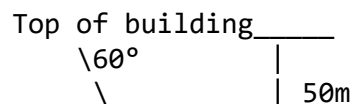
$$= \frac{\sqrt{3}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{3} \times \sqrt{2}}{2} = \frac{\sqrt{6}}{2}$$

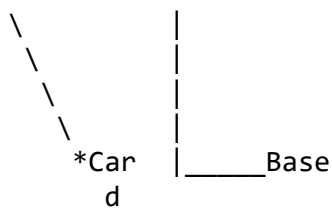
Answer: $\frac{\sqrt{6}}{2}$

Question 3b:

From the top of a building 50m high, the angle of depression to a car on the ground is 60° . Find the exact distance of the car from the base of the building.

Diagram:





Solution:

Step 1: Identify the problem type: Angle of depression problem.

Step 2: The angle of elevation from the car to the building top is also 60° (alternate angles).

Step 3: Set up the right triangle:

- Height (Opposite) = 50m
- Distance (Adjacent) = d
- Angle = 60°

Step 4: Choose the correct ratio: We have O and want A, so use TOA.

$$\tan 60^\circ = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{50}{d}$$

Step 5: Substitute $\tan 60^\circ = \sqrt{3}$:

$$\sqrt{3} = \frac{50}{d}$$

Step 6: Solve for d :

$$d = \frac{50}{\sqrt{3}}$$

Step 7: Rationalize:

$$d = \frac{50}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{50\sqrt{3}}{3}$$

Answer: The car is $\frac{50\sqrt{3}}{3}$ meters from the base (or approximately 28.87m).

Question 3c:

Given that $\cos \theta = -0.5$ and $0^\circ < \theta < 180^\circ$, find θ .

Solution:

Step 1: Recognize that $\cos \theta = -0.5$ means we need a negative cosine value.

Step 2: Recall from special angles that $\cos 60^\circ = 0.5$ (positive).

Step 3: Since cosine is negative, θ must be in Quadrant II (where $90^\circ < \theta < 180^\circ$).

Step 4: Using ASTC and the reference angle:

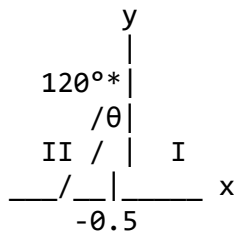
Reference angle = 60°

For Quadrant II: $\theta = 180^\circ - 60^\circ = 120^\circ$

Verification using special angles: $\cos 120^\circ = -\cos 60^\circ = -\frac{1}{2} = -0.5 \checkmark$

Answer: $\theta = 120^\circ$

Visual on Unit Circle:



Question 3d:

What is the period of the graph $y = \cos x$?

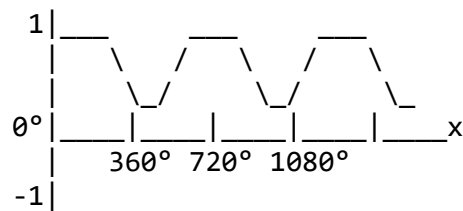
Answer:

The period of $y = \cos x$ is 360° (or 2π radians).

This means the graph completes one full cycle and repeats itself every 360° .

Visual Explanation:

$y = \cos x$



One complete cycle = 360°
Pattern repeats identically

4. ADDITIONAL PRACTICE PROBLEMS

Problem Set A: Mixed Constructions

Question 4:

Construct a rectangle $PQRS$ with $PQ = 8$ cm and $QR = 5$ cm.

a. Construct the locus of points 3cm from point Q . b. Construct the locus of points equidistant from PQ and QR . c. Mark the point(s) that satisfy both conditions.

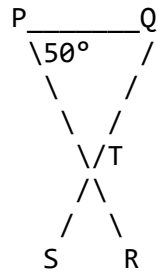
Expected Answer:

- Locus (a): Circle with center Q , radius 3cm
- Locus (b): Angle bisector of $\angle PQR$ (which is 45° since it's a rectangle)
- Intersection points: Two points where the 45° line cuts the circle

Problem Set B: Advanced Riders

Question 5:

In the diagram, $PQ \parallel RS$, $\angle QPT = 50^\circ$, and $\angle PTS = 130^\circ$. Find $\angle PTS$.



Solution Guide:

1. Use alternate angles: $\angle QPT = \angle RTP = 50^\circ$ (since $PQ \parallel RS$)
2. In triangle PTS , find angles
3. Apply angle sum property

Problem Set C: Advanced Trigonometry

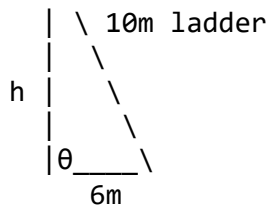
Question 6:

A ladder 10m long leans against a vertical wall. The foot of the ladder is 6m from the wall.

a. Calculate the angle the ladder makes with the ground (to 1 d.p.). b. How high up the wall does the ladder reach? c. If the ladder slips so that its foot is now 8m from the wall, by how much has the top of the ladder descended?

Diagram:





Solution:

Part (a):

We have:

- Hypotenuse = 10m
- Adjacent = 6m

Use CAH: $\cos \theta = \frac{6}{10} = 0.6$

$$\theta = \cos^{-1}(0.6) = 53.1^\circ$$

Part (b):

Use Pythagoras: $h^2 + 6^2 = 10^2$

$$h^2 = 100 - 36 = 64$$

$$h = 8\text{m}$$

Part (c):

New position: foot is 8m from wall

New height: $h_2^2 + 8^2 = 10^2$

$$h_2^2 = 100 - 64 = 36$$

$$h_2 = 6\text{m}$$

$$\text{Descent} = 8 - 6 = 2\text{m}$$

Answer: The ladder has descended by 2m.

5. EXAMINATION PREPARATION CHECKLIST

Construction Skills ✓

Can bisect any angle using compass only

Can construct 30° , 45° , 60° , 90° angles

Can copy an angle accurately
Can construct triangles given SSS, SAS, ASA
Can construct perpendicular bisectors
Understand all 4 fundamental loci
Can solve locus intersection problems

Theorem Knowledge ✓

Can prove: Sum of angles in triangle = 180°
Can prove: Exterior angle theorem
Know all parallel line angle rules (Alt, Corr, Co-int)
Know polygon angle formulas: $(n - 2) \times 180^\circ$
Understand all 4 congruence conditions
Know parallelogram properties
Can write formal proofs with reasons

Trigonometry Mastery ✓

Have memorized SOH CAH TOA
Know special angles table (30° , 45° , 60°) by heart
Can identify Opposite, Adjacent, Hypotenuse
Can find missing sides using trig ratios
Can find missing angles using inverse functions
Understand angles of elevation and depression
Know ASTC rule for all four quadrants
Can find reference angles
Understand trig graphs (sine, cosine, tangent)
Know periods and amplitudes

6. FINAL REVISION TIPS

The Night Before the Exam:

DO:

- ✓ Review your summary notes
- ✓ Practice drawing quick sketches of triangles
- ✓ Recite special angles table 5 times
- ✓ Check you have: compass, ruler, calculator, pencil, eraser
- ✓ Get a good night's sleep (at least 7-8 hours)

DON'T:

- ✗ Try to learn completely new topics
 - ✗ Stay up too late cramming
 - ✗ Panic if you don't remember everything
 - ✗ Forget to prepare your examination materials
-

During the Exam:

Time Management:

- Read all questions first (5 minutes)
- Answer easy questions first
- Allocate time per question based on marks
- Leave 10 minutes for review

Construction Questions:

- Draw lightly first, darken final answer
- Show all construction arcs
- Label all points clearly
- Don't erase construction marks

Proof Questions:

- Always start with "Given" and "To Prove"
- Give a reason for EVERY statement
- Draw clear diagrams
- Write neatly in two-column format

Trigonometry Questions:

- Draw a diagram for EVERY problem

- Label O, A, H on your diagram
- Show all working steps
- Include units in final answers
- Check if answer asks for exact or approximate value

7. COMMON MISTAKES TO AVOID

Construction Errors:

- ✗ Wrong:** Erasing construction arcs **✓ Right:** Leave all arcs visible as proof of method
- ✗ Wrong:** Using protractor for angle construction **✓ Right:** Use only compass and straightedge
- ✗ Wrong:** Not labeling points **✓ Right:** Clearly label all vertices and intersection points

Proof Errors:

- ✗ Wrong:** "Angle A = 60° because it looks like 60°" **✓ Right:** "Angle A = 60° because it's an equilateral triangle and all angles in an equilateral triangle are equal"
- ✗ Wrong:** Missing the "Given" and "To Prove" sections **✓ Right:** Always start with proper proof structure
- ✗ Wrong:** Statements without reasons **✓ Right:** Every statement must have a geometric reason

Trigonometry Errors:

- ✗ Wrong:** Confusing Opposite and Adjacent **✓ Right:** Always label from the perspective of the given angle
- ✗ Wrong:** $\sin 30^\circ = 30$ **✓ Right:** $\sin 30^\circ = 0.5$ (or $\frac{1}{2}$ for exact)
- ✗ Wrong:** Using degrees when answer needs exact values **✓ Right:** Use special angles and surds for exact values
- ✗ Wrong:** Forgetting to use inverse functions for finding angles **✓ Right:** Use $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$ when finding angles
- ✗ Wrong:** Angle of depression = angle in the triangle **✓ Right:** Angle of depression = alternate angle to angle of elevation
-

8. QUICK REFERENCE FORMULAS

Special Angles Table (MEMORIZE!):

Angle	sin	cos	tan
30°	1/2	$\sqrt{3}/2$	1/ $\sqrt{3}$
45°	$\sqrt{2}/2$	$\sqrt{2}/2$	1
60°	$\sqrt{3}/2$	1/2	$\sqrt{3}$

Polygon Formulas:

Sum of Interior Angles: $\text{Sum} = (n - 2) \times 180^\circ$

Each Interior Angle (Regular Polygon): $\text{Interior} = \frac{(n-2) \times 180^\circ}{n}$

Each Exterior Angle (Regular Polygon): $\text{Exterior} = \frac{360^\circ}{n}$

Sum of ALL Exterior Angles: Always 360°

Trigonometric Ratios:

SOH CAH TOA

$\sin \theta = \text{Opp/Hyp}$

$\cos \theta = \text{Adj/Hyp}$

$\tan \theta = \text{Opp/Adj}$

ASTC Rule (Quadrants):

II	I
S	A
T	C
III	IV

- **I (All):** All ratios positive
- **II (Sin):** Only sine positive
- **III (Tan):** Only tangent positive
- **IV (Cos):** Only cosine positive

Pythagoras Theorem:

$$a^2 + b^2 = c^2$$

(where c is the hypotenuse)

9. PRACTICE UNDER EXAM CONDITIONS

Mock Exam (90 minutes):

SECTION A: CONSTRUCTIONS (20 marks)

1. Construct a triangle ABC where AB = 7cm, $\angle A = 45^\circ$, $\angle B = 60^\circ$. [5 marks]
2. Draw a line XY = 10cm. Construct the locus of points that are:
 - a. 4cm from line XY [3 marks]
 - b. Equidistant from X and Y [3 marks]
 - c. Mark and label the points where these loci intersect [2 marks]

SECTION B: THEOREMS (25 marks)

3. Prove that the sum of angles in a triangle is 180° . [8 marks]
4. Find x if the angles of a triangle are $(2x + 10)^\circ$, $(3x - 5)^\circ$, and $(x + 25)^\circ$. [5 marks]
5. A regular polygon has an exterior angle of 40° . How many sides does it have? [4 marks]
6. In parallelogram PQRS, $\angle P = 65^\circ$. Find $\angle Q$, $\angle R$, and $\angle S$. [8 marks]

SECTION C: TRIGONOMETRY (35 marks)

7. Without a calculator, evaluate: $\sin 45^\circ \cos 45^\circ + \tan 60^\circ \sin 30^\circ$ [6 marks]
8. A flagpole casts a shadow 15m long when the angle of elevation of the sun is 60° . Find the exact height of the flagpole. [7 marks]
9. Find θ ($0^\circ < \theta < 360^\circ$) if:
 - a. $\sin \theta = -\frac{\sqrt{3}}{2}$ [4 marks]
 - b. $\cos \theta = \frac{1}{2}$ [4 marks]
10. In triangle PQR, $\angle R = 90^\circ$, PQ = 13cm, PR = 5cm.
 - a. Find QR [3 marks]
 - b. Find $\angle P$ (to 1 d.p.) [4 marks]
 - c. Find $\angle Q$ [2 marks]

11. Sketch the graph of $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$, showing clearly the maximum, minimum, and x-intercepts.
[5 marks]

10. ANSWER KEY FOR MOCK EXAM

Section A:

1. (Construction - check with teacher) 2a. Two parallel lines, 4cm on either side 2b. Perpendicular bisector at 5cm mark 2c. Four intersection points total

Section B: 3. (Full proof required - see Week 3) 4. $x = 30^\circ$ 5. $n = 9$ sides (nonagon) 6. $\angle Q = 115^\circ$, $\angle R = 65^\circ$, $\angle S = 115^\circ$

Section C: 7. $\frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{1+\sqrt{3}}{2}$ 8. $15\sqrt{3}$ meters 9a. $\theta = 240^\circ$, 300° 9b. $\theta = 60^\circ$, 300° 10a. QR = 12cm 10b. $\angle P \approx 67.4^\circ$ 10c. $\angle Q = 22.6^\circ$ 11. (Graph with correct wave pattern)

11. FINAL ENCOURAGEMENT

Remember:

"Success in Mathematics is not about being brilliant—it's about being persistent, practicing regularly, and understanding the fundamental principles."

Your preparation journey:

- ✓ You've learned geometric constructions
- ✓ You've mastered theorem proofs
- ✓ You've conquered trigonometry
- ✓ You've practiced numerous problems

You are ready!

Good luck with your examination!

END OF LESSON NOTES

Summary of Second Term Topics:

Week	Topic	Key Skills
1	Constructions I	Angle bisection, 60° , 90° , 30° , 45°
2	Constructions II	Copying angles, quadrilaterals, locus

Week	Topic	Key Skills
3	Proofs of Theorems I	Triangle angle sum, exterior angle
4	Proofs of Theorems II	Riders, parallel lines, polygons
5	Midterm Examination	Assessment
6	Trigonometric Ratios	SOH CAH TOA, finding sides and angles
7	Midterm Break	Rest and review
8	Special Angles	30° , 45° , 60° exact values
9	Applications	Elevation, depression, ASTC, quadrants
10	Trigonometric Graphs	Unit circle, sine, cosine, tangent graphs
11	Revision	Comprehensive review and practice

SS1 MATHEMATICS LESSON NOTES

THIRD TERM

WEEK 1: MENSURATION I

1. INTRODUCTION

Welcome to the Third Term! We will begin this term with **Mensuration**, which is the branch of mathematics that deals with the measurement of length, area, and volume of geometric shapes.

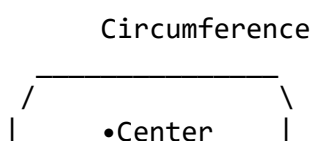
This week, we focus on the **circle**. A circle is a 2D shape where all points on the edge (the circumference) are the same distance from a central point.

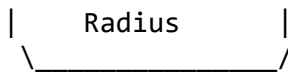
Historical Context:

The study of circles dates back to ancient civilizations. The Babylonians approximated π as 3.125, while the Egyptians used 3.16. The Greek mathematician Archimedes (287-212 BC) calculated π to remarkable accuracy and developed formulas for circular measurements that we still use today.

Key Formulas (Revision):

CIRCLE FUNDAMENTALS:





$$\text{Diameter } (d) = 2r$$

- **Radius (r):** The distance from the center to any point on the circumference.
- **Diameter (d):** The distance across the circle through the center ($d = 2r$).
- **Circumference (C):** The total distance around the circle. $C = 2\pi r$.
- **Area (A):** The space enclosed by the circle. $A = \pi r^2$.
- We will use the approximation $\pi \approx \frac{22}{7}$ or $\pi \approx 3.142$.

Quick Reference:

CIRCLE FORMULAS SUMMARY	
Diameter:	$d = 2r$
Circumference:	$C = 2\pi r = \pi d$
Area:	$A = \pi r^2$

2. CONTENT

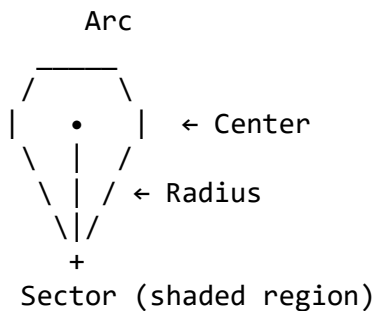
A. LENGTH OF ARCS OF CIRCLES

Definitions:

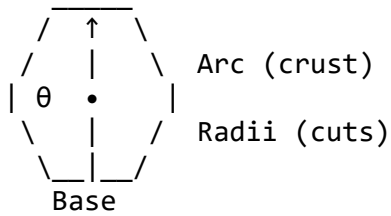
- **Arc:** An **arc** is any portion of the circumference of a circle.
- **Sector:** A **sector** is a "slice" of a circle bounded by two radii and the arc between them. It looks like a slice of pizza.

Visual Representation:

Parts of a Circle:

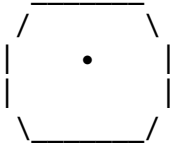


Pizza Slice Analogy:



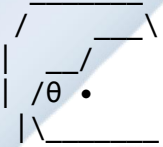
Detailed Arc Illustration:

Complete Circle:



Circumference = $2\pi r$

Sector with Arc:



Arc length = portion of circumference

Concept: The length of an arc is a *fraction* of the total circumference. The size of this fraction is determined by the angle θ (theta) that the arc subtends (forms) at the center of the circle.

The fraction is $\frac{\theta}{360^\circ}$ (since a full circle is 360°).

Formula:

$$\text{Length of Arc} = \frac{\theta}{360^\circ} \times \text{Circumference}$$

$$\text{Length of Arc} = \frac{\theta}{360^\circ} \times 2\pi r$$

Visual Derivation:

Angle Proportion:

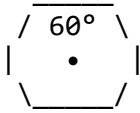
$360^\circ \rightarrow$ Full circumference ($2\pi r$)

$180^\circ \rightarrow$ Half circumference (πr)

$90^\circ \rightarrow$ Quarter circumference ($\pi r/2$)

$\theta^\circ \rightarrow (\theta/360) \times 2\pi r$

Example with 60° :



$$\begin{aligned}\text{Arc} &= (60/360) \times 2\pi r \\ &= (1/6) \times 2\pi r\end{aligned}$$

B. SECTORS AND SEGMENTS OF A CIRCLE

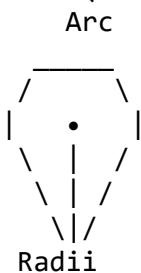
Key Definitions:

- **Chord:** A **chord** is a straight line segment whose endpoints both lie on the circle. (The diameter is the longest possible chord).
- **Segment:** A **segment** is the region of a circle bounded by a **chord** and its **arc**. It's the "crust" part of the pizza slice.
 - **Minor Segment:** The smaller region.
 - **Major Segment:** The larger region.

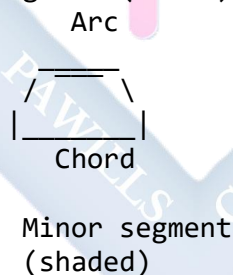
Comprehensive Diagram:

SECTOR vs SEGMENT:

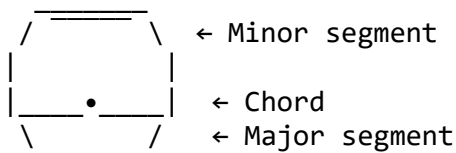
Sector (Pizza Slice):



Segment (Crust):



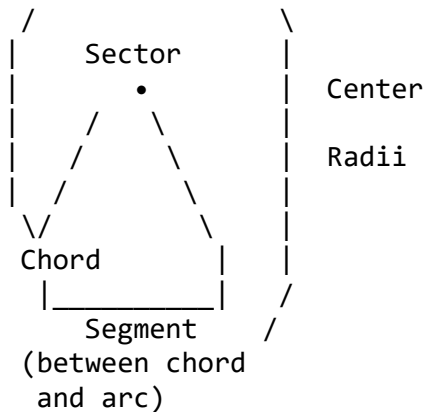
Major vs Minor Segments:



Detailed Component Breakdown:

Complete Circle Analysis:

Arc Length



Perimeter of a Sector:

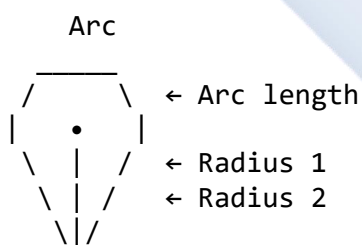
This is the total distance around the *sector*. It is the **Arc Length** plus the two **radii**.

Formula:

$$\begin{aligned}\text{Perimeter of Sector} &= \left(\frac{\theta}{360^\circ} \times 2\pi r \right) + 2r \\ &= \frac{\theta}{360^\circ} \times 2\pi r + 2r\end{aligned}$$

Visual Explanation:

Perimeter Components:



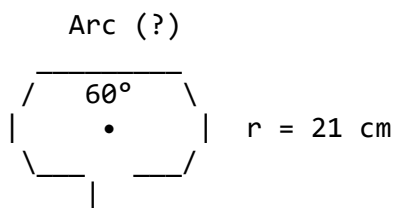
$$\begin{aligned}\text{Perimeter} &= \text{Arc} + r + r \\ &= \text{Arc} + 2r\end{aligned}$$

3. ROBUST WORKED EXAMPLES

Example 1.1: Find Arc Length

A circle has a radius of 21 cm. Find the length of an arc that subtends an angle of 60° at the center. (Take $\pi = \frac{22}{7}$).

Diagram:



Solution:

Given:

- $\theta = 60^\circ$
- $r = 21 \text{ cm}$
- $\pi = \frac{22}{7}$

Step 1: Write the formula Arc Length $= \frac{\theta}{360^\circ} \times 2\pi r$

Step 2: Substitute values Arc Length $= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21$

Step 3: Simplify the fraction $\frac{60}{360} = \frac{1}{6}$

Step 4: Calculate Arc Length $= \frac{1}{6} \times 2 \times \frac{22}{7} \times 21$
$$= \frac{1 \times 2 \times 22 \times 21}{6 \times 7}$$
$$= \frac{924}{42} = 22 \text{ cm}$$

Answer: The arc length is **22 cm**.

Visual Verification:

Full circle circumference:

$$C = 2\pi r = 2 \times (22/7) \times 21 = 132 \text{ cm}$$

60° is $1/6$ of 360°

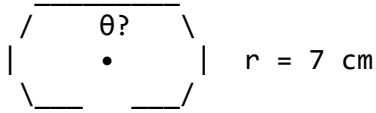
So arc $= 132/6 = 22 \text{ cm}$ ✓

Example 1.2: Find the Angle

An arc of length 11 cm is on a circle of radius 7 cm. What angle does the arc subtend at the center?

Diagram:

$$\text{Arc} = 11 \text{ cm}$$



Solution:

Given:

- Arc Length = 11 cm
- $r = 7 \text{ cm}$
- $\theta = ?$
- $\pi = \frac{22}{7}$

Step 1: Write the formula Arc Length = $\frac{\theta}{360} \times 2\pi r$

Step 2: Substitute known values $11 = \frac{\theta}{360} \times 2 \times \frac{22}{7} \times 7$

Step 3: Simplify $11 = \frac{\theta}{360} \times 2 \times 22$

$$11 = \frac{\theta}{360} \times 44$$

Step 4: Rearrange to find θ $\theta = \frac{11 \times 360}{44}$

$$\theta = \frac{3960}{44} = 90^\circ$$

Answer: The angle is 90° .

Visual Verification:

Quarter circle check:

$90^\circ = 1/4$ of circle

Arc = $1/4 \times$ Circumference

= $1/4 \times 2\pi(7)$

= $1/4 \times 44$

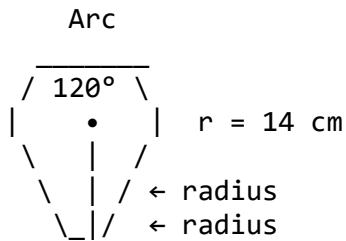
= 11 cm ✓

Example 1.3: Find Perimeter of a Sector

A sector of a circle has a radius of 14 cm and an angle of 120° . Find the perimeter of the sector.

Diagram:

Perimeter = Arc + 2 radii



Total distance around
the sector boundary

Solution:

The perimeter is the arc length *plus* two radii.

Part 1: Arc Length

$$\begin{aligned}\text{Arc} &= \frac{120}{360} \times 2 \times \frac{22}{7} \times 14 \\ &= \frac{1}{3} \times 2 \times \frac{22}{7} \times 14 \\ &= \frac{1}{3} \times 2 \times 22 \times 2 \\ &= \frac{1}{3} \times 88 = \frac{88}{3} \approx 29.33 \text{ cm}\end{aligned}$$

Part 2: Two Radii

$$2r = 2 \times 14 = 28 \text{ cm}$$

Part 3: Total Perimeter

$$\text{Perimeter} = 29.33 + 28 = 57.33 \text{ cm}$$

$$\text{Or exactly: } \frac{88}{3} + 28 = \frac{88 + 84}{3} = \frac{172}{3} \text{ cm}$$

Answer: The perimeter of the sector is **57.33 cm** (or $\frac{172}{3}$ cm exactly).

Visual Breakdown:

Components:

Arc length:	$88/3 \text{ cm}$
Radius 1:	14 cm
Radius 2:	14 cm

TOTAL :	172/3 cm
---------	----------

4. REVISION QUESTIONS (WEEK 1)

1. What is the formula for the circumference of a circle?
 2. What is the difference between an arc and a chord?
 3. What is the difference between a sector and a segment?
 4. What two components make up the perimeter of a sector?
 5. How many degrees are in a full circle?
-

5. ASSESSMENT (WEEK 1)

EVALUATION (IN-CLASS)

1. Define a "sector" of a circle.
 2. Define a "segment" of a circle.
 3. Write down the formula for the length of an arc.
 4. A circle has a radius of 10 cm. What is the length of an arc that subtends 90° ? (Leave π in your answer).
 5. A sector has a radius of 5 cm and an arc length of 10 cm. What is the perimeter of this sector?
-

ASSIGNMENT (TAKE-HOME)

1. Find the length of an arc of a circle with a radius of 28 cm, if the angle at the center is 45° . (Take $\pi = \frac{22}{7}$).
2. The perimeter of a sector of a circle of radius 7 cm is 25 cm.
 - a. Find the length of the arc of the sector.
 - b. Find the angle θ subtended by the arc at the center.
3. A bicycle wheel has a radius of 35 cm. If the wheel turns 100 times, what distance has the bicycle covered? (Hint: Find the circumference first).
4. The minute hand of a clock is 10 cm long. How far does the tip of the hand move in 20 minutes? (Hint: $20\text{minutes} = \frac{20}{60}$ of a full circle $\Rightarrow \theta = \frac{1}{3} \times 360^\circ = 120^\circ$).

WEEK 2: MENSURATION II

1. INTRODUCTION

Last week, we defined sectors and segments and found their *lengths*. This week, we will calculate their **Area** (the space they cover). We will also discover the direct and beautiful relationship between a flat sector and the curved surface area of a 3D cone.

Practical Applications:

Understanding sector and segment areas is essential in:

- Architecture (designing arches and domes)
- Engineering (calculating material needs)
- Agriculture (irrigation coverage)
- Manufacturing (cutting materials efficiently)

2. CONTENT

A. AREA OF SECTORS

Concept: Just like arc length, the area of a sector is a *fraction* of the total area of the circle. The fraction is again $\frac{\theta}{360^\circ}$.

Formula:

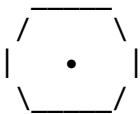
$$\text{Area of Sector} = \frac{\theta}{360^\circ} \times \text{Total Area}$$

$$\text{Area of Sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

Visual Explanation:

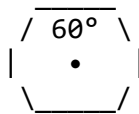
Area Proportion:

Full Circle:



$$\text{Area} = \pi r^2$$

Sector (60°):



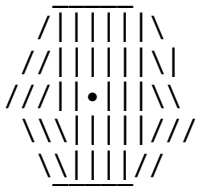
$$\begin{aligned}\text{Area} &= (60/360) \times \pi r^2 \\ &= (1/6) \times \pi r^2\end{aligned}$$

$360^\circ \rightarrow \text{Full area } (\pi r^2)$
 $180^\circ \rightarrow \text{Half area } (\pi r^2/2)$
 $90^\circ \rightarrow \text{Quarter area } (\pi r^2/4)$
 $\theta^\circ \rightarrow (\theta/360) \times \pi r^2$

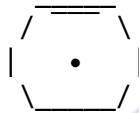
Sector Area Visualization:

Dividing the Pizza:

Full Pizza
(360° , πr^2)



Quarter Slice
(90° , $\pi r^2/4$)



Shaded area
= $1/4$ of total

B. AREA OF SEGMENTS

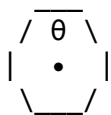
Concept: A segment is the "crust" of the pizza. To find its area, we take the full slice (the Sector) and subtract the triangular part.

$$\text{Area of Segment} = \text{Area of Sector} - \text{Area of Triangle}$$

Visual Breakdown:

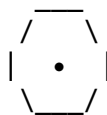
$$\text{Segment} = \text{Sector} - \text{Triangle}$$

Sector:



Full slice

Triangle:



Middle part

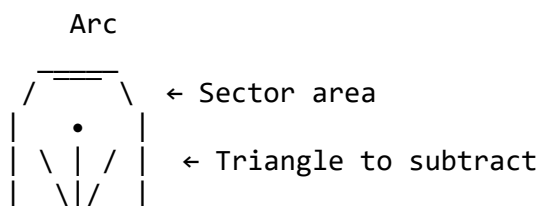
Segment:



Crust part

Detailed Illustration:

Finding Segment Area:



Chord

Segment (shaded)
= Sector - Triangle

Area of the Triangle:

The triangle inside the sector is formed by two radii (r) and the angle θ between them. We use the trigonometric formula for the area of a triangle:

$$\text{Area} = \frac{1}{2}ab\sin C$$

In our case:

- $a = r$ (one radius)
- $b = r$ (other radius)
- $C = \theta$ (angle between them)

$$\text{Area of Triangle} = \frac{1}{2} \times r \times r \times \sin \theta = \frac{1}{2}r^2 \sin \theta$$

Final Formula:

$$\text{Area of Segment} = \left(\frac{\theta}{360^\circ} \times \pi r^2 \right) - \left(\frac{1}{2}r^2 \sin \theta \right)$$

Component Diagram:

Segment Area Components:

SECTOR AREA: $(\theta/360) \times \pi r^2$
MINUS
TRIANGLE AREA: $(1/2)r^2 \sin \theta$
EQUALS
SEGMENT AREA

C. RELATIONSHIP BETWEEN SECTOR AND CONE

This is a critical concept in 3D mensuration. A **cone** is formed by folding a **sector** of a circle.

The Magic Transformation:

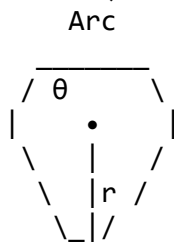
When you fold a sector into a cone:

1. The **Radius (r) of the Sector** becomes the **Slant Height (l) of the Cone**.
2. The **Arc Length of the Sector** becomes the **Base Circumference of the Cone** (which is $2\pi R$, where R is the cone's new base radius).
3. The **Area of the Sector** becomes the **Curved Surface Area (CSA) of the Cone**.

Visual Transformation:

FLAT SECTOR $\rightarrow$ FOLDED CONE

Before (Flat):



After (3D):



FOLD
 $\rightarrow$

r (sector radius)

Arc length $\rightarrow$

Sector area $\rightarrow$

l (slant height)

$2\pi R$ (base circumference)

CSA of cone

Detailed Mapping:

SECTOR PROPERTIES

$\rightarrow$

CONE PROPERTIES

Radius: r	$\rightarrow$	Slant height: l
Arc length: $(\theta/360) \times 2\pi r$	$\rightarrow$	Base circumference: $2\pi R$
Sector area: $(\theta/360) \times \pi r^2$	$\rightarrow$	Curved surface: $\pi R l$

Deriving the Cone Formula (CSA):

Step 1: Start with sector area $\text{Area of Sector} = \frac{\theta}{360^\circ} \times \pi r_{\text{sector}}^2$

Step 2: Replace r_{sector} with l (slant height): $\text{CSA} = \frac{\theta}{360^\circ} \times \pi l^2$

Step 3: From arc length relationship: $\frac{\theta}{360^\circ} \times 2\pi l = 2\pi R$

Step 4: Divide both sides by $2\pi l$: $\frac{\theta}{360^\circ} = \frac{R}{l}$

Step 5: Substitute back into CSA formula: $CSA = \left(\frac{R}{l}\right) \times \pi l^2$

$$CSA = R \times \pi \times l$$

$Curved\ Surface\ Area\ of\ a\ Cone\ (CSA) = \pi Rl$

Derivation Flowchart:

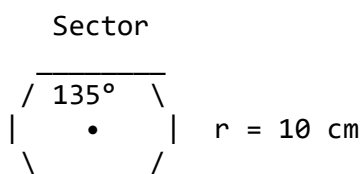
Sector area $(\theta/360) \times \pi l^2$ ↓ Replace ratio $\theta/360 = R/l$ ↓ Substitute back ↓ $CSA = (R/l) \times \pi l^2 = \pi Rl \quad \checkmark$	Arc = Circumference $(\theta/360) \times 2\pi l = 2\pi R$ ↓ Simplify $\theta/360 = R/l$ ↓
---	--

3. ROBUST WORKED EXAMPLES

Example 2.1: Area of a Sector

Find the area of a sector of a circle with radius 10 cm and a central angle of 135° . (Take $\pi = 3.142$).

Diagram:



Shaded area = ?

Solution:

Given:

- $\theta = 135^\circ$
- $r = 10\text{ cm}$
- $\pi = 3.142$

Step 1: Write formula $Area = \frac{\theta}{360^\circ} \times \pi r^2$

Step 2: Substitute values $\text{Area} = \frac{135}{360} \times 3.142 \times (10)^2$

Step 3: Simplify fraction $\frac{135}{360} = \frac{3}{8} = 0.375$

Step 4: Calculate $\text{Area} = 0.375 \times 3.142 \times 100$
 $= 0.375 \times 314.2$
 $= 117.825 \text{ cm}^2$

Answer: 117.825 cm^2 or 117.83 cm^2 (to 2 d.p.)

Visual Check:

135° is between 90° ($1/4$) and 180° ($1/2$)

Full circle area $= \pi(10)^2 = 314.2 \text{ cm}^2$

$135/360 = 3/8 = 0.375$

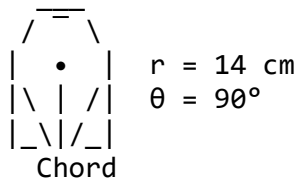
$0.375 \times 314.2 = 117.825 \text{ cm}^2$ ✓

Example 2.2: Area of a Segment

A chord subtends an angle of 90° at the center of a circle of radius 14 cm. Find the area of the minor segment. (Take $\pi = \frac{22}{7}$).

Diagram:

Minor Segment (shaded):



$\text{Area} = \text{Sector} - \text{Triangle}$

Solution:

$\text{Area of Segment} = \text{Area of Sector} - \text{Area of Triangle}$

Part 1: Area of Sector

Given: $\theta = 90^\circ$, $r = 14 \text{ cm}$

$$\begin{aligned}
 \text{Area}_{\text{sector}} &= \frac{90}{360} \times \frac{22}{7} \times (14)^2 \\
 &= \frac{1}{4} \times \frac{22}{7} \times 196 \\
 &= \frac{1}{4} \times \frac{22 \times 196}{7} \\
 &= \frac{1}{4} \times \frac{22 \times 28}{1} \\
 &= \frac{1}{4} \times 616 = 154\text{cm}^2
 \end{aligned}$$

Part 2: Area of Triangle

Given: $\theta = 90^\circ$, $r = 14$ cm

$$\begin{aligned}
 \text{Area}_{\text{triangle}} &= \frac{1}{2} r^2 \sin \theta \\
 &= \frac{1}{2} \times (14)^2 \times \sin(90^\circ)
 \end{aligned}$$

Note: $\sin 90^\circ = 1$

$$= \frac{1}{2} \times 196 \times 1 = 98\text{cm}^2$$

Part 3: Area of Segment

$$\text{Area}_{\text{segment}} = 154 - 98 = 56\text{cm}^2$$

Answer: The area of the segment is **56 cm²**.

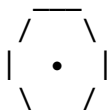
Visual Verification:

Sector (quarter circle):



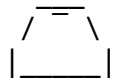
$$= 154 \text{ cm}^2$$

Triangle (inscribed):



$$= 98 \text{ cm}^2$$

Segment (difference):


$$= 56 \text{ cm}^2$$

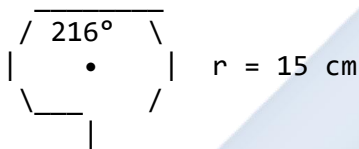
Example 2.3: Sector to Cone

A sector of a circle of radius 15 cm has an angle of 216° . It is folded to form a cone. Find:

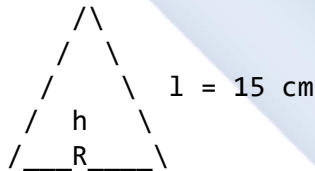
- a) The base radius of the cone (R).
- b) The vertical height of the cone (h).

Diagram:

BEFORE (Flat Sector):



AFTER (Cone):



Solution:

Given:

- Sector radius $r_s = 15 \text{ cm}$ (becomes cone slant height $l = 15 \text{ cm}$)
- Sector angle $\theta = 216^\circ$

a) Find Base Radius (R):

Relationship: Arc Length of Sector = Base Circumference of Cone

$$\frac{\theta}{360^\circ} \times 2\pi r_s = 2\pi R$$

Substitute values:

$$\frac{216}{360} \times 2\pi(15) = 2\pi R$$

Divide both sides by 2π :

$$\frac{216}{360} \times 15 = R$$

Simplify:

$$\frac{3}{5} \times 15 = R$$

$$R = 9\text{cm}$$

Answer: The base radius is **9 cm**.

b) Find Vertical Height (h):

The cone has $R = 9$ cm and $l = 15$ cm. h is the vertical height.

This forms a right-angled triangle (Pythagoras):

$$l^2 = R^2 + h^2$$

Substitute:

$$15^2 = 9^2 + h^2$$

$$225 = 81 + h^2$$

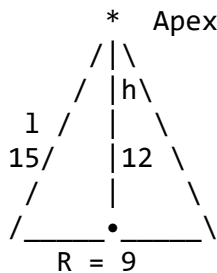
$$h^2 = 225 - 81 = 144$$

$$h = \sqrt{144} = 12\text{cm}$$

Answer: The vertical height is **12 cm**.

Pythagorean Triangle:

Cone Cross-Section:



Verify: $15^2 = 9^2 + 12^2$
 $225 = 81 + 144$
 $225 = 225 \checkmark$

4. REVISION QUESTIONS (WEEK 2)

1. What is the formula for the area of a circle?
 2. What is the formula for the area of a sector?
 3. What two components do you subtract to find the area of a segment?
 4. What is the trigonometric formula for the area of a triangle, given two sides and the angle between them?
 5. When a sector is folded into a cone, what does the arc length of the sector become?
 6. What does the radius of the sector become?
-

5. ASSESSMENT (WEEK 2)

EVALUATION (IN-CLASS)

1. Write down the formula for the area of a segment.
 2. A sector has a radius of r and an angle of 60° . What fraction of the circle's total area is it?
 3. Write down the formula for the curved surface area (CSA) of a cone in terms of its base radius R and slant height l .
 4. Find the area of a sector of radius 21 cm with an angle of 60° . (Use $\pi = \frac{22}{7}$).
 5. A sector with radius l and angle θ is folded into a cone of base radius R . Write the formula that connects θ , l , and R .
-

ASSIGNMENT (TAKE-HOME)

1. The arc of a circle of radius 10 cm subtends an angle of 72° at the center. Find:
 - a. The length of the arc.
 - b. The area of the sector.
 - c. The perimeter of the sector.
2. Find the area of the minor segment of a circle of radius 8 cm if the chord subtends an angle of 60° at the center. (Leave your answer in terms of π and $\sqrt{3}$).
3. A cone is formed from a sector of a circle of radius 25 cm with an angle of 144° .
 - a. Find the base radius of the cone.
 - b. Find the vertical height of the cone.
 - c. Find the curved surface area of the cone. (Hint: It's the area of the sector).

WEEK 3: SURFACE AREA OF SOLIDS

1. INTRODUCTION

This week, we move fully into 3D geometry. The **Surface Area (SA)** of a solid is the total area of all its "faces." It is the area of the "net" of the shape. Imagine you unfold a box to lay it flat; the surface area is the total area of that flat paper. It is measured in square units (cm^2 , m^2).

Real-World Applications:

Surface area calculations are essential for:

- **Painting:** How much paint to buy for walls
- **Packaging:** How much material to wrap a product
- **Manufacturing:** Metal sheet requirements
- **Construction:** Roofing and siding estimates

2. CONTENT

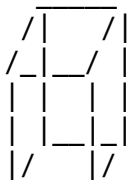
A. CUBE AND CUBOID

CUBE:

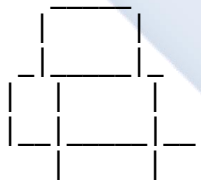
A 3D shape with 6 identical square faces.

Visual Representation:

Cube (3D):



Net (Unfolded):



All sides = 1

6 squares, each 1^2

Formula:

Let the side length be l .

- Area of one face = $l \times l = l^2$
- **Total Surface Area (TSA)** = $6 \times l^2 = 6l^2$

Visual Formula Derivation:

Cube has 6 faces:

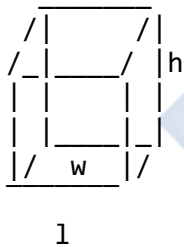
Top:	1^2
Bottom:	1^2
Front:	1^2
Back:	1^2
Left:	1^2
Right:	1^2
Total:	61^2

CUBOID (Rectangular Prism):

A 3D shape with 6 rectangular faces (3 pairs).

Visual Representation:

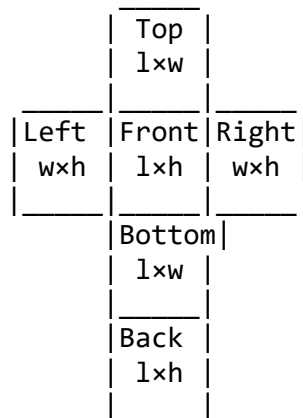
Cuboid (3D):



Dimensions: length (l), width (w), height (h)

Net Diagram:

Unfolded Cuboid Net:



Formula:

Let the dimensions be length (l), width (w), and height (h).

- Area of Top/Bottom faces = $2 \times (l \times w)$
- Area of Front/Back faces = $2 \times (l \times h)$
- Area of Left/Right faces = $2 \times (w \times h)$

$$\text{TSA} = 2lw + 2lh + 2wh = 2(lw + lh + wh)$$

Visual Breakdown:

CUBOID SURFACE AREA	
Top & Bottom:	$2lw$
Front & Back:	$2lh$
Left & Right:	$2wh$
TOTAL TSA:	$2(lw + lh + wh)$

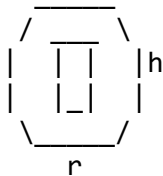
B. CYLINDER AND CONE

CYLINDER (Closed):

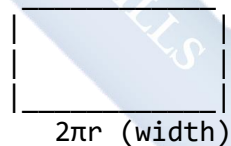
A solid with two circular ends and one curved side.

Visual Representation:

Cylinder (3D):



Unrolled curved surface:



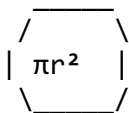
Components:

- 2 circular ends (top & bottom)
- 1 curved rectangular side

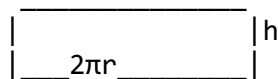
Detailed Component Diagram:

Cylinder Parts:

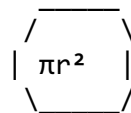
Top Circle



Curved Surface
(when unrolled)



Bottom Circle



$$\text{Area} = \pi r^2$$

$$\text{Area} = 2\pi rh$$

$$\text{Area} = \pi r^2$$

Formulas:

- Area of Top Circle = πr^2
- Area of Bottom Circle = πr^2
- **Curved Surface Area (CSA)** = The area of the side. If you unroll it, it's a rectangle with height h and width equal to the circumference ($2\pi r$).
- $\text{CSA} = 2\pi rh$
- **Total Surface Area (TSA)** = $2(\pi r^2) + 2\pi rh = 2\pi r(r + h)$

Visual Formula:

$$\begin{aligned}\text{TSA} &= 2\pi r^2 + 2\pi rh \\ &= 2\pi r(r + h)\end{aligned}$$

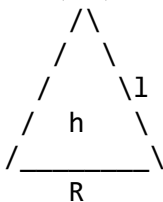
TSA Components:
2 circles: $2\pi r^2$
Curved side: $2\pi rh$
TOTAL: $2\pi r(r+h)$

CONE (Closed):

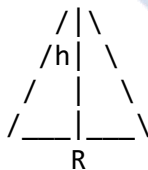
A solid with one circular base and one curved side.

Visual Representation:

Cone (3D):



Cross-section:



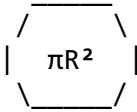
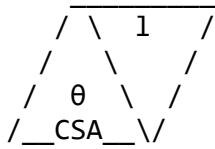
Slant height (l), vertical height (h), base radius (R)

Net Diagram:

Cone Unfolded:

Sector
(curved surface)

Circular Base



$$CSA = \pi R l$$

$$Base = \pi R^2$$

Formulas:

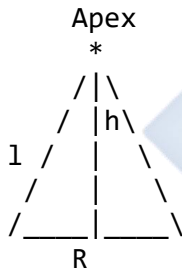
- Area of Base = πR^2
- **Curved Surface Area (CSA)** = $\pi R l$ (where l is the **slant height**)
- **Total Surface Area (TSA)** = $\pi R^2 + \pi R l = \pi R(R + l)$

CRITICAL Triangle Relationship:

The slant height l , vertical height h , and base radius R form a right-angled triangle:

$$l^2 = h^2 + R^2$$

Pythagorean Triangle in Cone:



Right triangle inside cone:

$$l^2 = h^2 + R^2$$

Example:

If $R = 3$, $h = 4$

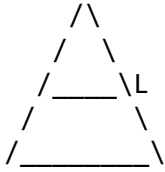
Then $l = \sqrt{9+16} = \sqrt{25} = 5$

C. PRISM

Definition: A prism is a solid with a *uniform cross-section*. This means it has the same shape (e.g., a triangle, a trapezoid) running all the way through it. A cylinder is a circular prism.

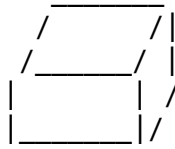
Visual Examples:

Triangular Prism:



Same triangle
throughout

Rectangular Prism:



Same rectangle
throughout

Total Surface Area (TSA):

$$\text{TSA} = 2(\text{Base Area}) + (\text{Base Perimeter} \times \text{Length})$$

$$\text{TSA} = 2A_{\text{base}} + P_{\text{base}} \times L$$

Component Breakdown:

Prism TSA Components:

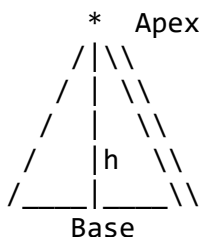
2 identical end faces: $2 \times (\text{Area of cross-section})$
Rectangular sides: $(\text{Perimeter}) \times (\text{Length})$
TOTAL TSA

D. PYRAMID

Definition: A solid with a polygon base (e.g., square, triangle) and triangular faces that meet at a single point (the apex).

Visual Representation:

Square-based Pyramid:



Components:

- 1 square base
- 4 triangular faces

Detailed Diagram:

Pyramid Parts:



Triangular face
(slant height = s)



Square base
(side = l)

Slant height (s) $\neq$ Vertical height (h)

Total Surface Area (TSA):

TSA = Area of Base + Area of all Triangular Faces

For a **square-based pyramid**:

$$\begin{aligned} \text{TSA} &= l^2 + 4 \times \left(\frac{1}{2} \times l \times s \right) \\ &= l^2 + 2ls \end{aligned}$$

where s is the slant height of the triangular face.

Visual Formula:

Square Pyramid TSA:

Base: l^2

4 Triangles: $4 \times \left(\frac{1}{2} \times \text{base} \times \text{slant height} \right)$
 $= 4 \times \left(\frac{1}{2} \times l \times s \right)$
 $= 2ls$

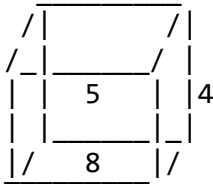
Total = $l^2 + 2ls$

3. ROBUST WORKED EXAMPLES

Example 3.1: Cuboid

Find the total surface area of a cuboid with length 8 cm, width 5 cm, and height 4 cm.

Diagram:



$l = 8 \text{ cm}, w = 5 \text{ cm}, h = 4 \text{ cm}$

Solution:

Given: $l = 8, w = 5, h = 4$

Formula: $TSA = 2(lw + lh + wh)$

Substitute: $TSA = 2((8 \times 5) + (8 \times 4) + (5 \times 4))$
 $= 2(40 + 32 + 20)$
 $= 2(92) = 184\text{cm}^2$

Answer: 184 cm^2

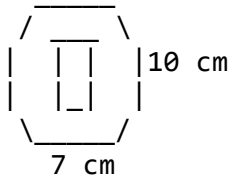
Visual Breakdown:

Face Areas:	
Top/Bottom:	$8 \times 5 = 40$
(2 faces):	$2 \times 40 = 80$
Front/Back:	$8 \times 4 = 32$
(2 faces):	$2 \times 32 = 64$
Left/Right:	$5 \times 4 = 20$
(2 faces):	$2 \times 20 = 40$
TOTAL:	$80 + 64 + 40$ $= 184 \text{ cm}^2$

Example 3.2: Cylinder

A closed metal tin is a cylinder of radius 7 cm and height 10 cm. Find the total surface area of the tin. (Take $\pi = \frac{22}{7}$).

Diagram:



$$r = 7 \text{ cm}, h = 10 \text{ cm}$$

Solution:

Given: $r = 7, h = 10, \pi = \frac{22}{7}$

Formula: $TSA = 2\pi r(r + h)$

Substitute: $TSA = 2 \times \frac{22}{7} \times 7 \times (7 + 10)$

$$= 2 \times 22 \times (17)$$

$$= 44 \times 17 = 748 \text{ cm}^2$$

Answer: 748 cm^2

Step-by-Step Calculation:

$$\begin{aligned} TSA &= 2\pi r(r + h) \\ &= 2 \times (22/7) \times 7 \times (17) \\ &= 2 \times 22 \times 17 \\ &= 44 \times 17 \\ &= 748 \text{ cm}^2 \end{aligned}$$

Verification:

$$2 \text{ circles: } 2 \times \pi \times 7^2 = 2 \times 22 \times 7 = 308 \text{ cm}^2$$

$$\text{Curved: } 2 \times \pi \times 7 \times 10 = 2 \times 22 \times 10 = 440 \text{ cm}^2$$

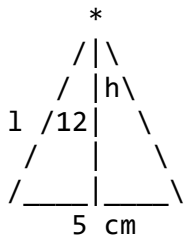
$$\text{Total: } 308 + 440 = 748 \text{ cm}^2 \checkmark$$

Example 3.3: Cone (Advanced)

Find the total surface area of a cone with a base radius of 5 cm and a vertical height of 12 cm. (Leave π in your answer).

Diagram:

Cone with given h:



Need to find l first!

Solution:

Given: $R = 5$, $h = 12$. We *need* the slant height l .

Step 1: Find slant height using Pythagoras:

$$l^2 = h^2 + R^2$$

$$l^2 = 12^2 + 5^2 = 144 + 25 = 169$$

$$l = \sqrt{169} = 13\text{cm}$$

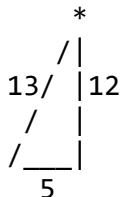
Step 2: Find TSA:

$$\begin{aligned} \text{TSA} &= \pi R(R + l) \\ &= \pi \times 5 \times (5 + 13) \\ &= \pi \times 5 \times 18 \\ &= 90\pi\text{cm}^2 \end{aligned}$$

Answer: $90\pi\text{ cm}^2$ (or approximately 282.74 cm^2)

Pythagorean Triangle:

Right triangle in cone:



Verify: $13^2 = 12^2 + 5^2$

$$169 = 144 + 25$$

$$169 = 169 \checkmark$$

Famous 5-12-13 Pythagorean triple!

4. REVISION QUESTIONS (WEEK 3)

1. What is the "net" of a solid?
 2. What is the formula for the SA of a cube?
 3. What are the three components of the surface area of a closed cylinder?
 4. What is the difference between vertical height (h) and slant height (l) in a cone?
 5. What formula connects R , h , and l in a cone?
-

5. ASSESSMENT (WEEK 3)

EVALUATION (IN-CLASS)

1. Write the formula for the total surface area of a cuboid.
 2. Write the formula for the curved surface area (CSA) of a cylinder.
 3. Write the formula for the total surface area (TSA) of a cone.
 4. A cube has a side length of 5 cm. What is its total surface area?
 5. A cone has $R = 3$, $h = 4$. What is its slant height l ?
-

ASSIGNMENT (TAKE-HOME)

1. A room (cuboid) has a length of 10 m, a width of 8 m, and a height of 3 m. Find the total area of the four walls and the ceiling. (Note: Do not include the floor).
2. A cylinder has a radius of 5 cm. Its *curved surface area* is $100\pi\text{cm}^2$.
 - a. Find the height (h) of the cylinder.
 - b. Find the total surface area of the cylinder.
3. A square-based pyramid has a base side of 10 cm and a *slant height* (the height of the triangular face) of 12 cm. Find the total surface area of the pyramid.
4. A cone has a slant height of 10 cm and a base radius of 6 cm.
 - a. Find its vertical height (h).
 - b. Find its total surface area (in terms of π).

WEEK 4: VOLUME OF SOLIDS

1. INTRODUCTION

This week, we move from surface area (a 2D measure) to **Volume** (a 3D measure). The **Volume** of a solid is the amount of 3D space it occupies. It is the "capacity" of the object—how much water, sand, or air it can hold. It is measured in cubic units (cm^3 , m^3).

Real-World Applications:

Volume calculations are essential for:

- **Construction:** Concrete needed for foundations
- **Medicine:** Drug dosages and IV fluids
- **Cooking:** Recipe measurements
- **Shipping:** Container capacity
- **Agriculture:** Grain storage silos
- **Engineering:** Fuel tank capacity

Understanding Volume:

Area (2D):



Square units (cm^2)
Covers surface

Volume (3D):



Cubic units (cm^3)
Fills space

2. CONTENT

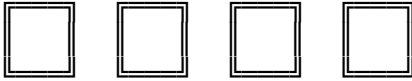
There are two main families of solids: **Prisms** and **Pyramids**.

A. PRISMS (AND CYLINDERS)

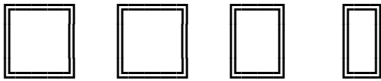
Definition: A **Prism** is a solid with a uniform (identical) cross-section from one end to the other. If you slice it anywhere along its length, the new face is identical to the base.

Visual Concept:

Prism has constant cross-section:



Same shape throughout → PRISM



Different shapes → NOT a prism

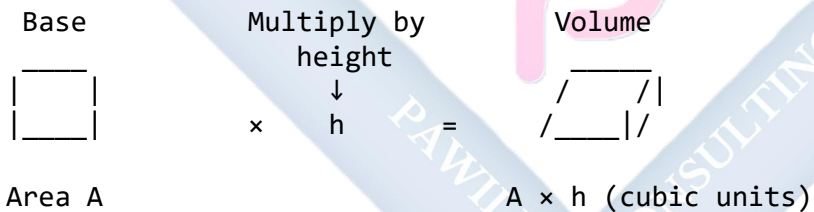
General Formula:

Volume of Prism = Area of Base (Cross-section) × Height (or Length)

$$V = A_{\text{base}} \times h$$

Visual Formula:

Volume = Base Area × Height



SPECIFIC PRISM FORMULAS:

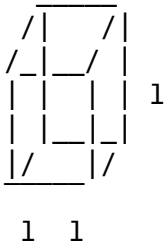
1. CUBE:

Base is a square (l^2). Height is l .

$$V = l^2 \times l = l^3$$

Diagram:

Cube:



$$V = 1 \times 1 \times 1 = 1^3$$

Example: $l = 3 \text{ cm}$

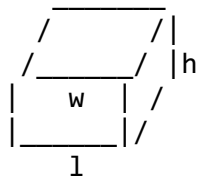
$$V = 3^3 = 27 \text{ cm}^3$$

2. CUBOID:Base is a rectangle ($l \times w$). Height is h .

$$V = (l \times w) \times h = lwh$$

Diagram:

Cuboid:



$$V = l \times w \times h$$

Example: $l=4, w=3, h=2$

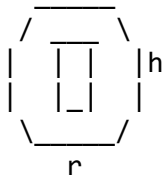
$$V = 4 \times 3 \times 2 = 24 \text{ cm}^3$$

3. CYLINDER:Base is a circle (πr^2). Height is h .

$$V = (\pi r^2) \times h = \pi r^2 h$$

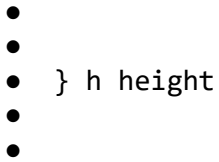
Diagram:

Cylinder:



$$V = \pi r^2 \times h$$

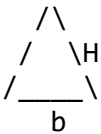
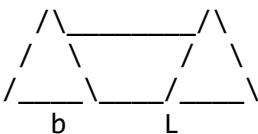
Think: Stack of circles

Each circle: πr^2 Total: $\pi r^2 \times h$ **4. TRIANGULAR PRISM:**Base is a triangle (Area = $\frac{1}{2}bH$). Length is L .

$$V = \left(\frac{1}{2}bH\right) \times L$$

Diagram:

Triangular Prism:

↓ Extend length L 

$$V = \left(\frac{1}{2}bH\right) \times L$$

Example: $b=6$, $H=4$, $L=10$

$$\begin{aligned} V &= \left(\frac{1}{2} \times 6 \times 4\right) \times 10 \\ &= 12 \times 10 = 120 \text{ cm}^3 \end{aligned}$$

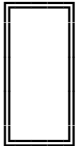
B. PYRAMIDS (AND CONES)

Definition: A **Pyramid** is a solid with a polygon base that rises to a single point (the apex).

Visual Concept:

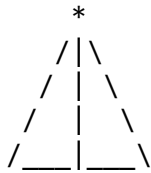
Pyramid vs Prism:

PRISM:



Constant
cross-section

PYRAMID:



Tapers to
single point

General Formula:

The volume of *any* pyramid is exactly **one-third** the volume of the *prism* with the same base and height.

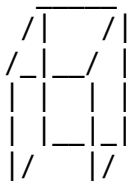
$$\text{Volume of Pyramid} = \frac{1}{3} \times \text{Area of Base} \times \text{Height}$$

$$V = \frac{1}{3} A_{\text{base}} \times h$$

Why One-Third?

Visual Proof (Informal):

Take a cube:



Divide into 3 pyramids:



Cube = 3 pyramids

∴ 1 pyramid = 1/3 cube

SPECIFIC PYRAMID FORMULAS:

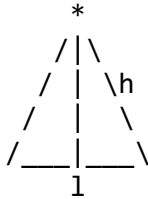
1. SQUARE-BASED PYRAMID:

Base is a square (l^2). Height is h .

$$V = \frac{1}{3} \times l^2 \times h = \frac{1}{3} l^2 h$$

Diagram:

Square Pyramid:



$$V = (1/3) \times l^2 \times h$$

Example: $l=6$, $h=9$

$$V = (1/3) \times 36 \times 9 \\ = 108 \text{ cm}^3$$

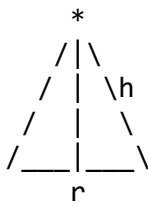
2. CONE:

Base is a circle (πr^2). Height is h .

$$V = \frac{1}{3} \times (\pi r^2) \times h = \frac{1}{3} \pi r^2 h$$

Diagram:

Cone:



$$V = (1/3) \times \pi r^2 \times h$$

Cone vs Cylinder:

Same base & height:

Cylinder: $\pi r^2 h$

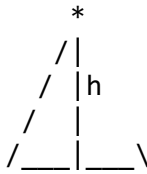
Cone: $(1/3) \pi r^2 h$

Cone = $1/3$ Cylinder

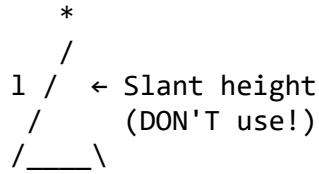
IMPORTANT NOTE:

The height h must be the **vertical height**, not the slant height.

Correct height:



Wrong height:



Use vertical $\uparrow$ Not slant $\nearrow$

VOLUME FORMULA SUMMARY:

VOLUME FORMULAS CHART	
PRISMS (Full height):	
• Cube:	$V = l^3$
• Cuboid:	$V = lwh$
• Cylinder:	$V = \pi r^2 h$
• Prism:	$V = (\text{Base Area}) \times h$
PYRAMIDS (1/3 rule):	
• Pyramid:	$V = (1/3)(\text{Base})h$
• Cone:	$V = (1/3)\pi r^2 h$
• Any:	$V = (1/3)(\text{Area})h$

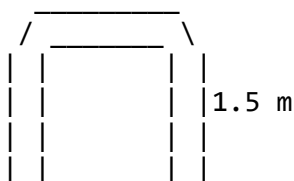
3. ROBUST WORKED EXAMPLES

Example 4.1: Cylinder

A cylindrical oil drum has a radius of 0.7 m and a height of 1.5 m. How many litres of oil can it hold? (Take $\pi = \frac{22}{7}$ and $1\text{m}^3 = 1000\text{L}$).

Diagram:

Oil Drum (Cylinder):



$$\sqrt{\quad\quad\quad}$$

0.7 m

Find volume, then convert to litres

Solution:

Given: $r = 0.7$ m, $h = 1.5$ m, $\pi = \frac{22}{7}$

Step 1: Find volume in m^3

$$\begin{aligned} V &= \pi r^2 h \\ &= \frac{22}{7} \times (0.7)^2 \times 1.5 \\ &= \frac{22}{7} \times 0.49 \times 1.5 \end{aligned}$$

Note: $0.7 = \frac{7}{10}$, so $0.7^2 = \frac{49}{100} = 0.49$

$$\begin{aligned} &= \frac{22}{7} \times \frac{49}{100} \times 1.5 \\ &= 22 \times \frac{7}{100} \times 1.5 \\ &= 22 \times 0.07 \times 1.5 \\ &= 1.54 \times 1.5 = 2.31\text{m}^3 \end{aligned}$$

Step 2: Convert to litres

$$2.31\text{m}^3 \times 1000\text{L/m}^3 = 2310\text{L}$$

Answer: It can hold **2310 litres**.

Visual Calculation:

$$\begin{aligned} V &= \pi r^2 h \\ &= (22/7) \times (0.7)^2 \times 1.5 \end{aligned}$$

Break down:

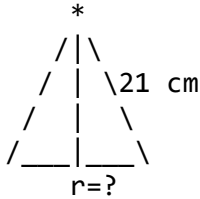
$$\begin{aligned} &(22/7) \times (7/10)^2 \times (3/2) \\ &= (22/7) \times (49/100) \times (3/2) \\ &= 22 \times 7 \times 3 / (7 \times 100 \times 2) \\ &= 22 \times 3 / 200 \\ &= 66/200 \times 7.7 \\ &= 2.31 \text{ m}^3 \\ &= 2310 \text{ L} \end{aligned}$$

Example 4.2: Cone (Working Backwards)

A cone has a volume of 110cm^3 and a height of 21 cm. Find its base radius. (Take $\pi = \frac{22}{7}$).

Diagram:

Cone with known V and h :



Given: $V = 110\text{ cm}^3$

Find: r

Solution:

Given: $V = 110\text{ cm}^3$, $h = 21\text{ cm}$, $r = ?$, $\pi = \frac{22}{7}$

Step 1: Write formula $V = \frac{1}{3}\pi r^2 h$

Step 2: Substitute known values $110 = \frac{1}{3} \times \frac{22}{7} \times r^2 \times 21$

Step 3: Simplify right side $110 = \frac{1}{3} \times \frac{22}{7} \times 21 \times r^2$

$$110 = \frac{22 \times 21}{3 \times 7} \times r^2$$

$$110 = \frac{22 \times 21}{21} \times r^2$$

$$110 = 22 \times r^2$$

Step 4: Solve for r^2 $r^2 = \frac{110}{22} = 5$

Step 5: Find r $r = \sqrt{5}\text{cm}$

Answer: The radius is $\sqrt{5}\text{ cm}$ (or approximately 2.24 cm).

Verification:

$$\begin{aligned}\text{Check: } V &= \left(\frac{1}{3}\right)\pi r^2 h \\ &= \left(\frac{1}{3}\right) \times \left(\frac{22}{7}\right) \times 5 \times 21 \\ &= \left(\frac{1}{3}\right) \times 22 \times 5 \times 3 \\ &= 22 \times 5 \\ &= 110\text{ cm}^3 \quad \checkmark\end{aligned}$$

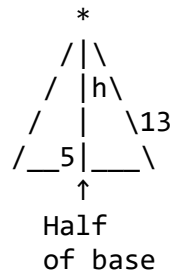
Example 4.3: Pyramid (Advanced)

A pyramid has a square base of side 10 cm. Its slant height (the height of a triangular face) is 13 cm. Find the volume of the pyramid.

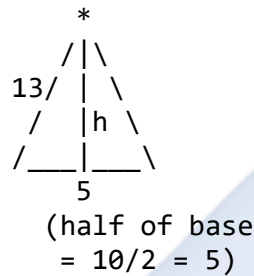
Diagram:

Finding vertical height:

Side view:



3D view:



Need h to find volume!

Solution:

Given: Base side = 10 cm, Slant height of face = 13 cm

Step 1: Find the vertical height (h)

We need the **vertical height** (h), not the slant height ($l_s = 13$).

Imagine a right-angled triangle *inside* the pyramid, formed by:

- The vertical height h (from apex to center of base)
- The "apothem" of the base (from center to midpoint of a side) $= \frac{10}{2} = 5$ cm
- The slant height l_s (13 cm), which is the hypotenuse

Step 2: Use Pythagoras $l_s^2 = h^2 + (\text{apothem})^2$

$$13^2 = h^2 + 5^2$$

$$169 = h^2 + 25$$

$$h^2 = 169 - 25 = 144$$

$$h = \sqrt{144} = 12\text{cm}$$

Step 3: Calculate volume $V = \frac{1}{3} \times \text{Base Area} \times h$

$$= \frac{1}{3} \times (10 \times 10) \times 12$$

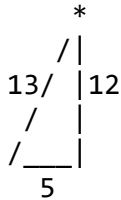
$$= \frac{1}{3} \times 100 \times 12$$

$$= \frac{1200}{3} = 400 \text{ cm}^3$$

Answer: The volume is **400 cm³**.

Visual Triangle:

Right triangle in pyramid:



5-12-13 Pythagorean triple!
(Related to 3-4-5)

Verify: $13^2 = 12^2 + 5^2$
 $169 = 144 + 25$
 $169 = 169 \checkmark$

4. REVISION QUESTIONS (WEEK 4)

1. What is the general formula for the volume of any prism?
2. What is the general formula for the volume of any pyramid?
3. What is the relationship between the volume of a cone and a cylinder if they have the same base radius and vertical height?
4. To find the volume of a cone, do you need the slant height or the vertical height?
5. How many cm³ are in 1 litre? (1000 cm³ = 1 L).

5. ASSESSMENT (WEEK 4)

EVALUATION (IN-CLASS)

1. Write the formula for the volume of a cuboid.
2. Write the formula for the volume of a cylinder.
3. Write the formula for the volume of a cone.
4. A cube has a side of 4 cm. What is its volume?
5. A cone and a cylinder have the same radius and height. If the cylinder's volume is 30 m³, what is the cone's volume?

ASSIGNMENT (TAKE-HOME)

1. A triangular prism has a base that is a right-angled triangle with sides 5 cm and 12 cm. The length of the prism is 20 cm. Find the volume of the prism. (Hint: Find the area of the triangular base first).
2. Find the volume of a cone with radius 7 cm and slant height 25 cm. (Take $\pi = \frac{22}{7}$).
3. A cylindrical container is 20 cm deep. If it holds 3.08 litres of water, calculate the radius of the container. (Hint: $1\text{L} = 1000\text{cm}^3$. $V = 3080\text{cm}^3$).
4. A square-based pyramid has a base side of 18 cm and a vertical height of 12 cm. Find its volume.



WEEK 5: MIDTERM EXAMINATION

This week is for your **midterm examination**, which will cover all content from Weeks 1-4 (Arcs, Sectors, Segments, Surface Area, and Volume of all basic solids).

Examination Coverage:

MIDTERM EXAM TOPICS
Week 1: Arc lengths, sectors
Week 2: Areas, cone-sector link
Week 3: Surface areas of solids
Week 4: Volumes of solids

Preparation Checklist:

Review all formulas

Practice arc and sector calculations

Master the sector-to-cone transformation

Know surface area formulas (cube, cuboid, cylinder, cone, prism, pyramid)

Know volume formulas (all prisms and pyramids)

Practice mixed problems

Understand when to use vertical vs. slant height

Key Formulas to Memorize:

CIRCLES:

- Arc: $(\theta/360) \times 2\pi r$
- Sector area: $(\theta/360) \times \pi r^2$
- Segment: Sector - Triangle

SURFACE AREAS:

- Cube: $6l^2$
- Cuboid: $2(lw+lh+wh)$
- Cylinder: $2\pi r(r+h)$
- Cone: $\pi R(R+l)$

VOLUMES:

- Cube: l^3
- Cuboid: lwh
- Cylinder: $\pi r^2 h$
- Cone: $(1/3)\pi r^2 h$

- Prism: $(\text{Base Area}) \times h$
- Pyramid: $(1/3)(\text{Base Area}) \times h$



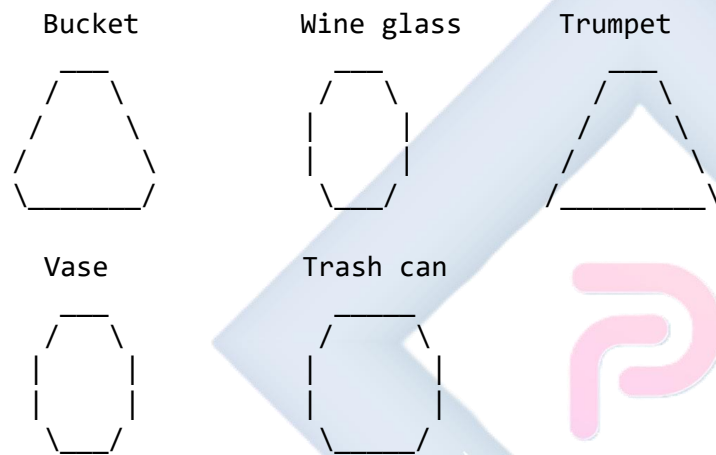
WEEK 6: AREA AND VOLUME OF FRUSTUM

1. INTRODUCTION

This week, we study a special composite shape called a **Frustum** (plural: **frusta**). A frustum is the part of a cone or pyramid that is left when you "cut off" the top with a slice parallel to the base.

Real-World Examples:

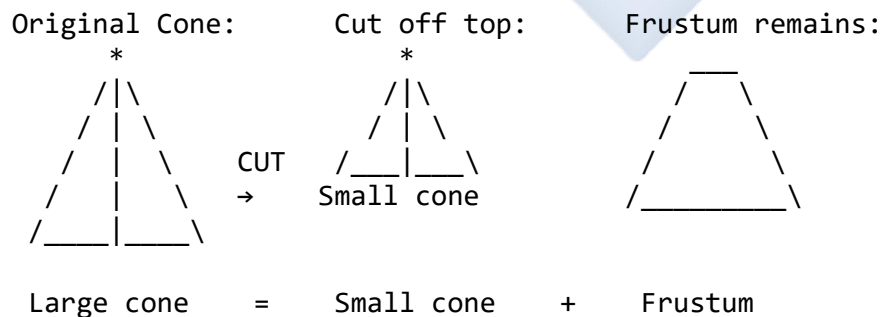
Frustums in everyday life:



The method for solving frustum problems is "**Large Cone - Small Cone**" or "**Large Pyramid - Small Pyramid**".

Visual Concept:

Creating a Frustum:



2. CONTENT

A. FINDING THE "MISSING" HEIGHT

To find the volume or area, you must first find the height of the original cone/pyramid. We use **Similar Triangles**.

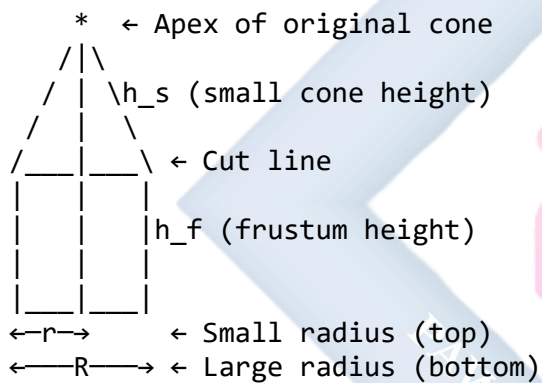
Key Variables:

- R = radius of the large base (bottom)
- r = radius of the small top
- H = height of the *original* large cone
- h_s = height of the *removed* small cone
- h_f = height of the *frustum* itself

Relationship: $H = h_s + h_f$

Visual Diagram:

Complete Setup:



$$H = h_s + h_f \text{ (total height)}$$

Similar Triangles Ratio:

$$\frac{\text{small radius}}{\text{large radius}} = \frac{\text{small height}}{\text{large height}}$$

$$\frac{r}{R} = \frac{h_s}{H}$$

Since $H = h_s + h_f$, we can write:

$$\frac{r}{R} = \frac{h_s}{h_s + h_f}$$

Solving for heights:

From $\frac{r}{R} = \frac{h_s}{H}$.

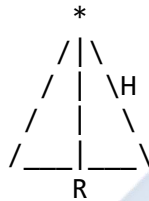
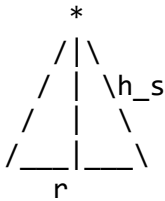
$$h_s = \frac{r \cdot H}{R}$$

Or: $H = \frac{R \cdot h_s}{r}$

Visual Similar Triangles:

Similar Triangles Principle:

Small triangle: Large triangle:



Ratio: $r/R = h_s/H$
(Corresponding sides proportional)

B. VOLUME OF A FRUSTUM

Formula:

Volume = Volume of Large Cone – Volume of Small Cone

$$\begin{aligned} V_{\text{frustum}} &= \frac{1}{3} \pi R^2 H - \frac{1}{3} \pi r^2 h_s \\ &= \frac{1}{3} \pi (R^2 H - r^2 h_s) \end{aligned}$$

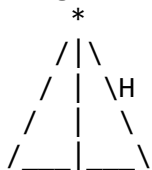
This also works for pyramids:

$$V_{\text{frustum}} = \frac{1}{3} A_{\text{Base}} H - \frac{1}{3} A_{\text{top}} h_s$$

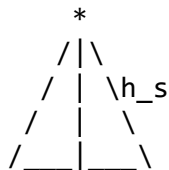
Visual Subtraction:

Volume Calculation:

Large Cone



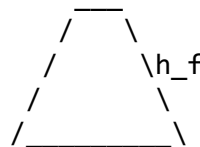
Small Cone



-

=

Frustum



$$\begin{array}{ccc} R & & r \\ V_{\text{large}} & - & V_{\text{small}} \\ (1/3)\pi R^2 H & & (1/3)\pi r^2 h_s \end{array} = \begin{array}{c} V_{\text{frustum}} \\ \text{Large} - \text{Small} \end{array}$$

C. SURFACE AREA OF A FRUSTUM

The total surface area is the sum of:

1. Area of the top circle
2. Area of the bottom circle
3. Curved surface area (CSA)

Curved Surface Area (CSA) of Frustum:

$$\text{CSA} = \text{CSA of Large Cone} - \text{CSA of Small Cone}$$

$$\text{CSA} = \pi RL - \pi r l_s$$

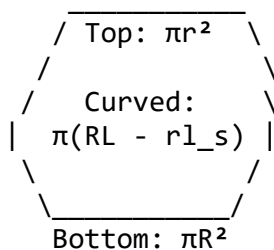
where L and l_s are the slant heights of the large and small cones.

Total Surface Area (TSA):

$$\begin{aligned} \text{TSA} &= \pi r^2 + \pi R^2 + (\pi RL - \pi r l_s) \\ &= \pi r^2 + \pi R^2 + \pi(RL - r l_s) \end{aligned}$$

Component Diagram:

TSA Components:



$$\text{Total} = \pi r^2 + \pi R^2 + \pi(RL - r l_s)$$

D. COMPOUND SHAPES

Definition: These are shapes made by joining two or more simple solids.

Rules:

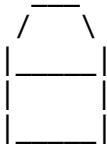
- **Total Volume:** Find the volume of each part and **ADD** them.

- **Total Surface Area:** Find the surface area of each part and **ADD** them, but be careful to **SUBTRACT** any surfaces that are *hidden* by the join.

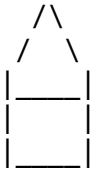
Examples:

Common Compound Shapes:

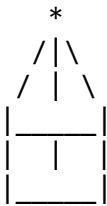
1. Hemisphere on Cylinder:



2. Cone on Cylinder:

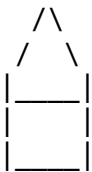


3. Pyramid on Cube:



Surface Area Caution:

Hidden Surfaces:



When joined:
Don't count
the base of cone
OR top of cylinder
(they're hidden!)

Only count exposed surfaces!

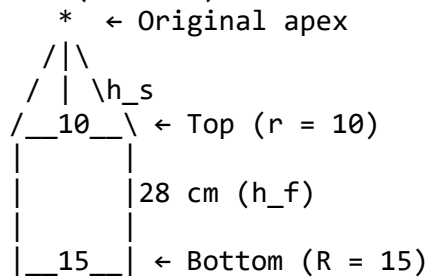
3. ROBUST WORKED EXAMPLES

Example 6.1: Volume of a Cone Frustum

A bucket is 28 cm high. Its top and bottom radii are 15 cm and 10 cm respectively. Find the volume of the bucket. (Take $\pi = \frac{22}{7}$).

Diagram:

Bucket (Frustum):

Given: $h_f=28$, $R=15$, $r=10$

Find: Volume

Solution:**Step 1: Given Information**

- $h_f = 28$ cm (frustum height)
- $R = 15$ cm (bottom radius)
- $r = 10$ cm (top radius)
- $\pi = \frac{22}{7}$

Step 2: Find Heights (H and h_s)Let h_s be the height of the small cone.

$$H = h_s + 28 \text{ (total height)}$$

Using similar triangles: $\frac{r}{R} = \frac{h_s}{H}$

$$\frac{10}{15} = \frac{h_s}{h_s + 28}$$

$$\frac{2}{3} = \frac{h_s}{h_s + 28}$$

$$\text{Cross-multiply: } 2(h_s + 28) = 3h_s$$

$$2h_s + 56 = 3h_s$$

$$56 = 3h_s - 2h_s$$

$$h_s = 56 \text{ cm}$$

$$\text{Therefore: } H = 56 + 28 = 84 \text{ cm}$$

Step 3: Calculate Large Cone Volume

$$\begin{aligned}V_{\text{Large}} &= \frac{1}{3}\pi R^2 H \\&= \frac{1}{3} \times \frac{22}{7} \times (15)^2 \times 84 \\&= \frac{1}{3} \times \frac{22}{7} \times 225 \times 84 \\&= \frac{22 \times 225 \times 84}{3 \times 7} \\&= \frac{22 \times 225 \times 12}{7} \\&= 22 \times 225 \times \frac{12}{7}\end{aligned}$$

Since $84 = 12 \times 7$:

$$\begin{aligned}&= \frac{22 \times 225 \times 84}{21} \\&= 22 \times 225 \times 4 = 19,800\text{cm}^3\end{aligned}$$

Step 4: Calculate Small Cone Volume

$$\begin{aligned}V_{\text{Small}} &= \frac{1}{3}\pi r^2 h_s \\&= \frac{1}{3} \times \frac{22}{7} \times (10)^2 \times 56 \\&= \frac{1}{3} \times \frac{22}{7} \times 100 \times 56 \\&= \frac{22 \times 100 \times 56}{21} \\&= \frac{22 \times 100 \times 8}{3} \\&= \frac{17,600}{3} \approx 5,866.67\text{cm}^3\end{aligned}$$

Step 5: Find Frustum Volume

$$V_{\text{frustum}} = V_{\text{Large}} - V_{\text{Small}}$$

$$= 19,800 - 5,866.67$$

$$= 13,933.33 \text{ cm}^3$$

Answer: $13,933.33 \text{ cm}^3$ or $\frac{41,800}{3} \text{ cm}^3$ exactly.

Visual Summary:

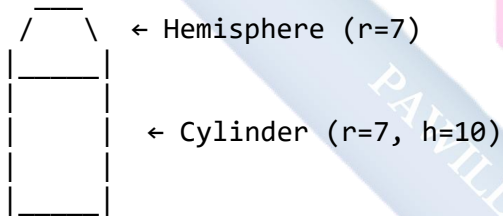
VOLUME CALCULATION:	
Large cone:	$19,800 \text{ cm}^3$
Small cone:	$5,867 \text{ cm}^3$
Frustum:	$13,933 \text{ cm}^3$

Example 6.2: Compound Shape (Volume)

A shape is formed by a hemisphere of radius 7 cm placed on top of a cylinder of radius 7 cm and height 10 cm. Find the total volume.

Diagram:

Compound Shape:



$$\text{Total Volume} = V_{\text{hemisphere}} + V_{\text{cylinder}}$$

Solution:

Formula for sphere: $V = \frac{4}{3}\pi r^3$

Formula for hemisphere: $V = \frac{2}{3}\pi r^3$

Part 1: Volume of Cylinder

$$V_{\text{Cylinder}} = \pi r^2 h$$

$$= \frac{22}{7} \times (7)^2 \times 10$$

$$\begin{aligned}
 &= \frac{22}{7} \times 49 \times 10 \\
 &= 22 \times 7 \times 10 \\
 &= 1,540 \text{ cm}^3
 \end{aligned}$$

Part 2: Volume of Hemisphere

$$\begin{aligned}
 V_{\text{Hemisphere}} &= \frac{2}{3} \pi r^3 \\
 &= \frac{2}{3} \times \frac{22}{7} \times (7)^3 \\
 &= \frac{2}{3} \times \frac{22}{7} \times 343 \\
 &= \frac{2 \times 22 \times 343}{3 \times 7} \\
 &= \frac{2 \times 22 \times 49}{3} \\
 &= \frac{2,156}{3} \approx 718.67 \text{ cm}^3
 \end{aligned}$$

Part 3: Total Volume

$$\begin{aligned}
 V_{\text{Total}} &= 1,540 + 718.67 \\
 &= 2,258.67 \text{ cm}^3
 \end{aligned}$$

Answer: 2,258.67 cm³ (or 2,259 cm³ to nearest cm³)

Calculation Breakdown:

VOLUME COMPONENTS:	
Cylinder:	1,540.00 cm ³
Hemisphere:	718.67 cm ³
TOTAL:	2,258.67 cm ³

4. REVISION QUESTIONS (WEEK 6)

1. What is a frustum?
 2. What is the key mathematical principle used to find the "missing" height of the original cone/pyramid?
 3. Write the formula for the volume of a cone frustum using V_{Large} and V_{Small} .
 4. Write the formula for the TSA of a cone frustum.
 5. When finding the Total Surface Area of a compound shape (e.g., cone on a cylinder), what must you be careful *not* to include?
-

5. ASSESSMENT (WEEK 6)

EVALUATION (IN-CLASS)

1. Draw a frustum of a cone and label $R, r, h_f, H,$ and h_s .
 2. Write the similar triangle ratio for the heights and radii of a cone frustum.
 3. How do you find the volume of a compound shape?
 4. A hemisphere is placed on a cube. How do you find the total volume?
 5. A small cone is cut from a large cone. How do you find the volume of the remaining part?
-

ASSIGNMENT (TAKE-HOME)

1. A frustum of a cone has a top radius of 5 cm, a bottom radius of 10 cm, and a vertical height of 12 cm.
 - a. Find the height of the small cone that was cut off.
 - b. Find the volume of the frustum.
 2. A frustum is cut from a square-based pyramid. The large base has sides 15 cm, the small base has sides 5 cm. The frustum height is 6 cm.
 - a. Find the height of the original pyramid.
 - b. Find the volume of the frustum.
 3. A solid object is made of a cylinder of height 10 cm and radius 3 cm, with a hemisphere of radius 3 cm on top. Find:
 - a. The total volume of the object.
 - b. The total surface area of the object. (Hint: It's the CSA of the cylinder + Area of the *base* circle + CSA of the hemisphere).
-

WEEK 7: MIDTERM BREAK

This week is for your **midterm break**.

Suggested Activities:

- Review mensuration formulas
- Practice frustum problems
- Work on compound shapes
- Prepare for statistics topics



WEEK 8: STATISTICS I

1. INTRODUCTION

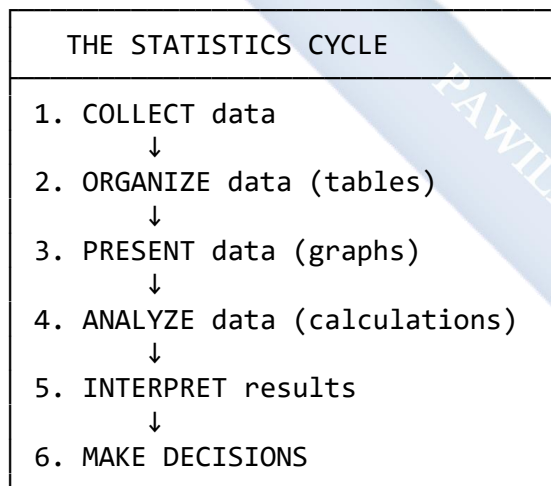
This week, we begin our study of **Statistics**. Statistics is the science of **collecting, organizing, analyzing, and interpreting** data to make informed decisions. It's how we find patterns in the world, from polling to medicine to finance.

Why Study Statistics?

Statistics is everywhere in modern life:

- Social media analytics
- Medical research and drug trials
- Election polls and predictions
- Business decisions and market research
- Climate change data
- Educational assessments
- Economic indicators

The Statistical Process:



2. CONTENT

A. COLLECTION OF MEANINGFUL DATA

Key Definitions:

Data: A set of facts, figures, or information (e.g., test scores, heights, colors).

Meaningful Data: The data must be relevant to the problem you are studying.

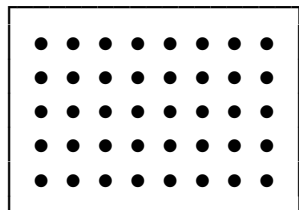
Population: The *entire* group you are interested in (e.g., all SS1 students in Nigeria).

Sample: A *small, representative subgroup* of the population, used when the population is too large to study.

Visual Explanation:

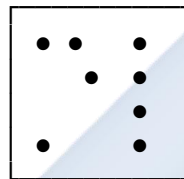
POPULATION vs SAMPLE:

Population (ALL):



All 5 million
SS1 students

Sample (SOME):



Selected subset
500 people
(representative)

SOURCES OF DATA

1. PRIMARY DATA

Information you collect *yourself*, firsthand.

Methods:

- **Questionnaires:** Written surveys
- **Interviews:** Face-to-face questions
- **Experiments:** Controlled tests
- **Observation:** Watching and recording

Advantages (Pros):

- ✓ Specific to your research question
- ✓ Accurate and up-to-date
- ✓ You control the quality
- ✓ Original and reliable

Disadvantages (Cons):

- ✗ Time-consuming
- ✗ Expensive

- ✗ Requires expertise
- ✗ May need large sample

Example:

Primary Data Collection:

Student conducts survey:
"How many hours do you
study per day?"

Goes to 100 classmates
Records their answers
→ PRIMARY DATA

2. SECONDARY DATA

Information that *someone else* has already collected.

Methods:

- Textbooks and journals
- Government reports (e.g., National Bureau of Statistics)
- Company records
- Websites and databases
- Newspapers and magazines

Advantages (Pros):

- ✓ Cheap or free
- ✓ Quick to obtain
- ✓ Large datasets available
- ✓ Historical data accessible

Disadvantages (Cons):

- ✗ May be outdated
- ✗ May not fit your exact needs
- ✗ Potential bias
- ✗ Unknown collection methods

Example:**Secondary Data Use:**

Student reads report from:
"National Bureau of
Statistics Nigeria"

Uses published data on
school enrollment rates
→ SECONDARY DATA

COMPARISON TABLE:

Aspect	Primary Data	Secondary
Source	You collect it yourself	Already collected
Cost	Expensive	Cheap/Free
Time	Slow	Quick
Relevance	Very high	May vary
Control	Full control	No control

TYPES OF DATA**1. QUALITATIVE (CATEGORICAL) DATA**

Data that describes a quality or category. It is **non-numerical**.

Characteristics:

- Describes attributes
- Cannot be measured numerically
- Can be named or labeled
- Used for classification

Examples:

- Favorite food (Rice, Beans, Yam)

- Gender (Male, Female)
- Color of car (Red, Blue, Black)
- State of origin (Lagos, Kano, Rivers)
- Marital status (Single, Married)
- Blood type (A, B, AB, O)

Visual:

Qualitative Data Examples:

Food preferences:
Pizza | Jollof | Suya | Amala

Clothing sizes:
Small | Medium | Large | XL

Colors:
Red | Green | Blue | Yellow

2. QUANTITATIVE (NUMERICAL) DATA

Data that consists of **numbers**.

A. DISCRETE DATA

Can be **counted**. Has finite, distinct values (often integers).

Characteristics:

- Countable items
- Whole numbers (usually)
- Gaps between values
- Cannot have fractions/decimals in context

Examples:

- Number of students in a class (25, 30, 35...)
- Shoe size (38, 39, 40, 41...)
- Number of goals scored (0, 1, 2, 3...)
- Number of children (0, 1, 2, 3...)
- Age in years (15, 16, 17...)
- Number of cars in a parking lot

Visual:

Discrete Data (Countable):

Students: ● ● ● ● ● = 5 students
(can't have 5.3 students!)

Cars: = 3 cars
(not 3.7 cars)

Goals: ⚽ ⚽ = 2 goals
(not 2.5 goals)

B. CONTINUOUS DATA

Can be **measured**. Can take any value within a range.

Characteristics:

- Measured quantities
- Can have decimals/fractions
- Infinite possible values
- Depends on precision of measurement

Examples:

- Height (1.72m, 1.721m, 1.7215m...)
- Temperature (36.5°C, 36.52°C...)
- Time (10.5 seconds, 10.53s...)
- Weight (65.3kg, 65.34kg...)
- Distance (100.5m, 100.52m...)
- Speed (80.7 km/h...)

Visual:

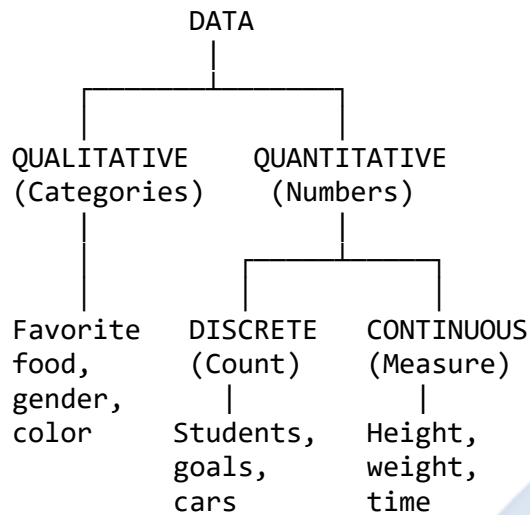
Continuous Data (Measured):

Height: |-----●-----| 1.72m
Could be 1.721m or 1.7215m
(depends on measurement tool)

Time: |-----●-----| 10.5s
Could be 10.53s or 10.534s

Weight: [====●=====] 65.3kg
Could be 65.34kg or 65.345kg

DATA TYPES FLOWCHART:



B. TABULATION AND PRESENTATION (UNGROUPEd DATA)

Raw Data: Data that has just been collected and is in no particular order.

Example: Scores of 20 students:

5, 7, 8, 6, 7, 5, 9, 8, 7, 7,
9, 5, 6, 8, 7, 10, 6, 8, 7, 5

This is messy and hard to understand!

Tabulation: The process of organizing raw data into a **Frequency Distribution Table**.

Key Terms:

- **Tally:** A quick way of counting using strokes
 - | = 1
 - || = 2
 - ||| = 3
 - |||| = 4
 - |||| = 5 (crossed)
- **Frequency (*f*):** The total count for each item

Visual Tally System:

Counting with Tallies:

1: |
2: ||
3: |||
4: ||||
5: ||||| (crossed diagonally)
6: |||||
7: |||||
8: |||||
9: |||||
10: |||||

3. ROBUST WORKED EXAMPLE

Example 8.1: Creating a Frequency Table

The raw data below represents the shoe sizes of 20 students. Organize this data into a frequency distribution table.

Raw Data:

7, 8, 9, 7, 8, 10, 6, 8, 9, 7,
8, 8, 7, 6, 8, 9, 7, 8, 10, 8

Solution:

Step 1: Identify the smallest and largest values

- Smallest = 6
- Largest = 10

Step 2: Create table structure with columns:

- "Shoe Size (x)"
- "Tally"
- "Frequency (f)"

Step 3: Go through the raw data one by one, making a tally mark for each value

Step 4: Count the tallies to get frequency

Final Frequency Distribution Table:

Shoe Size (x)	Tally	Frequency (f)
6		
7		
8		
9		
10		
Total		$\Sigma f = 20$

Visual Process:

Raw Data (messy):

7,8,9,7,8,10,6,8,9,7,8,8,7,6,8,9,7,8,10,8

↓ ORGANIZE ↓

Frequency Table (clear):

Size	Frequency
6	(2)
7	(5)
8	(8) ← Most common
9	(3)
10	(2)

Interpretation:

✓ This table is much clearer than raw data ✓ We can immediately see that:

- Size 8 is most popular (8 students)
- Sizes 6 and 10 are least common (2 each)
- Total students = 20 ($\Sigma f = 20$)

Benefits of Frequency Tables:

WHY USE FREQUENCY TABLES?

- ✓ Easy to read and understand
- ✓ Shows patterns clearly
- ✓ Identifies most/least common
- ✓ Compact representation
- ✓ Ready for graphing
- ✓ Easy to calculate statistics

4. REVISION QUESTIONS (WEEK 8)

1. Define Statistics.
 2. What is the difference between Primary and Secondary data?
 3. What is the difference between Qualitative and Quantitative data?
 4. Give an example of Discrete data and Continuous data.
 5. What is the purpose of a Tally column in a frequency table?
-

5. ASSESSMENT (WEEK 8)

EVALUATION (IN-CLASS)

1. A researcher gets data from the National Bureau of Statistics. Is this primary or secondary data?
 2. A student counts the number of cars passing the school gate. Is this primary or secondary data?
 3. Classify the following as qualitative, discrete, or continuous:
 - a. The brands of phones in the class.
 - b. The number of siblings each student has.
 - c. The time it takes to run 100m.
 4. What is "raw data"?
 5. What does $\sum f$ mean?
-

ASSIGNMENT (TAKE-HOME)

1. Ask 20 friends or family members for their favorite color from this list: Red, Blue, Green, Black, Other.
 - a. What type of data is this (qualitative, discrete, continuous)?
 - b. Is this primary or secondary data?
 - c. Present your findings in a frequency distribution table (with Tally and Frequency).
2. The following are the marks (out of 10) of 30 students in a Mathematics quiz:
5, 8, 6, 7, 7, 10, 9, 4, 5, 6,
7, 8, 9, 7, 6, 5, 8, 7, 7, 9,
4, 6, 8, 7, 6, 7, 8, 9, 5, 7
 - a. What is the range of the scores?
 - b. Construct a frequency distribution table for this data.
 - c. What is the modal score (the score with the highest frequency)?

WEEK 9: STATISTICS II

1. INTRODUCTION

Last week, we organized raw data into simple frequency tables. But what if our data is continuous or very spread out (e.g., heights of 100 people)? We need to **group** the data into **class intervals**. We will also learn how to represent this data visually using **Bar Charts** and **Histograms**.

Why Group Data?

Ungrouped (100 different heights):

150, 151, 152, 153, 154, 155, 156...

- ✗ Too many individual values!
- ✗ Hard to see patterns!

Grouped (into ranges):

150-154: 15 people

155-159: 25 people

160-164: 35 people

165-169: 20 people

170-174: 5 people

✓ Much clearer!

✓ Easy to see pattern!

2. CONTENT

A. FREQUENCY DISTRIBUTION (GROUPED DATA)

Key Terms:

1. Class Interval:

We group the data into "bins" or ranges.

Examples:

- 10-19, 20-29, 30-39 (ages)
- 0-4, 5-9, 10-14 (scores)
- 150-159, 160-169 (heights in cm)

Visual:

Class Intervals:

10-19 [===]
20-29 [===]
30-39 [===]
40-49 [===]

Each bracket = one class interval

2. Class Boundaries:

To avoid gaps (e.g., where does 19.5 go?), we create "true" boundaries.

Rule: Subtract 0.5 from the lower limit and add 0.5 to the upper limit.

Examples:

- 10-19 becomes **9.5-19.5**
- 20-29 becomes **19.5-29.5**
- 30-39 becomes **29.5-39.5**

Visual Explanation:

Class Intervals with Boundaries:

Stated: 10-19 20-29 30-39
 [===] [===] [===]

True 9.5 — 19.5 — 29.5 — 39.5
Boundaries: [] [] []
 No gaps - continuous!

Where does 19.5 belong?

→ Between 10-19 and 20-29

→ Boundaries solve this!

3. Class Width (Size):

The size of the interval.

Formula: Class Width = Upper Boundary – Lower Boundary

Example:

- Class 10-19 has boundaries 9.5-19.5
 - Width = $19.5 - 9.5 = 10$
-

4. Class Midpoint (Mark):

The middle value of the interval.

Formula: $\text{Midpoint} = \frac{\text{Lower Limit} + \text{Upper Limit}}{2}$

Example:

- Class 10-19: $\text{Midpoint} = \frac{10+19}{2} = 14.5$
- Class 20-29: $\text{Midpoint} = \frac{20+29}{2} = 24.5$

Visual:

Class Midpoint:

10 ————— 19
↓
14.5 (middle)

20 ————— 29
↓
24.5 (middle)

COMPREHENSIVE EXAMPLE:

Grouping the 30 test scores from Week 8 assignment.

Raw Data:

5, 8, 6, 7, 7, 10, 9, 4, 5, 6,
7, 8, 9, 7, 6, 5, 8, 7, 7, 9,
4, 6, 8, 7, 6, 7, 8, 9, 5, 7

Step 1: Find range

- Minimum = 4
- Maximum = 10
- Range = $10 - 4 = 6$

Step 2: Choose class intervals: 4-5, 6-7, 8-9, 10-11

Grouped Frequency Table:

Class Interval	Tally	Frequency (f)
4-5		
6-7		
8-9		

Class Interval	Tally	Frequency (f)
10-11		
Total		30

With Additional Columns:

Class	Tally	f	Boundaries	Midpoint
4-5				
6-7				
8-9				
10-11			1	9.5-11.5
Σ		30		

B. GRAPHS

Graphs are visual representations of data.

1. LINEAR GRAPH

A simple graph using lines to connect points.

Used for: Showing *trends over time*

Examples:

- Temperature changes during the day
- Company profit over 5 years
- Student enrollment by year
- Stock prices over time

Visual Example:

Temperature Throughout Day:



Line shows continuous change

2. BAR CHART

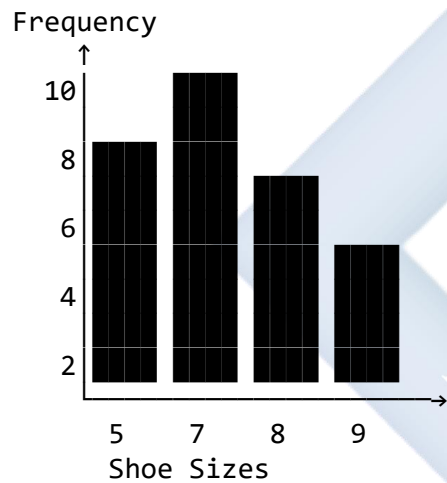
Used for: Qualitative or Discrete data

Key Features:

- Has *gaps* between the bars
- X-axis lists the *categories* or discrete values
- Y-axis represents the *frequency*
- Bars can be vertical or horizontal
- All bars same width

Visual Structure:

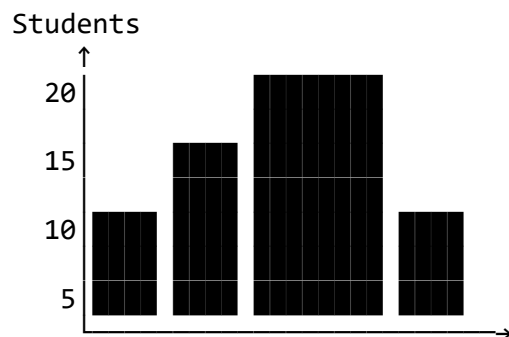
Bar Chart (Discrete Data):



Notice the GAPS between bars!
(Shows discrete categories)

Example:

Favorite Subjects:



Gaps show different categories

3. HISTOGRAM

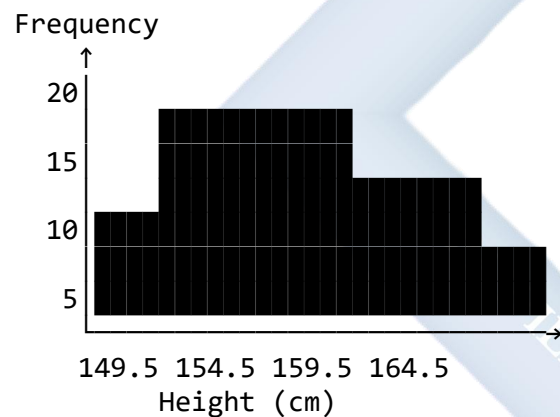
Used for: Continuous, grouped data

Key Features:

- Bars **must touch** (no gaps)
- X-axis marked with **Class Boundaries**
- Y-axis represents *frequency*
- Area of bar = frequency (usually)
- Shows continuous data distribution

Visual Structure:

Histogram (Continuous Data):



NO GAPS - bars touch!
(Shows continuous data)

Bar Chart vs Histogram:

Feature	Bar Chart	Histogram
Data Type	Discrete Qualitative	Continuous Grouped
Gaps	YES (spaces)	NO (touching)
X-axis	Categories	Class

	or values	Boundaries
Bars can be rearranged?	YES	Usually NO

3. ROBUST WORKED EXAMPLE

Example 9.1: Histogram

The table below shows the heights of 50 SS1 students. Represent this data on a histogram.

Height (cm)	150-154	155-159	160-164	165-169	170-174
Frequency (f)	6	12	18	10	4

Solution:

Step 1: Find Class Boundaries

We must close the gaps between intervals.

Class	Boundaries
150-154	149.5-154.5
155-159	154.5-159.5
160-164	159.5-164.5
165-169	164.5-169.5
170-174	169.5-174.5

Step 2: Set Up Axes

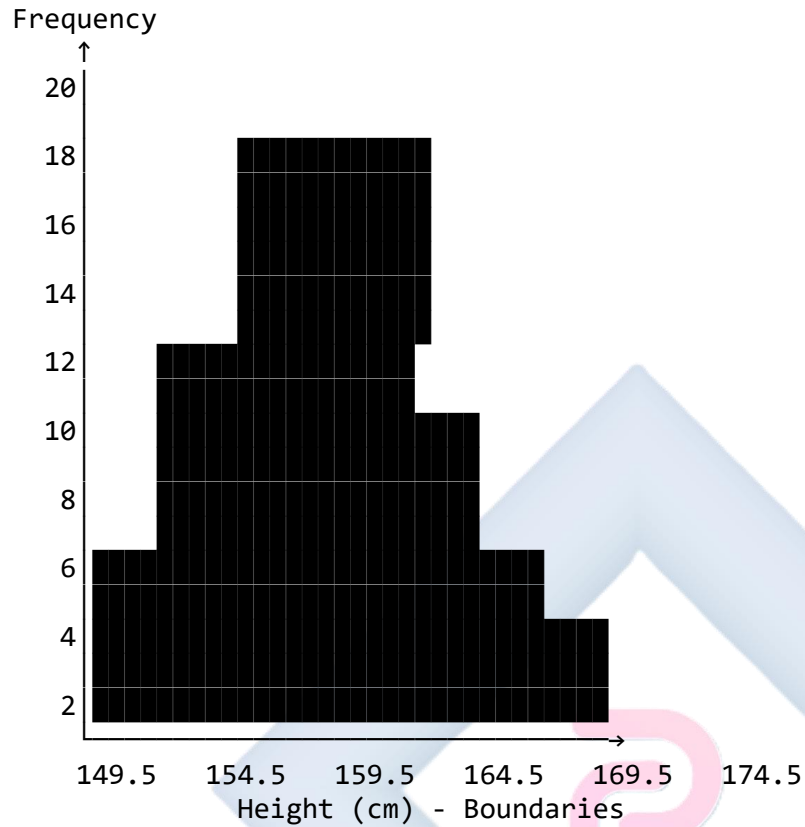
- **Y-axis:** Frequency (0 to 18, or slightly higher)
- **X-axis:** Height (cm), marked with boundaries

Step 3: Draw Bars

Each bar starts at lower boundary and ends at upper boundary, with height equal to frequency.

Histogram Diagram:

Histogram of Student Heights



Key features:

- ✓ Bars touch (continuous data)
- ✓ Boundaries on x-axis
- ✓ Tallest bar at 160-164 (modal class)
- ✓ Distribution roughly bell-shaped

Step-by-Step Drawing:

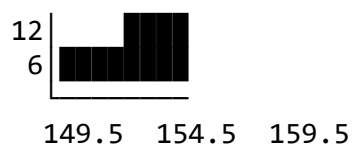
1. First bar (149.5 to 154.5):

Height = 6



2. Second bar (154.5 to 159.5):

Height = 12



3. Continue for all classes...

Interpretation:

From the histogram we can see:

- ✓ Most students (18) are 160-164 cm
- ✓ Few students (4) are 170-174 cm
- ✓ Distribution is roughly normal
- ✓ Mean height likely around 160 cm

4. REVISION QUESTIONS (WEEK 9)

1. Why do we use "grouped" frequency tables?
2. What is the main physical difference between a bar chart and a histogram?
3. What type of data is a bar chart used for?
4. What type of data is a histogram used for?
5. What is plotted on the horizontal (x) axis of a histogram?

5. ASSESSMENT (WEEK 9)

EVALUATION (IN-CLASS)

1. For the class interval 20-29, what is the:
 - a) class midpoint
 - b) class boundary
2. What does a gap between bars on a chart signify?
3. What is a linear graph most suitable for showing?
4. If you have data for "Number of children per family", would you use a bar chart or a histogram? Explain.
5. If you have data for "Height of students", would you use a bar chart or a histogram? Explain.

ASSIGNMENT (TAKE-HOME)

1. The following are the ages of 30 people at a party:
25, 30, 18, 45, 42, 28, 19, 20, 22, 33,
35, 40, 38, 29, 27, 22, 25, 30, 31, 26,
41, 37, 26, 28, 30, 33, 39, 21, 19, 44
 - a. Create a grouped frequency distribution table for this data. Use class intervals: 15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49.

- b. Draw a **Histogram** to represent this data. (Remember to use class boundaries!)



WEEK 10: STATISTICS III

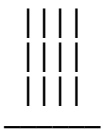
1. INTRODUCTION

This week, we learn the final two methods of data presentation: **Pie Charts** (for showing proportions) and **Frequency Polygons** (for showing the shape of a distribution).

Visual Overview:

Statistical Graphs Summary:

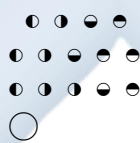
BAR CHART
(discrete)



HISTOGRAM
(continuous)



PIE CHART
(proportions)



FREQUENCY
POLYGON
(distribution)



2. CONTENT

A. PIE CHARTS

Definition: A circular chart divided into **sectors**, where each sector represents a *proportion* of the whole.

Use: Best for showing:

- Market share
- Percentage breakdown
- Parts of a whole
- Budget allocation
- Survey results (proportions)

Concept:

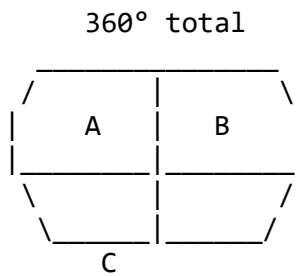
The whole circle (360°) represents the total frequency (or 100%).

Formula:

$$\text{Angle of Sector} = \frac{\text{Frequency of item}}{\text{Total Frequency}} \times 360^\circ$$

Visual Concept:

Pie Chart Structure:



Each slice (sector) = proportion of whole

If Total = 100 students

Sector A = 40 students

Angle A = $(40/100) \times 360^\circ = 144^\circ$

Example 10.1: Pie Chart

In an SS1 class of 80 students, 30 are in the Arts stream, 40 are in Science, and 10 are in Commercial. Represent this on a pie chart.

Solution:

Step 1: Find Total Frequency

Total = $30 + 40 + 10 = 80$ students

Step 2: Calculate the angle for each sector

Arts: Angle = $\frac{30}{80} \times 360^\circ = \frac{3}{8} \times 360^\circ = 3 \times 45^\circ = 135^\circ$

Science: Angle = $\frac{40}{80} \times 360^\circ = \frac{1}{2} \times 360^\circ = 180^\circ$

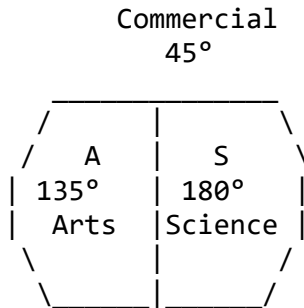
Commercial: Angle = $\frac{10}{80} \times 360^\circ = \frac{1}{8} \times 360^\circ = 45^\circ$

Step 3: Verification

Check that angles add to 360° : $135^\circ + 180^\circ + 45^\circ = 360^\circ \checkmark$

Step 4: Draw the Pie Chart

Complete Pie Chart:



Legend:

Arts: 30 students (135°) - Light shade

Science: 40 students (180°) - Medium shade

Commercial: 10 students (45°) - Dark shade

Drawing Steps:

Step 1: Draw circle



Step 2: Draw radius



Step 3: Measure 180° (Science)



Step 4: Measure 135° (Arts)



Step 5: Remaining 45° (Commercial)
Complete!

Calculation Table:

Stream	Frequency	Calculation	Angle
Arts	30	$(30/80) \times 360^\circ$	135°
Science	40	$(40/80) \times 360^\circ$	180°
Commercial	10	$(10/80) \times 360^\circ$	45°
Total	80		360°

B. FREQUENCY POLYGONS

Definition: A line graph formed by joining the **midpoints** of the tops of the bars of a histogram.

Use:

- Shows the *shape* of the distribution more clearly
- Good for comparing two datasets on the same graph
- Emphasizes trends and patterns
- Simpler visual than histogram

Plotting:

- **X-axis:** Class Midpoints (Marks)
- **Y-axis:** Frequency

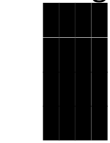
"Grounding" the Polygon:

To make it a closed shape ("polygon"), we always add an "imaginary" class with 0 frequency at the beginning and at the end.

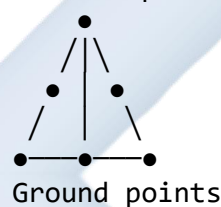
Visual Concept:

From Histogram to Frequency Polygon:

Histogram:



Mark midpoints:



Connect points:



3. ROBUST WORKED EXAMPLE

Example 10.2: Frequency Polygon

Use the student height data from Week 9 to draw a frequency polygon.

Height (cm)	150-154	155-159	160-164	165-169	170-174
Frequency (f)	6	12	18	10	4

Solution:

Step 1: Find Class Midpoints

Class	Calculation	Midpoint
150-154	$(150+154)/2$	152
155-159	$(155+159)/2$	157
160-164	$(160+164)/2$	162
165-169	$(165+169)/2$	167

Class	Calculation	Midpoint
170-174	$(170+174)/2$	172

Step 2: Add "Grounding" Points

Before first class (145-149):

- Midpoint = 147
- Frequency = 0

After last class (175-179):

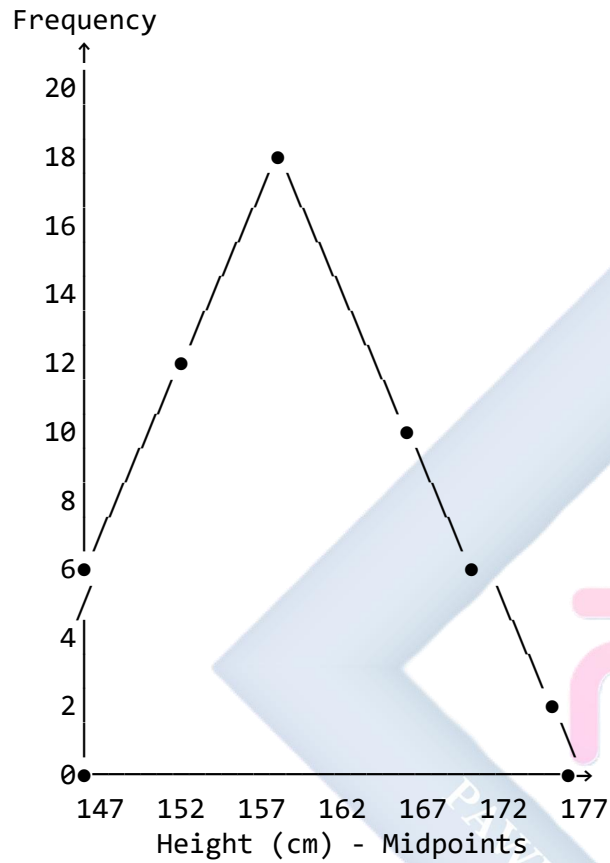
- Midpoint = 177
- Frequency = 0

Step 3: Create Plotting Table

Midpoint (x)	Frequency (f)
147	0 (ground)
152	6
157	12
162	18
167	10
172	4
177	0 (ground)

Step 4: Plot and Connect

Frequency Polygon



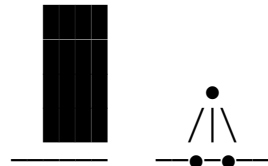
Key features:

- ✓ Starts and ends at ground (0)
- ✓ Connects midpoints
- ✓ Shows distribution shape
- ✓ Peak at 162 cm (modal class)

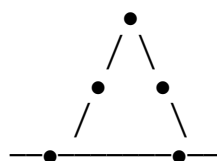
Comparison: Histogram vs Frequency Polygon

Same data, different views:

HISTOGRAM:



FREQUENCY POLYGON:



Bars show
amounts

Line shows
trend/shape

4. REVISION QUESTIONS (WEEK 10)

1. What is the main purpose of a pie chart?
 2. What is the formula for calculating the angle of a sector in a pie chart?
 3. What is plotted on the x-axis of a frequency polygon?
 4. What is plotted on the x-axis of a histogram?
 5. Why must a frequency polygon be "grounded" at 0 frequency?
-

5. ASSESSMENT (WEEK 10)

EVALUATION (IN-CLASS)

1. If a sector on a pie chart represents 20 people out of a total of 80, what is the angle of that sector?
 2. What is the main advantage of a frequency polygon over a histogram?
 3. How can you draw a frequency polygon *using* a histogram?
 4. A pie chart is used to show a student's budget. "Food" takes up 120° . What *percentage* of the budget is food?
 5. For the class interval 30-39, what is the class midpoint?
-

ASSIGNMENT (TAKE-HOME)

1. A family spends its monthly income as follows:
 - Food: ₦30,000
 - Rent: ₦20,000
 - School Fees: ₦15,000
 - Transport: ₦10,000
 - Savings: ₦15,000

a. What is the total monthly income? b. Calculate the angle for *each* item. c. Draw a neat, labeled **Pie Chart** to represent this data.
2. Use the data from the **Week 9 Assignment** (ages of 30 people at a party): a. Create a table showing the **Class Midpoints** and **Frequencies**. b. Draw a **Frequency Polygon** to represent this data. (Remember to ground it!)

WEEK 11: REVISION

1. INTRODUCTION

This final week is dedicated to revising the two major themes of this term: **Mensuration** and **Statistics**. The examination will test your ability to recall formulas, apply them accurately, and present data clearly.

Term Overview:

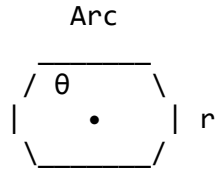
THIRD TERM TOPICS SUMMARY
MENSURATION (Weeks 1-6): • Arcs, sectors, segments • Sector-cone relationship • Surface areas • Volumes • Frustums & compound shapes
STATISTICS (Weeks 8-10): • Data collection & types • Frequency tables • Bar charts & histograms • Pie charts & frequency polygons

2. MENSURATION REVISION (FORMULA SHEET)

CIRCLES:

CIRCLE FORMULAS
Circumference: $C = 2\pi r$ Area: $A = \pi r^2$ Arc Length: $L = (\theta/360) \times 2\pi r$ Sector Area: $A = (\theta/360) \times \pi r^2$ Segment: Sector - $\frac{1}{2}r^2 \sin \theta$ Sector Perimeter: Arc + $2r$

Visual Reminder:



$$\text{Arc} = (\theta/360) \times 2\pi r$$

$$\text{Sector} = (\theta/360) \times \pi r^2$$

CONE FROM SECTOR:

Sector → Cone Transformation:

Sector	Cone
r (radius)	l (slant)
Arc length	$2\pi R$ (base)
Sector area	$CSA = \pi Rl$

Key formula: $(\theta/360) \times 2\pi r = 2\pi R$

SURFACE AREAS:

SURFACE AREA FORMULAS	
Cube:	$SA = 6l^2$
Cuboid:	$SA = 2(lw + lh + wh)$
Cylinder:	$CSA = 2\pi rh$ $TSA = 2\pi r(r + h)$
Cone:	$CSA = \pi Rl$ $TSA = \pi R(R + l)$ $l^2 = h^2 + R^2$
Prism:	$2(\text{Base}) + (\text{Perimeter} \times L)$
Pyramid:	Base + Triangular faces

VOLUMES:

VOLUME FORMULAS	
PRISMS (Full):	
• Cube:	$V = l^3$
• Cuboid:	$V = lwh$
• Cylinder:	$V = \pi r^2 h$

- Prism: $V = (\text{Base Area}) \times h$

PYRAMIDS (1/3 rule):

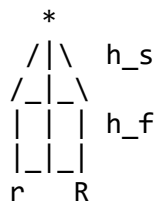
- Pyramid: $V = \frac{1}{3}(\text{Base Area}) \times h$
- Cone: $V = \frac{1}{3}\pi r^2 h$

FRUSTUM:

Frustum Strategy:

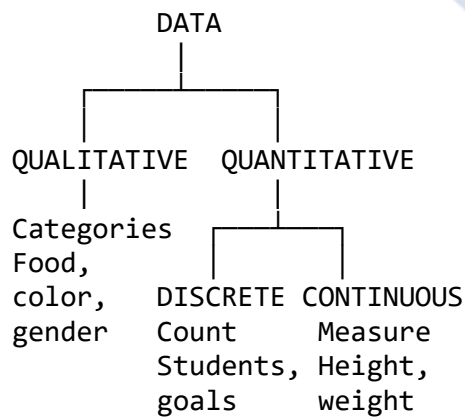
1. Use similar triangles:
 $r/R = h_s/H$
2. $V = V_{\text{large}} - V_{\text{small}}$
3. $SA = \text{Top} + \text{Bottom} + CSA$

Visual:



3. STATISTICS REVISION (PROCESS SHEET)

DATA TYPES:



DATA SOURCES:

Aspect	Primary	Secondary
--------	---------	-----------

Source Cost Time Control	You High Slow Full	Others Low Quick None
-----------------------------------	-----------------------------	--------------------------------

GRAPHS SUMMARY:

Graph	Data	Features
Bar Chart	Discrete Qualitative	Gaps Categories
Histogram	Continuous Grouped	No gaps Boundaries
Pie Chart	Any	Proportions 360° total
Frequency Polygon	Continuous Grouped	Midpoints Grounded

Pie Chart Formula: $\text{Angle} = \frac{\text{Frequency}}{\text{Total}} \times 360^\circ$

4. COMPREHENSIVE REVISION ASSIGNMENT (WEEK 11)

SECTION A: MENSURATION

Question 1: A sector of a circle with radius 21 cm has an angle of 150°.

- a. Find the length of the arc.
- b. Find the area of the sector.
- c. Find the area of the minor segment. ($\sin 150^\circ = \sin 30^\circ = 0.5$)

Question 2: A solid cylinder has a height of 10 cm and a base radius of 7 cm. Find:

- a. Its total surface area.
- b. Its volume.

Question 3: A cone is made by folding a sector of a circle with a 120° angle and a radius of 18 cm. Find:

- a. The base radius (R) of the cone.

- b. The vertical height (h) of the cone.
- c. The volume of the cone.

SECTION B: STATISTICS

Question 4: The scores of 40 students in a test are given in the table below.

Score (%)	21-30	31-40	41-50	51-60	61-70	71-80
Frequency	3	5	11	10	8	3

- a. Draw a **Histogram** for this data.
 - b. Draw a **Frequency Polygon** on the same graph (or a new one).
 - c. Calculate the angles needed to represent this data on a **Pie Chart**. (Do not draw)
-

5. FINAL EXAMINATION TIPS

Mensuration Tips:

- ✓ Always identify what's given and what's needed
 - ✓ Draw clear diagrams
 - ✓ Label all measurements
 - ✓ Check if you need slant or vertical height
 - ✓ Use Pythagoras for height conversions
 - ✓ Remember: Cone volume = $\frac{1}{3}$ Cylinder volume
 - ✓ Show all working steps
-

Statistics Tips:

- ✓ Identify data type first (qualitative/quantitative)
 - ✓ Choose correct graph for data type
 - ✓ Use boundaries for histograms
 - ✓ Use midpoints for frequency polygons
 - ✓ Always check pie chart angles = 360°
 - ✓ Label all axes and provide titles
 - ✓ Show calculations clearly
-